Protocole de comparaison des librairies d'analyse des trajectoires : Sequenzo

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1. Introduction

L'objectif de ce document est de proposer une vue d'ensemble des possibilités qu'offre la librairie `Sequenzo` en Python pour l'analyse de séquences, et plus généralement de trajectoires. Ce document a été pensé pour être mis en parallèle avec son équivalent dédié aux packages `TraMineR` (+ `TraMineRextras`) en R et `TanaT` en Python également, afin de comparer les trois outils et de déterminer le plus adapté selon la situation à traiter.

Pour faciliter cette comparaison, les trois documents suivent le même plan :

- Présentation des formats de données acceptés par les packages.
- Exploration des jeux de données intégrés et méthodes de simulation.
- Introduction des données "fil rouge" utilisées pour illustrer la suite du document.
- Manipulation des données et sélection des séquences selon différents critères.
- Description statistique et visualisation des séquences.
- Construction de typologies (calcul des dissimilarités et algorithmes de clustering), incluant le cas multidimensionnel (*multichannel*).
- Autres analyses (divergence, survie, test entre groupes, formatage pour autres analyses)
- Autres possibilités (séquences d'événements et séquences d'intervalles).

Nous traiterons dans ce document essentiellement de l'analyse de séquences d'**états**. Les autres types (événements et intervalles) seront spécifiquement abordés dans la dernière section.

In [1]:
```
import sequenzo
import time
import pandas as pd
import numpy as np
import random
from datetime import datetime
import datetime as dt
from scipy.stats import entropy
from scipy.spatial.distance import pdist, squareform
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from functools import reduce
```

2. Format de données

Le seul format de données accepté par `Sequenzo` est le format STS avec une ligne par individu et une colonne par point dans le temps.

Prenons l'exemple avec deux séquences X et Y, où a, b et c représentent les états possibles (l'alphabet):

$$X = (a, a, a, b, b, c, c) \quad \text{et} \quad Y = (a, a, a, c, c, c, c)$$

In [2]:
```
df_exemple = pd.DataFrame(
    {
        "id":["X","Y"],
        "t1":["a","a"],
        "t2":["a","a"],
        "t3":["a","a"],
        "t4":["b","c"],
        "t5":["b","c"],
        "t6":["c","c"],
        "t7":["c","c"]
    }
)
df_exemple
```

Out[2]:

	id	t1	t2	t3	t4	t5	t6	t7
0	X	a	a	a	b	b	c	c
1	Y	a	a	a	c	c	c	c

Ainsi, la librairie est assez restrictive vis-à-vis du format accepté pour la création d'un objet sequence. Cependant, elle met à disposition deux fonctions permettant de convertir les données du format STS au format SPELL et inversement.

In [3]:
```
df_long = sequenzo.wide_to_long_format_data(
    df_exemple,
    id_col="id",
    time_cols=["t1","t2","t3","t4","t5","t6","t7"],
    var_name="time_point",
    value_name="state"
)
df_long.sort_values("id")
```

Out[3]:

	id	time_point	state
0	X	t1	a
2	X	t2	a
4	X	t3	a
6	X	t4	b
8	X	t5	b
10	X	t6	c
12	X	t7	c
1	Y	t1	a
3	Y	t2	a
5	Y	t3	a
7	Y	t4	c
9	Y	t5	c
11	Y	t6	c
13	Y	t7	c

In [4]:
```
df_wide = sequenzo.long_to_wide_format_data(
    df_long,
    id_col="id",
    time_col="time_point",
    value_col="state"
)
df_wide
```

Out[4]:

	id	state_t1	state_t2	state_t3	state_t4	state_t5	state_t6	state_t7
0	X	a	a	a	b	b	c	c
1	Y	a	a	a	c	c	c	c

Pour les autres formats de données, il sera nécessaire à l'utilisateur de procéder lui-même à une conversion à partir de librairies de transformation de données comme `pandas` .

Définition d'une séquence à partir des données au format STS :

In [5]:
```
sequence = sequenzo.SequenceData(
    data=df_exemple,
    time=["t1","t2","t3","t4","t5","t6","t7"],
    states=['a','b','c'],
    labels=['a','b','c'],
    id_col='id'
)
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 2
[>] Number of time points: 7
[>] Min/Max sequence length: 7 / 7
[>] States: ['a', 'b', 'c']
[>] Labels: ['a', 'b', 'c']
[>] Weights: Not provided
```

3. Données exemples et simulations

Jeux de données exemple

La librairie Sequenzo met à disposition plusieurs jeux de données pour illustrer ses fonctionnalités :

Jeu de données	Type de séquence	Thématique	Individus	Longueur	États	Covariables
biofam + version binaire pour les états (left, married et child)	Etats	Statut familial (Swiss Household Panel)	2000	16	8	7
Chinese Territories' Colonial History	Etats	Statut de souveraineté de territoires chinois	11	111	7	0
Country child mortality	Etats	Mortalité infantile dans les pays du monde	194	302	10	0
Country GPD per capita	Etats	Emissions de CO2	193	223	5	0
CO_2 Emissions Dataset (découpage en déciles ou quintiles / découpage local ou global)	Etats	Emissions de CO2	193	223	5 ou 10	0
Country life expectancy	Etats	Espérance de vie dans les	194	302	10	0

Jeu de données	Type de séquence	Thématique	Individus	Longueur	États	Covariables
		pays du monde				
dyadic (children)	Etats	Formation de la famille de 15 à 39 ans, enfants	60	24	6	1
dyadic (parents)	Etats	Formation de la famille de 15 à 39 ans, parents	60	24	6	1
mvad	Etats	Transition école-travail	712	72	6	13
pairfam activity (par mois)	Etats	Trajectoires d'emploi / d'activité	1027	264	8	16
pairfam activity (par années)	Etats	Trajectoires d'emploi / d'activité	1027	22	8	0
pairfam family (par mois)	Etats	Formation de la famille	1027	264	9	16
pairfam family (par années)	Etats	Formation de la famille	1027	22	9	0
political science aid shock	Etats	Sciences politiques	165	36	2	0
political science donor fragmentation	Etats	Sciences politiques	156	36	3	0
College Students' stress	Etats	Evolution du stress d'étudiants	76	8	3	5

`Sequenzo` propose plusieurs jeux de données multichannel : `dyadic` , `pairfam` et `political science` , à chaque fois avec deux domaines.

Simulations de données

La librairie n'offre pas la possibilité de simuler des données. Toutefois, il est possible pour l'utilisateur de le faire par lui-même en utilisant des librairies telles que `random` et `numpy` .

In [6]:
```
import numpy as np
```

```
import pandas as pd

def seqgen(
    n: int,
    length,  # int ou array-like de longueur n
    alphabet: list,
    p: list = None,
    id_prefix: str = "seq",
    missing: str = None,
    p_missing: float = 0.0,
    pad: str = "NA",
    seed: int = None
) -> pd.DataFrame:
    """
    Génère n séquences aléatoires au format STS (State Sequence).

    Paramètres
    ----------
    n : int
        Nombre de séquences à générer.
    length : int ou array-like de longueur n
        Longueur de chaque séquence. Si un entier est fourni, toutes les
        séquences ont la même longueur. Si un vecteur est fourni, chaque
        séquence i a la longueur length[i].
    alphabet : list
        Liste des états possibles.
    p : list, optional
        Probabilités associées à chaque état (doit sommer à 1).
        Si None, distribution uniforme.
    id_prefix : str, optional
        Préfixe pour les identifiants de séquences (défaut : "seq").
    missing : str, optional
        Valeur représentant les données manquantes (ex : "*" comme dans TraMineR).
        Si None, pas de données manquantes générées.
    p_missing : float, optional
        Probabilité qu'un état soit manquant (entre 0 et 1). Ignoré si missing=None
    pad : str, optional
        Valeur utilisée pour remplir les positions inexistantes des séquences
        plus courtes que la longueur maximale (défaut : "NA").
    seed : int, optional
        Graine pour la reproductibilité.

    Retourne
    --------
    pd.DataFrame
        DataFrame au format STS : colonne 'id' + colonnes 't1' à 't{max_length*'.
        Les positions hors longueur individuelle sont remplies avec `pad`.
    """
    rng = np.random.default_rng(seed)

    # Normalisation de length en vecteur
    if np.isscalar(length):
        lengths = np.full(n, length, dtype=int)
    else:
        lengths = np.asarray(length, dtype=int)
        if len(lengths) != n:
```

```
            raise ValueError("'length' doit être un scalaire ou un vecteur de longu
        if np.any(lengths <= 0):
            raise ValueError("Toutes les valeurs de 'length' doivent être stricteme

    max_length = lengths.max()

    # Validation des probabilités
    if p is not None:
        p = np.array(p, dtype=float)
        if len(p) != len(alphabet):
            raise ValueError("'p' doit avoir la même longueur que 'alphabet'.")
        if not np.isclose(p.sum(), 1.0):
            raise ValueError("Les probabilités dans 'p' doivent sommer à 1.")
    else:
        p = np.ones(len(alphabet)) / len(alphabet)

    if not (0.0 <= p_missing < 1.0):
        raise ValueError("'p_missing' doit être compris entre 0 (inclus) et 1 (excl

    # Génération séquence par séquence (longueurs variables)
    sequences = []
    for i in range(n):
        seq = rng.choice(alphabet, size=lengths[i], p=p).astype(object)

        # Injection de valeurs manquantes
        if missing is not None and p_missing > 0.0:
            mask = rng.random(size=lengths[i]) < p_missing
            seq[mask] = missing

        # Padding jusqu'à max_length
        padding = np.full(max_length - lengths[i], pad, dtype=object)
        sequences.append(np.concatenate([seq, padding]))

    # Construction du DataFrame STS
    ids = [f"{id_prefix}_{i+1}" for i in range(n)]
    col_names = [f"t{j+1}" for j in range(max_length)]
    df = pd.DataFrame(sequences, columns=col_names)
    df.insert(0, "id", ids)

    return df
```

In [7]:
```
# Exemple de 10 séquences de longueur 12 avec pour alphabet {a, b, c} avec une répa
seqgen(n=3, length=[8,10,7], alphabet=["a","b","c"],p=[0.75,0.1,0.15],missing="Miss
```

Out[7]:

	id	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10
0	seq_1	b	a	a	a	a	a	a	a	NA	NA
1	seq_2	a	b	a	c	a	c	a	a	a	a
2	seq_3	c	b	a	a	a	c	Missing	NA	NA	NA

4. Données fil-rouge

Pour réaliser cette comparaison systématique, nous utilisons les mêmes jeux de données dans les trois environnements. Pour cela, nous nous appuyons sur le Swiss Household Panel (SHP), une étude longitudinale d'envergure nationale lancée en 1999 qui recueille annuellement des informations sociodémographiques et économiques auprès d'un échantillon représentatif de ménages suisses. Pour les analyses de séquences unidimensionnelles, nous nous appuyons sur le jeu de données *_Family life states* `biofam`, notamment proposé comme jeu de données exemple par `TraMineR` et `sequenzo`. Pour les analyses nécessitant des séquences multidimensionnelles avec plusieurs thématiques, nous travaillons à partir d'un extrait plus large du Swiss Household Panel `extract_shp` que nous avons réalisé nous-mêmes. Nous décrivons plus en détail ces deux jeux-de données ci-après.

Family life states `biofam`

Le jeu de données *Family life states* (`biofam`) est issu de l'enquête biographique rétrospective menée en 2002 dans le cadre du panel suisse de ménages. Il documente les trajectoires de formation familiale d'un échantillon de 2 000 individus sur une période de seize années, sans données manquantes, retraçant l'évolution de leur situation familiale de l'âge de 15 ans à l'âge de 30 ans.

Ces séquences sont initialement structurées au format **STS** et donc sont directement exploitables par `TraMineR`. Une version convertie au format **SPELL** est utilisée pour `TanaT`. L'alphabet des séquences décrites dans ce jeu de données est composé de huit statuts familiaux décrivant à la fois le lieu de résidence, le statut marital et la parentalité. Les individus sont ainsi catégorisés à chaque pas de temps selon qu'ils vivent au domicile parental (*Parent*), qu'ils l'ont quitté (*Left*), qu'ils ont un enfant (*Child*), qu'ils sont mariés (*Married*) qu'ils ont quitté le domicile familial et qu'ils sont mariés (*Left+Marr*), qu'ils ont quitté le domicile familial et ont un enfant (*Left+Child*), qu'ils soient hors du domicile familial, mariés et avec un enfant (*Left+Marr+Child*) ou qu'ils soient divorcés (*Divorced*).

En complément de la séquence d'états, le jeu de données inclut plusieurs covariables transversales: caractéristiques socio-démographiques de l'individu (sexe, année de naissance, première nationalité, langue de réponse au questionnaire) et culturelles (religion et participation religieuse), ainsi que sur l'origine sociale avec le statut social du père et de la mère. Cependant, certaines de ces variables sont manquantes chez certains individus. Nous utilisons uniquement la variable sexe (disponible pour l'ensemble des individus) dans nos comparaisons.

Enfin, ce jeu de données propose deux sets de poids : l'un ajusté sur l'ensemble de la population suisse et l'autre ramené à la taille de l'échantillon. Cependant, ils ne sont fournis qu'à titre illustratif, puisque ce jeu de données est un sous-échantillon des réponses globales du panel et donc les poids proposés ne sont plus adaptés.

Extrait du Swiss Household Panel `extract_shp`

Le jeu de données `extract_shp` est un extrait des réponses à l'enquête longitudinale menée auprès d'un panel de ménages suisses entre 1999 et 2018 inclus. Nous avons sélectionné uniquement les ménages sans données manquantes et ayant participé à l'enquête chaque année, ce qui conduit à 20 observations pour 1032 foyers. Six variables ont été retenues afin de constituer trois thématiques, comprenant chacune une à trois variables évoluant au fil du temps :

Thématique familiale : Pour cette thématique, nous considérons uniquement le nombre de personnes composant le ménage. Cette variable prend des valeurs allant de 1 à 12, que nous regroupons en cinq catégories : 1, 2, 3, 4, et 5 habitants ou plus au sein du foyer.

Thématique résidentielle : Pour cette thématique, nous nous intéressons à la région d'habitation, au type de commune d'habitation et au statut résidentiel. La région d'habitation comporte sept modalités, les cantons correspondants étant précisés entre parenthèses : Lac Léman (VD, VS, GE), Middleland (BE, FR, SO, NE, JU), Suisse Nord-Ouest (BS, BL, AG), Zurich, Suisse Est (GL, SH, AR, AI, SG, GR, TG), Suisse Centrale (LU, UR, SZ, OW, NW, ZG) et Tessin.

La variable décrivant le type de commune d'habitation comporte neuf modalités : centres, communes de banlieue, communes aisées, communes urbaines périphériques, communes touristiques, communes industrielles et tertiaires, communes rurales de banlieue, communes mixtes agricoles et communes agricoles périphériques.

Enfin, le statut résidentiel comporte trois modalités : locataire, propriétaire ou copropriétaire, et locataire non payant.

Thématique qualité de vie : Cette thématique est composée d'une variable décrivant l'état du lieu de vie du foyer et d'une variable mesurant la satisfaction à l'égard de la situation financière du foyer. L'état du lieu de vie comporte trois modalités : en mauvais état, en bon état mais n'ayant pas été rénové récemment, et neuf ou récemment rénové.

La satisfaction à l'égard de la situation financière du foyer prend des valeurs entières allant de 0 (pas du tout satisfait) à 10 (totalement satisfait). Nous regroupons cette variable en cinq catégories : 0-2, 3-4, 5-6, 7-8 et 9-10.

Cette base de données est proposée au format **SPELL**, compatible avec `TraMineR` et `TanaT`. Une version au format **STS** est utilisée pour `sequenzo`.

Dans le cadre de ce document, nous avons choisi de sélectionner ces variables et de regrouper certaines modalités afin de disposer d'un support permettant d'illustrer les possibilités des trois packages étudiés. Ce document n'a pas pour objectif de produire une véritable analyse et, dans un contexte applicatif réel, ces choix ne conduiraient pas

nécessairement à des analyses pertinentes.

In [8]:
```
# Préparation du dataset biofam
biofam = pd.read_csv("../biofam.csv").dropna(axis=0,subset="id")
biofam_static = pd.read_csv("../biofam_sts.csv")

# Préparation du dataset extract_shp
extract_shp_long = pd.read_csv("../data_SHP.csv")
extract_shp_sts_region = sequenzo.long_to_wide_format_data(
    extract_shp_long,
    id_col="idhous",
    time_col="year",
    value_col="region"
)
extract_shp_sts_community = sequenzo.long_to_wide_format_data(
    extract_shp_long,
    id_col="idhous",
    time_col="year",
    value_col="com2_"
)
extract_shp_sts_type_logement = sequenzo.long_to_wide_format_data(
    extract_shp_long,
    id_col="idhous",
    time_col="year",
    value_col="hh29"
)
extract_shp_sts_nbpers = sequenzo.long_to_wide_format_data(
    extract_shp_long,
    id_col="idhous",
    time_col="year",
    value_col="nbpers"
)
extract_shp_sts_condlieuvie = sequenzo.long_to_wide_format_data(
    extract_shp_long,
    id_col="idhous",
    time_col="year",
    value_col="hh14"
)
extract_shp_sts_situatuionfinance = sequenzo.long_to_wide_format_data(
    extract_shp_long,
    id_col="idhous",
    time_col="year",
    value_col="hi30"
)
dfs = [
    extract_shp_sts_region,
    extract_shp_sts_community,
    extract_shp_sts_type_logement,
    extract_shp_sts_nbpers,
    extract_shp_sts_condlieuvie,
    extract_shp_sts_situatuionfinance
]
extract_shp_sts = reduce(lambda left, right: pd.merge(left, right, on='idhous', how
extract_shp_sts.to_csv("C:/Users/malnoeg/Documents/Stage_M2_SequenceAnalysis/Protoc
sequenzo.clean_time_columns_auto(extract_shp_sts_region, prefix_patterns=["region_"
```

```
sequenzo.clean_time_columns_auto(extract_shp_sts_community, prefix_patterns=["com2_
sequenzo.clean_time_columns_auto(extract_shp_sts_type_logement, prefix_patterns=["h
sequenzo.clean_time_columns_auto(extract_shp_sts_nbpers, prefix_patterns=["nbpers_"
sequenzo.clean_time_columns_auto(extract_shp_sts_condlieuvie, prefix_patterns=["hh1
sequenzo.clean_time_columns_auto(extract_shp_sts_situatuionfinance, prefix_patterns
```

5. Gestion de données

Pour l'analyse de séquence il peut se réveler très utile de filtrer et sélectionner des séquences sur certains critères comme la sélection d'un échantillon ou le respect d'une contrainte de temporelle. La librairie Python `TanaT` sépare le filtrage et la sélection des données en plusieurs catégories :

- **Query** : sélection par requête sur les états (ex. : présence de l'état A).
- **Static** : filtrage via les covariables des individus (ex. : sélection par genre).
- **Time** : restriction temporelle ou sélection sur une fenêtre de temps donnée (ex. : trajectoires entre 2001 et 2005).
- **Pattern** : recherche de motifs séquentiels précis (ex. : état A → état B → état D).
- **Length** : filtrage selon la longueur des séquences (ex. : taille strictement supérieure à 5).

Nous allons illuster ces différents filtrages par des exemples sur le jeu de données `biofam` .

Nous commençons par définir l'objet séquence avec `SequenceData` .

In [9]:
```
biofam = sequenzo.load_dataset("biofam")
biofam = biofam.reset_index()
biofam.head(5)
```

Out[9]:

	index	idhous	sex	birthyr	nat_1_02	plingu02	p02r01	p02r04	
0	0	66891.0	man	1943	Switzerland	german	Protestant or Reformed Church	religious celebrations and family events	ot e
1	1	28621.0	man	1935	Switzerland	german	Roman Catholic	once a week	ot e
2	2	57711.0	woman	1946	Switzerland	french	no denomination or religion	only for family ceremonies	ot e
3	3	17501.0	man	1918	Switzerland	german	Roman Catholic	once a week	mana
4	4	147701.0	man	1946	Switzerland	german	Roman Catholic	only for family ceremonies	

5 rows × 28 columns

In [10]:
```
timepoints_columns = ["15","16","17","18","19","20","21","22","23","24","25","26","
states_biofam =[0,1,2,3,4,5,6,7]
labels_biofam =['Parent','Left','Married','Left+Marr','Child','Left+Child','Left+Ma
color_list=["#40b15c","#a89cec","#ffbb62","#f5ec73","#3760e9","#ee4091","#916237","
biofam_seq = sequenzo.SequenceData(
    data=biofam,
    time= timepoints_columns,
    states=states_biofam,
    labels=labels_biofam,
    id_col='index',
    custom_colors=color_list
)
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 2000
[>] Number of time points: 16
[>] Min/Max sequence length: 16 / 16
[>] States: [0, 1, 2, 3, 4, 5, 6, 7]
[>] Labels: ['Parent', 'Left', 'Married', 'Left+Marr', 'Child', 'Left+Child', 'Left
+Marr+Child', 'Divorced']
[>] Weights: Not provided
```

Query

Exemple : ne sélectionner que les séquences passant au moins une fois par l'état 1 (Left) au cours du temps.

In [11]:
```
# Récupération de l'index des individus ayant au moins l'état 1 (Left) dans leur sé
index_left = (biofam == 1).any(axis=1)
seq_left = biofam[index_left]

# Récréation d'un objet séquence à partir du df filtré
seq_filtered = sequenzo.SequenceData(
    data=seq_left,
    time=timepoints_columns,
    id_col="index",
    states=states_biofam,
    labels=labels_biofam
)
```

```
[i] Note: Some provided states are not observed in this dataset: [2, 4]
    Provided states are allowed to be a superset of observed values.
    Unique values actually in data: [0, 1, 3, 5, 6, 7]

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 896
[>] Number of time points: 16
[>] Min/Max sequence length: 16 / 16
[>] States: [0, 1, 2, 3, 4, 5, 6, 7]
[>] Labels: ['Parent', 'Left', 'Married', 'Left+Marr', 'Child', 'Left+Child', 'Left
+Marr+Child', 'Divorced']
[>] Weights: Not provided
```

Static

Exemple : sélection des personnes nées strictement après 1950.

In [12]:
```
# Récupération de l'index des individus ayant au moins l'état 1 (Left) dans leur sé
index_left = biofam["birthyr"] >1950
seq_left = biofam[index_left]

# Récréation d'un objet séquence à partir du df filtré
seq_filtered = sequenzo.SequenceData(
    data=seq_left,
    time=timepoints_columns,
    id_col="index",
    states=states_biofam,
    labels=labels_biofam
)
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 569
[>] Number of time points: 16
[>] Min/Max sequence length: 16 / 16
[>] States: [0, 1, 2, 3, 4, 5, 6, 7]
[>] Labels: ['Parent', 'Left', 'Married', 'Left+Marr', 'Child', 'Left+Child', 'Left
+Marr+Child', 'Divorced']
[>] Weights: Not provided
```

Time

Exemple : sélectionner les séquences comprises (en entiereté) entre le temps 1 et le temps

8.

In [13]:
```
# Récupération de l'index des individus ayant une longueur inférieure ou égale à 8
sequences = biofam_seq.sequences
index_left = [len(sequences[i])<=8 for i in range(0,len(sequences))]
seq_left = biofam[index_left]

# Récréation d'un objet séquence à partir du df filtré
seq_filtered = sequenzo.SequenceData(
    data=seq_left,
    time=timepoints_columns,
    id_col="index",
    states=states_biofam,
    labels=labels_biofam
)
```

```
[i] Note: Some provided states are not observed in this dataset: [0, 1, 2, 3, 4, 5,
6, 7]
    Provided states are allowed to be a superset of observed values.
    Unique values actually in data: []

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 0
[>] Number of time points: 16
[>] Min/Max sequence length: nan / nan
[>] States: [0, 1, 2, 3, 4, 5, 6, 7]
[>] Labels: ['Parent', 'Left', 'Married', 'Left+Marr', 'Child', 'Left+Child', 'Left
+Marr+Child', 'Divorced']
[>] Weights: Not provided
```

Pattern

Exemple : présence du pattern Parent -> Left + Married.

In [14]:
```
# Construction de biofam avec uniquement les indices des personnes avec la transiti
arr = biofam[timepoints_columns].values
from_state = arr[:, :-1]
to_state   = arr[:, 1:]
index_left = ((from_state == 0) & (to_state == 3)).any(axis=1)
seq_left = biofam[index_left]

# Récréation d'un objet séquence à partir du df filtré
seq_filtered = sequenzo.SequenceData(
    data=seq_left,
    time=timepoints_columns,
    id_col="index",
    states=states_biofam,
    labels=labels_biofam
)
```

```
[i] Note: Some provided states are not observed in this dataset: [1, 2, 4, 5]
    Provided states are allowed to be a superset of observed values.
    Unique values actually in data: [0, 3, 6, 7]

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 506
[>] Number of time points: 16
[>] Min/Max sequence length: 16 / 16
[>] States: [0, 1, 2, 3, 4, 5, 6, 7]
[>] Labels: ['Parent', 'Left', 'Married', 'Left+Marr', 'Child', 'Left+Child', 'Left
+Marr+Child', 'Divorced']
[>] Weights: Not provided
```

Length

Exemple : sélection des séquences de longueur strictement supérieure à 10.

In [15]:
```
# Récupération de l'index des individus ayant une longueur inférieure ou égale à 8
sequences = biofam_seq.sequences
index_left = [len(sequences[i])>10 for i in range(0,len(sequences))]
seq_left = biofam[index_left]

# Récréation d'un objet séquence à partir du df filtré
seq_filtered = sequenzo.SequenceData(
    data=seq_left,
    time=timepoints_columns,
    id_col="index",
    states=states_biofam,
    labels=labels_biofam
)
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 2000
[>] Number of time points: 16
[>] Min/Max sequence length: 16 / 16
[>] States: [0, 1, 2, 3, 4, 5, 6, 7]
[>] Labels: ['Parent', 'Left', 'Married', 'Left+Marr', 'Child', 'Left+Child', 'Left
+Marr+Child', 'Divorced']
[>] Weights: Not provided
```

Changement de granularité : aggrégation des états

Dans le même esprit de préparation des données, il est parfois utile de procéder à un **changement de granularité**, par exemple dans un contexte où les différents individus n'auraient pas répondu à la même fréquence mais que l'on souhaite les étudier ensemble.

`sequenzo` ne met à disposition de fonction ou de méthode pour le faire automatiquement mais il est possible avec `pandas` de modifier au préalable les données.

Exemple : aggrégation des états 4 à 4 en gardant l'état le + fréquent.

In [16]:
```
from scipy import stats
```

```
arr = biofam[timepoints_columns].values
n_periods = arr.shape[1]

# Pour chaque fenêtre de 4, garder l'état le plus fréquent
aggregated = np.array([
    stats.mode(arr[:, i*4:(i+1)*4], axis=1).mode
    for i in range(n_periods)
]).T

df_aggregated = pd.DataFrame(aggregated, columns=[f"period_{i+1}" for i in range(n_
df_aggregated.head(4)
```

```
C:\Users\malnoeg\AppData\Local\Temp\ipykernel_1060\1211512956.py:8: SmallSampleWarni
ng: All axis-slices of one or more sample arguments are too small; all elements of r
eturned arrays will be NaN. See documentation for sample size requirements.
  stats.mode(arr[:, i*4:(i+1)*4], axis=1).mode
```

Out[16]:

	period_1	period_2	period_3	period_4	period_5	period_6	period_7	period_8	period_
0	0.0	0.0	6.0	6.0	NaN	NaN	NaN	NaN	Na
1	1.0	1.0	1.0	6.0	NaN	NaN	NaN	NaN	Na
2	0.0	0.0	1.0	6.0	NaN	NaN	NaN	NaN	Na
3	0.0	1.0	1.0	1.0	NaN	NaN	NaN	NaN	Na

6. Description des séquences (statistiques et visualisations)

Alphabet

L'alphabet qui décrit les séquences peut s'obtenir par la méthode `alphabet`, les labels associés par `labels` et les dictionnaires entre labels et états s'obtient avec `label_to_state` ou `state_to_label`.

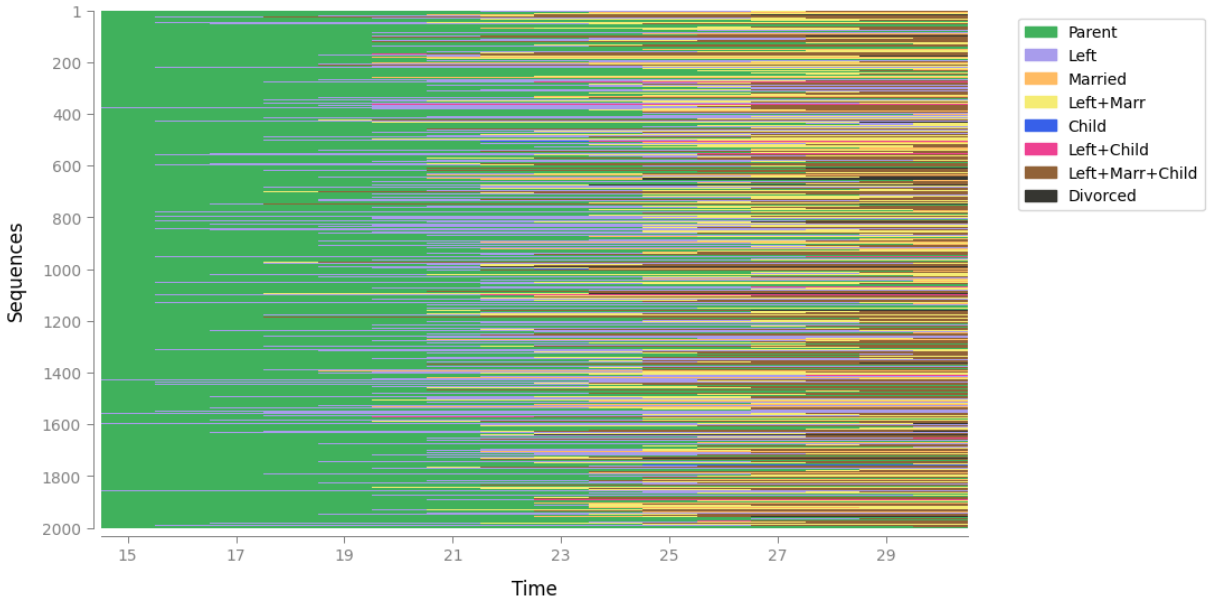
In [17]:
```
print(biofam_seq.alphabet)
print(biofam_seq.labels)
print(biofam_seq.label_to_state)
print(biofam_seq.state_to_label)
```

```
[0, 1, 2, 3, 4, 5, 6, 7]
['Parent', 'Left', 'Married', 'Left+Marr', 'Child', 'Left+Child', 'Left+Marr+Child',
'Divorced']
{'Parent': 0, 'Left': 1, 'Married': 2, 'Left+Marr': 3, 'Child': 4, 'Left+Child': 5,
'Left+Marr+Child': 6, 'Divorced': 7}
{0: 'Parent', 1: 'Left', 2: 'Married', 3: 'Left+Marr', 4: 'Child', 5: 'Left+Child',
6: 'Left+Marr+Child', 7: 'Divorced'}
```

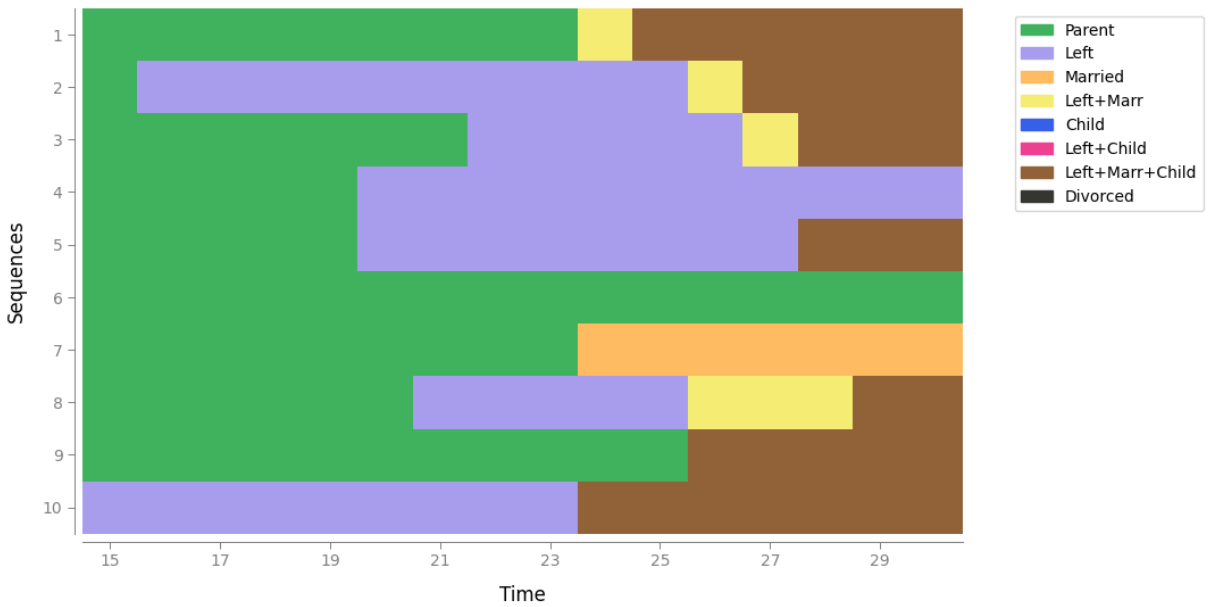
Visualisation des séquences

`sequenzo` propose une fonction permettant de visualiser les trajectoires brutes : `plot_sequence_index` . Elle permet à la fois de visualiser quelques séquences ou l'ensemble du jeu de données.

In [18]:
```
sequenzo.plot_sequence_index(
    biofam_seq,
    sort_by="unsorted"
)
```



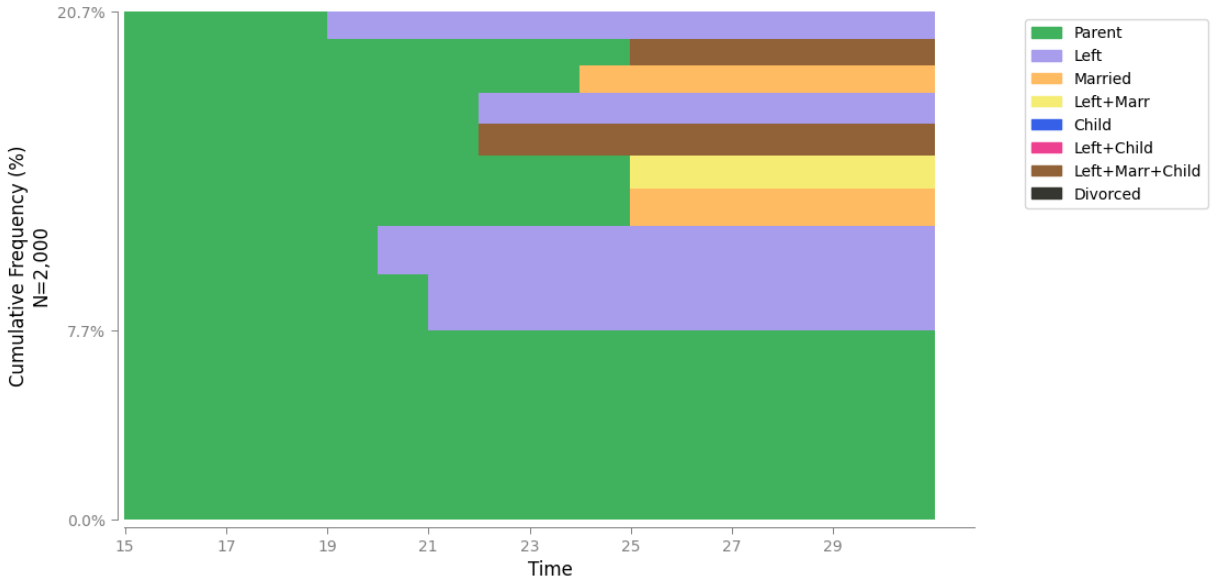
In [19]:
```
sequenzo.plot_sequence_index(
    biofam_seq,
    sort_by="unsorted",
    sequence_selection="first_n",
    n_sequences=10
)
```



Fréqeunces des séquences

sequenzo propose uniquement de visualiser le graphe des fréquences cumulatives des séquences les plus fréquentes mais pas la table directement pour l'ensemble des séquences.

In [20]:
```
sequenzo.plot_most_frequent_sequences(
    biofam_seq,
    top_n=10,          # how many sequences to show
    save_as=None,      # e.g., "top_sequences.png"
    dpi=200
)
```



In [21]:
```
# Fréquences
sequences_tuples = [tuple(row) for row in arr]
from collections import Counter
counts = Counter(sequences_tuples)
df_freq = pd.DataFrame([
    {"sequence": list(seq), "frequency": freq/len(arr)}
    for seq, freq in counts.most_common()
])

print(df_freq.sort_values("frequency").tail(10))
```

```
                                             sequence  frequency
7   [0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 2, 2, 2, 2, 2, 2]     0.0110
10  [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 3, 6, 6, 6, 6, 6]     0.0110
9   [0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]     0.0110
6   [0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1]     0.0125
5   [0, 0, 0, 0, 0, 0, 0, 6, 6, 6, 6, 6, 6, 6, 6, 6]     0.0130
4   [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 3, 3, 3, 3, 3, 3]     0.0135
3   [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 2, 2, 2, 2, 2]     0.0150
2   [0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]     0.0200
1   [0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]     0.0225
0   [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]     0.0770
```

Fréquences des états co-occurents

`sequenzo` ne propose pas de fonction permettant d'obtenir les fréquences des états co-occurents.

In [22]:
```
from collections import Counter

# For each individual, get the frozenset of unique states visited
state_sets = [frozenset(row) for row in arr]

counts = Counter(state_sets)

df_stocc = pd.DataFrame([
    {"states": sorted(seq), "frequency": freq/len(arr)}
    for seq, freq in counts.most_common()
])

print(df_stocc.sort_values("frequency").tail(10))
```

```
        states  frequency
9     [0, 3, 7]     0.0105
8     [0, 1, 6]     0.0585
7           [0]     0.0770
6        [0, 6]     0.0850
5     [0, 1, 3]     0.0890
4        [0, 3]     0.0930
3        [0, 2]     0.1025
2  [0, 1, 3, 6]     0.1295
1        [0, 1]     0.1350
0     [0, 3, 6]     0.1465
```

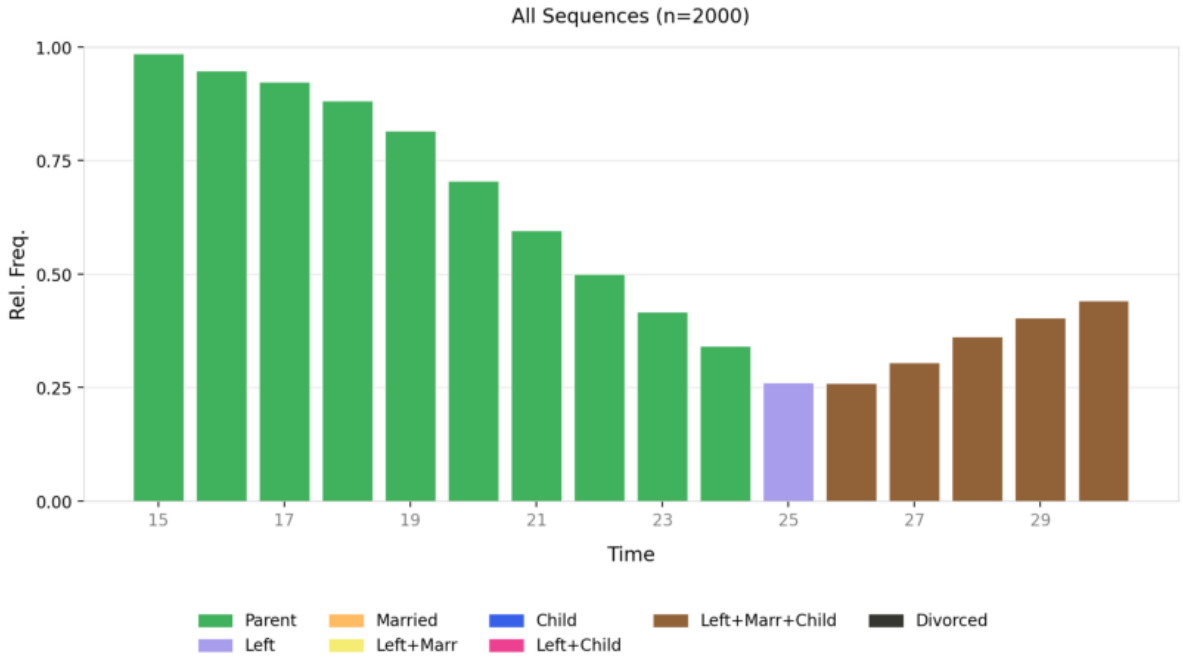
Séquence représentative

`sequenzo` ne met pas à disposition de fonction permettant d'extraire la séquence représentative (la séquence composée de l'état le plus populaire à chaque point de temps) mais propose une visualisation.

In [23]:
```
from scipy import stats
modal_states = stats.mode(arr, axis=0).mode
print("-".join(str(s) for s in modal_states))
```

```
0-0-0-0-0-0-0-0-0-0-1-6-6-6-6-6
```

In [24]:
```
sequenzo.plot_modal_state(
    biofam_seq
)
```



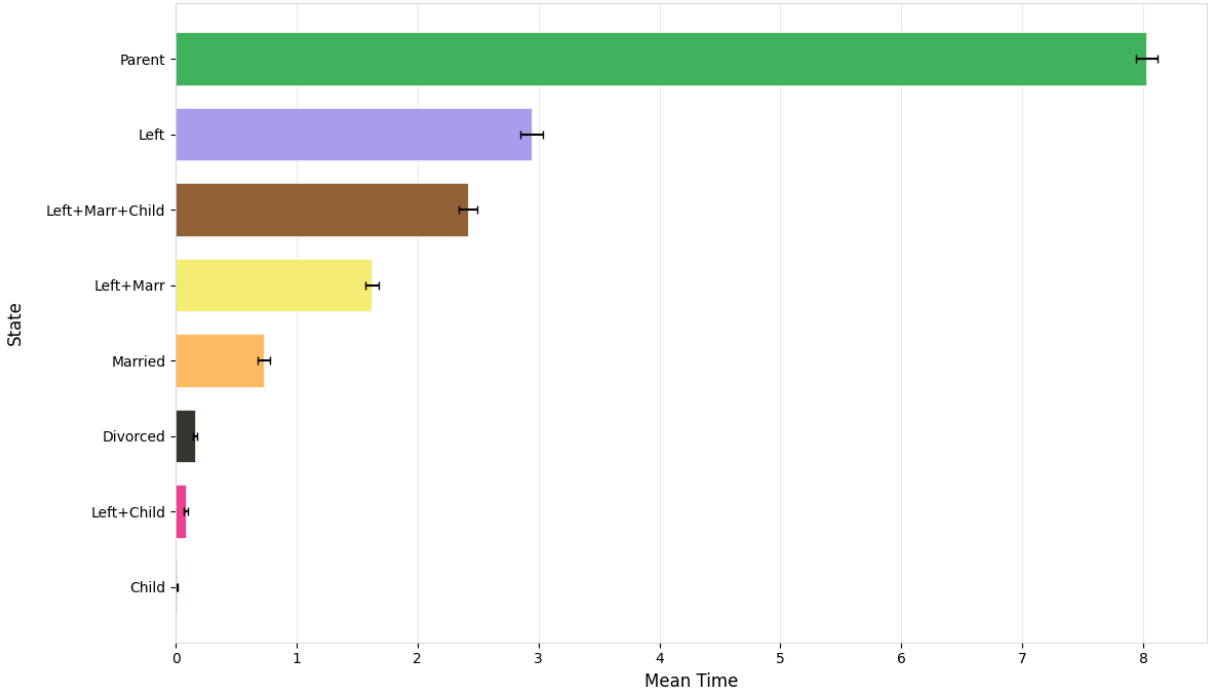
Temps moyen dans chaque état

De nouveau, `sequenzo` ne permet pas d'obtenir directement la table contenant le temps moyen passé dans chaque état mais propose une visualisation.

In [25]:
```
all_states = sorted(biofam[timepoints_columns].stack().unique())
mean_time = pd.DataFrame(
    {state: (arr == state).sum(axis=1) for state in all_states}
).mean()
print(mean_time)
```

```
0    8.0280
1    2.9440
2    0.7330
3    1.6220
4    0.0095
5    0.0835
6    2.4190
7    0.1610
dtype: float64
```

In [26]:
```
sequenzo.plot_mean_time(biofam_seq)
```



Note: Some bars may appear as zero, but actually have small non-zero values.

Tableau des transitions

A nouveau, l'utilisateur n'a pas accès directement à la table de transitions entre les états mais `sequenzo` propose une visualisation.

In [27]:
```
from_states = arr[:, :-1].flatten()
to_states   = arr[:, 1:].flatten()

transition_counts = pd.crosstab(
    pd.Series(from_states, name="from"),
    pd.Series(to_states, name="to")
)

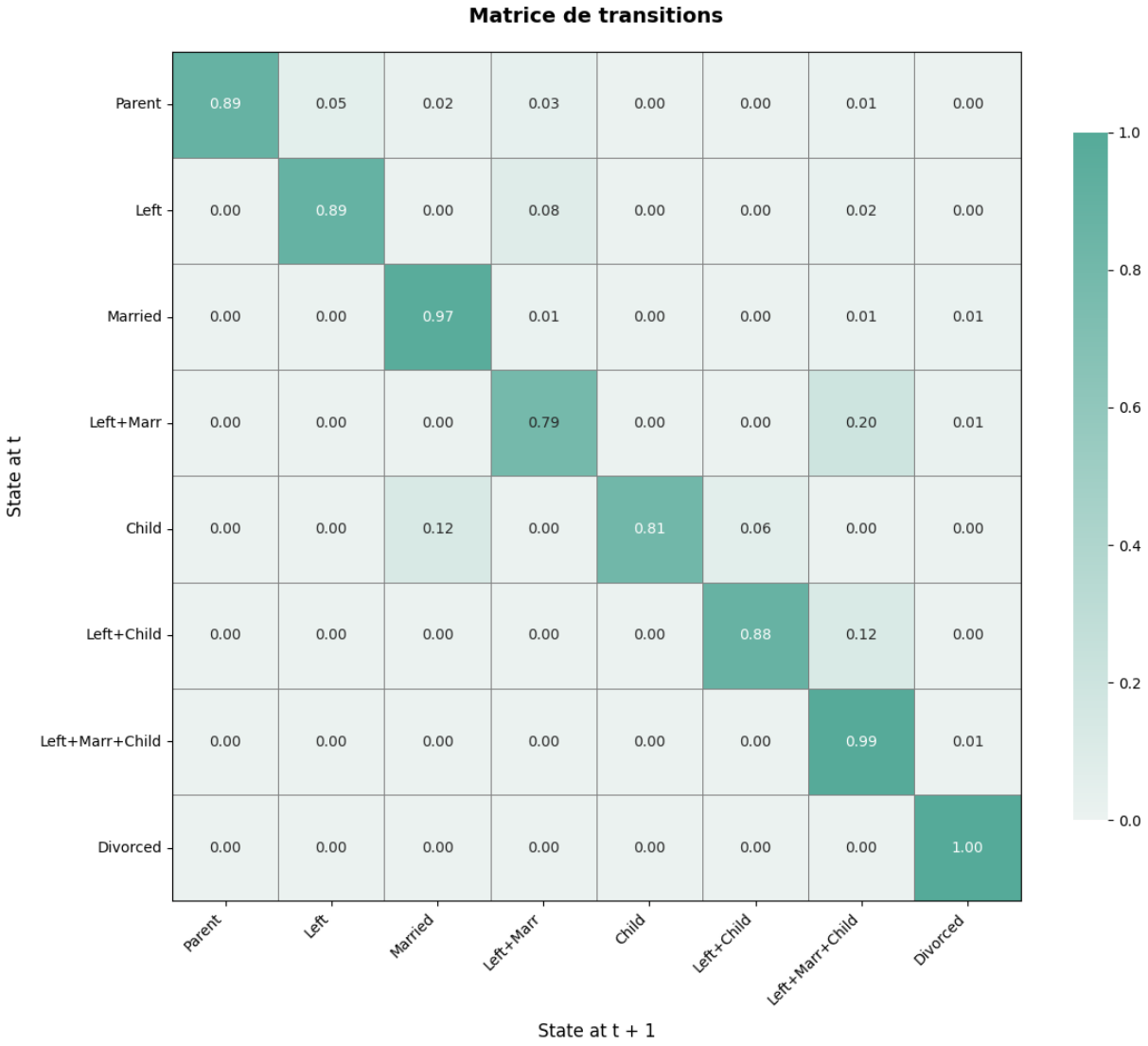
# Normalize by row to get probabilities
transition_probs = transition_counts.div(transition_counts.sum(axis=1), axis=0)

print(transition_probs.round(3))
```

```
to        0      1      2      3      4      5      6      7
from
0     0.886  0.055  0.015  0.032  0.000  0.001  0.011  0.000
1     0.000  0.890  0.000  0.083  0.000  0.004  0.023  0.000
2     0.000  0.000  0.969  0.010  0.000  0.000  0.011  0.010
3     0.000  0.000  0.000  0.787  0.000  0.000  0.199  0.014
4     0.000  0.000  0.125  0.000  0.812  0.062  0.000  0.000
5     0.000  0.000  0.000  0.000  0.000  0.882  0.118  0.000
6     0.000  0.000  0.000  0.000  0.000  0.000  0.994  0.006
7     0.000  0.000  0.000  0.000  0.000  0.000  0.000  1.000
```

In [28]:
```
sequenzo.plot_transition_matrix(
    biofam_seq,
    weights="auto",        # sequence weights; "auto" uses seqdata.weights if avail
```

```
    title="Matrice de transitions",
    fontsize=12
)
```



Distribution transversale

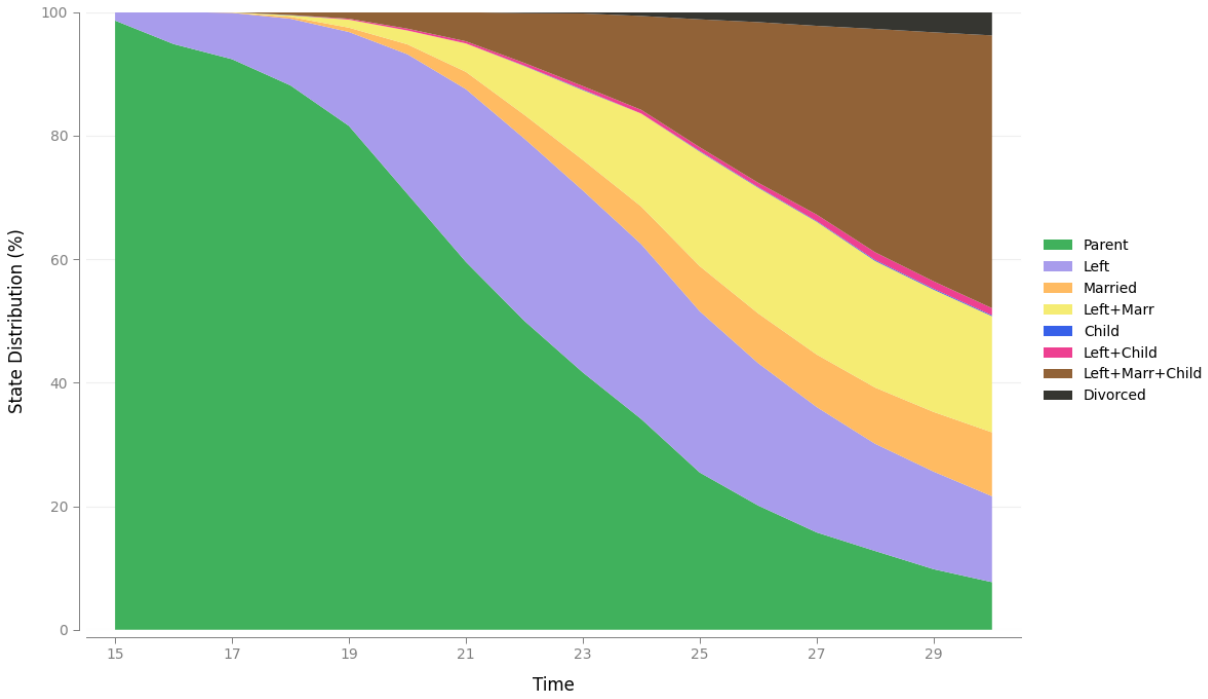
sequenzo ne permet pas d'accèder à la table contenant la distribution transversale de chaque état mais propose une visualisation.

In [29]:
```
# Pour chaque pas de temps, la proportion de chaque état
cross_dist = pd.DataFrame(
    {t: pd.Series(arr[:, i]).value_counts(normalize=True) for i, t in enumerate(tim
).T.fillna(0).sort_index()

cross_dist.columns = [biofam_seq.state_to_label[i] for i in sorted(cross_dist.colum
print(cross_dist.round(3))
```

	Parent	Left	Married	Left+Marr	Child	Left+Child	Left+Marr+Child	Divorced
15	0.986	0.014	0.000	0.000	0.000	0.000	0.000	0.000
16	0.948	0.052	0.000	0.000	0.000	0.000	0.000	0.000
17	0.924	0.075	0.000	0.000	0.000	0.000	0.000	0.000
18	0.881	0.108	0.003	0.002	0.000	0.000	0.005	0.000
19	0.816	0.152	0.007	0.013	0.000	0.002	0.011	0.000
20	0.706	0.226	0.016	0.022	0.000	0.003	0.027	0.000
21	0.596	0.280	0.028	0.046	0.000	0.004	0.047	0.000
22	0.500	0.295	0.038	0.079	0.000	0.004	0.080	0.002
23	0.416	0.294	0.050	0.113	0.001	0.006	0.118	0.002
24	0.341	0.283	0.061	0.150	0.000	0.006	0.152	0.006
25	0.254	0.261	0.073	0.186	0.001	0.006	0.207	0.012
26	0.201	0.230	0.080	0.204	0.001	0.007	0.260	0.016
27	0.158	0.203	0.085	0.215	0.001	0.010	0.306	0.022
28	0.128	0.174	0.091	0.205	0.002	0.012	0.362	0.028
29	0.098	0.158	0.096	0.198	0.002	0.012	0.403	0.033
30	0.077	0.139	0.104	0.188	0.002	0.012	0.442	0.038

In [30]:
```
sequenzo.plot_state_distribution(biofam_seq)
```



Entropie transversale

In [31]:
```
sequenzo.get_cross_sectional_entropy(biofam_seq)
```

```
================================================================================
Cross-Sectional Entropy Summary
================================================================================
[>] Number of states: 8
[>] Number of time points: 16
[>] On average, the most common state accounts for 36.1% of cases
[>] Entropy is highest at time point 27
[>] Entropy is lowest at time point 15
[>] Average normalized entropy: 0.709 (range: 0 = fully concentrated, 1 = evenly dis
tributed)
================================================================================

c:\Users\malnoeg\Documents\Stage_M2_SequenceAnalysis\venv-sequenzo\Lib\site-packages
\sequenzo\sequence_characteristics_indicators\overall_cross_sectional_entropy.py:10
9: UserWarning: 'where' used without 'out', expect unitialized memory in output. If
this is intentional, use out=None.
  P = np.divide(freq_counts, N_valid, where=(N_valid>0))  # Frequencies
```

Out[31]:

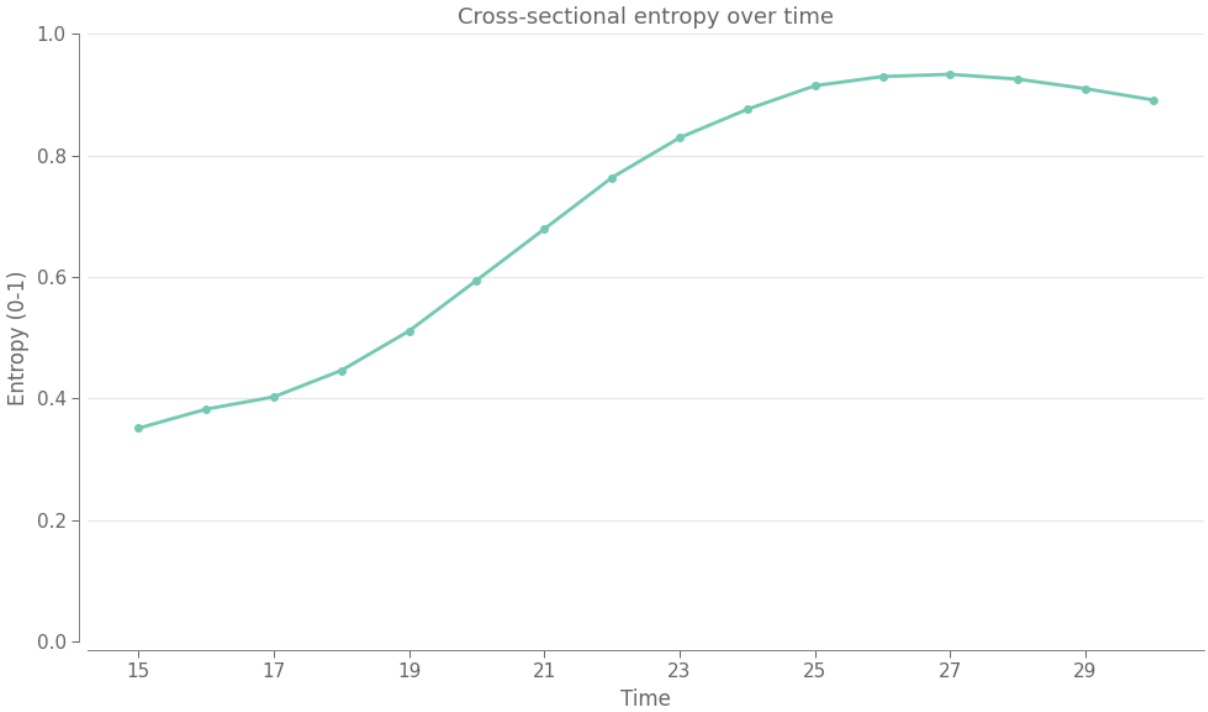
	time	state	freq	per_time_entropy_norm	N_valid	Effective States	rank	is_top
0	15	1	0.500000	0.351046	4000.0	2.075037	1	True
1	15	0	0.493000	0.351046	4000.0	2.075037	2	False
2	15	2	0.007000	0.351046	4000.0	2.075037	3	False
3	15	3	0.000000	0.351046	4000.0	2.075037	4	False
4	15	4	0.000000	0.351046	4000.0	2.075037	5	False
...	...	...	...	...	...	...	...	...
123	30	2	0.123598	0.891322	3924.0	6.381812	4	False
124	30	1	0.110092	0.891322	3924.0	6.381812	5	False
125	30	4	0.096585	0.891322	3924.0	6.381812	6	False
126	30	0	0.039246	0.891322	3924.0	6.381812	7	False
127	30	5	0.006626	0.891322	3924.0	6.381812	8	False

128 rows × 8 columns

In [32]:
```
sequenzo.plot_cross_sectional_characteristics(biofam_seq)
```

```
======================================================================
Cross-Sectional Entropy Summary
======================================================================
[>] Number of states: 8
[>] Number of time points: 16
[>] On average, the most common state accounts for 36.1% of cases
[>] Entropy is highest at time point 27
[>] Entropy is lowest at time point 15
[>] Average normalized entropy: 0.709 (range: 0 = fully concentrated, 1 = evenly dis
tributed)
======================================================================

c:\Users\malnoeg\Documents\Stage_M2_SequenceAnalysis\venv-sequenzo\Lib\site-packages
\sequenzo\sequence_characteristics_indicators\overall_cross_sectional_entropy.py:10
9: UserWarning: 'where' used without 'out', expect unitialized memory in output. If
this is intentional, use out=None.
  P = np.divide(freq_counts, N_valid, where=(N_valid>0))  # Frequencies
```

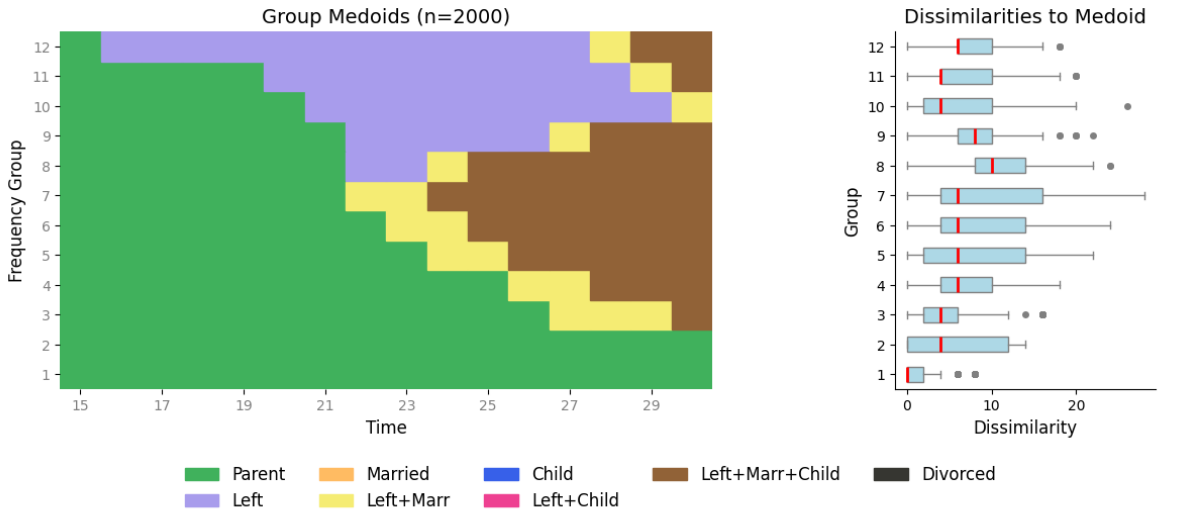


Frequence relative

In [33]:
```
distance_matrix = sequenzo.get_distance_matrix(biofam_seq,method="OM",sm="constant"
sequenzo.plot_relative_frequency(
    biofam_seq,
    distance_matrix
)
```

```
[>] Processing 2000 sequences with 8 unique states.
  - Creating 9x9 substitution-cost matrix using 2 as constant value
[>] generated an indel of type number

[>] Identified 537 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
```



Representation quality: Pseudo/medoid-based R^2 = 126.43, F statistic = 24537.50, p = 0.00e+00 *** (*: p < 0.05, **: p < 0.01, ***: p < 0.001)

Single medoid

In [35]:
```
# Ne fonctionne pas mais est dans la documentation
#sequenzo.plot_single_medoid(
#    biofam_seq,
#    distance_matrix=distance_matrix,
#    show_legend=True
#)
```

Descripteurs individuels

In [36]:
```
sequenzo.get_subsequences_all_sequences(biofam_seq)
```

Out[36]:

	Subseq.
0	8
1	16
2	16
3	4
4	8
...	...
1995	8
1996	16
1997	2
1998	4
1999	4

2000 rows × 1 columns

In [37]:
```
sequenzo.get_number_of_transitions(biofam_seq)
```

Out[37]:

	ID	Transitions
0	0	2
1	1	3
2	2	3
3	3	1
4	4	2
...	...	...
1995	1995	2
1996	1996	3
1997	1997	0
1998	1998	1
1999	1999	1

2000 rows × 2 columns

In [38]:
```
sequenzo.get_spell_duration_variance(biofam_seq)
```

Out[38]:
```
{'meand':         ID      meand
 0        0   5.333333
 1        1   4.000000
 2        2   4.000000
 3        3   8.000000
 4        4   5.333333
 ...    ...        ...
 1995  1995   5.333333
 1996  1996   4.000000
 1997  1997  16.000000
 1998  1998   8.000000
 1999  1999   8.000000

 [2000 rows x 2 columns],
 'result':         ID  var_spell_dur
 0        0      10.888889
 1        1      13.500000
 2        2       5.000000
 3        3       9.000000
 4        4       4.222222
 ...    ...            ...
 1995  1995       2.888889
 1996  1996       2.000000
 1997  1997       0.000000
 1998  1998      25.000000
 1999  1999       4.000000

 [2000 rows x 2 columns],
 'vmax':         ID    var_max
 0        0  37.555556
 1        1  27.000000
 2        2  27.000000
 3        3  49.000000
 4        4  37.555556
 ...    ...        ...
 1995  1995  37.555556
 1996  1996  27.000000
 1997  1997   0.000000
 1998  1998  49.000000
 1999  1999  49.000000

 [2000 rows x 2 columns]}
```

In [39]:
```
sequenzo.get_state_freq_and_entropy_per_seq(biofam_seq)
```

```
[>] Computing state distribution for 2000 sequences and 8 states ...
```

Out[39]:

	ID	Parent	Left	Married	Left+Marr	Child	Left+Child	Left+Marr+Child	Divorce
0	0	9.0	0.0	0.0	1.0	0.0	0.0	6.0	0
1	1	1.0	10.0	0.0	1.0	0.0	0.0	4.0	0
2	2	7.0	5.0	0.0	1.0	0.0	0.0	3.0	0
3	3	5.0	11.0	0.0	0.0	0.0	0.0	0.0	0
4	4	5.0	8.0	0.0	0.0	0.0	0.0	3.0	0
...	...	...	...	...	...	...	...	...	
1995	1995	6.0	3.0	0.0	0.0	0.0	0.0	7.0	0
1996	1996	2.0	6.0	0.0	4.0	0.0	0.0	4.0	0
1997	1997	16.0	0.0	0.0	0.0	0.0	0.0	0.0	0
1998	1998	3.0	0.0	13.0	0.0	0.0	0.0	0.0	0
1999	1999	6.0	0.0	0.0	0.0	0.0	0.0	10.0	0

2000 rows × 9 columns

In [40]:
```
sequenzo.get_within_sequence_entropy(biofam_seq)
```

Out[40]:

	ID	Entropy
0	0	0.415852
1	1	0.474598
2	2	0.583000
3	3	0.298679
4	4	0.492406
...	...	...
1995	1995	0.501747
1996	1996	0.635213
1997	1997	0.000000
1998	1998	0.232071
1999	1999	0.318145

2000 rows × 2 columns

In [41]:
```
sequenzo.get_complexity_index(biofam_seq)
```

Out[41]:

	ID	Complexity Index
0	0	0.235472
1	1	0.308090
2	2	0.341467
3	3	0.141110
4	4	0.256231
...	...	...
1995	1995	0.258650
1996	1996	0.356430
1997	1997	0.000000
1998	1998	0.124384
1999	1999	0.145635

2000 rows × 2 columns

In [42]:
```
sequenzo.get_turbulence(biofam_seq)
```

```
[!] One or more missing values were found after calculating the number of distinct s
ubsequences. They have been replaced with a large number of 1e15 to ensure the calcu
lation continues.
```

Out[42]:

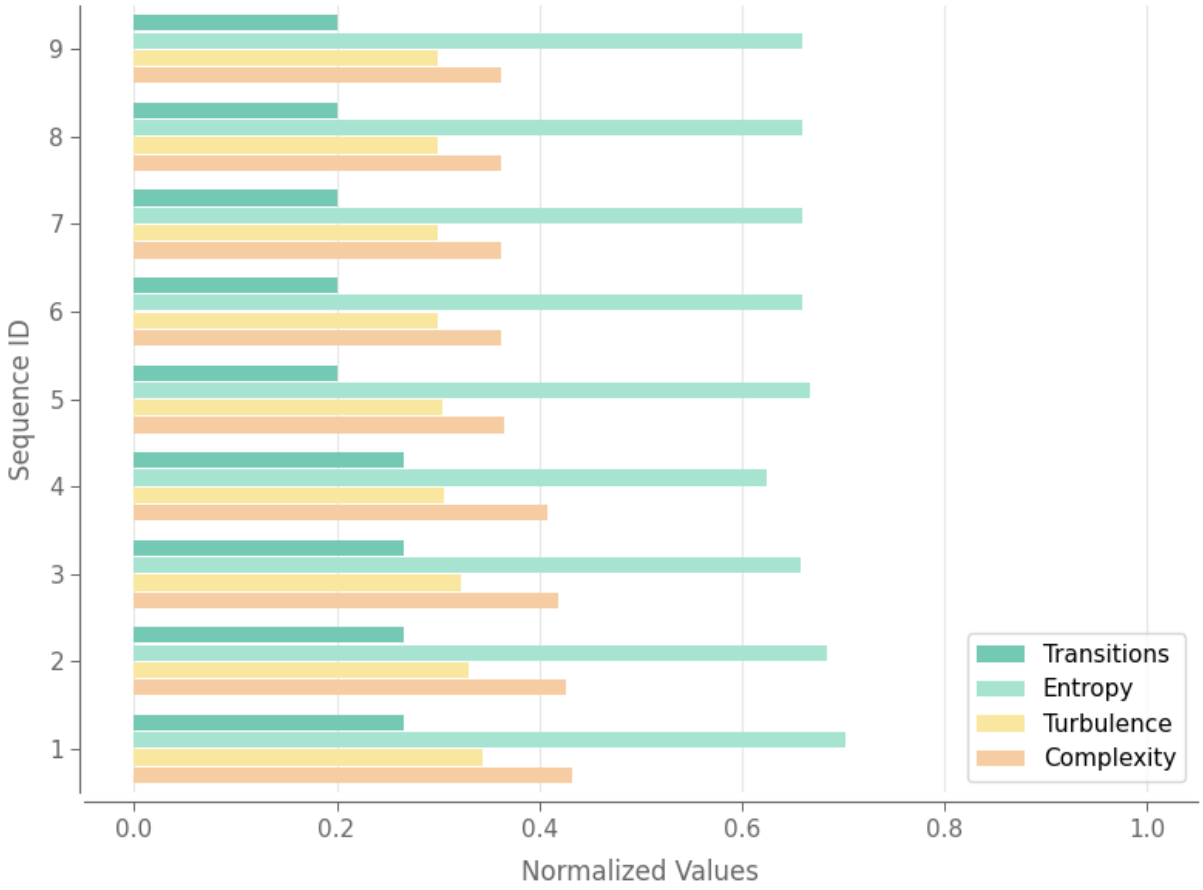
	ID	Turbulence
0	0	4.697325
1	1	4.949374
2	2	6.222392
3	3	4.321928
4	4	5.884203
...	...	...
1995	1995	6.309509
1996	1996	7.222392
1997	1997	1.000000
1998	1998	2.943416
1999	1999	5.321928

2000 rows × 2 columns

In [44]:
```
# Ne fonctionne pas mais est dans la documentation
#sequenzo.get_positive_negative_indicators(biofam_seq,  pos_states=[0, 1, 2, 3, 4,
#sequenzo.get_integration_index(biofam_seq)
#sequenzo.get_badness_index(biofam_seq)
#sequenzo.get_degradation_index(biofam_seq, method="RANK", penalized="BOTH")
#sequenzo.get_precarity_index(biofam_seq, otto=0.2, a=1.0, b=1.2, method="TRATEDSS"
#sequenzo.get_insecurity_index(biofam_seq, method="RANK", pow=1.0, bound=False)
```

In [45]:
```
sequenzo.plot_longitudinal_characteristics(biofam_seq)
```

```
[!] One or more missing values were found after calculating the number of distinct s
ubsequences. They have been replaced with a large number of 1e15 to ensure the calcu
lation continues.
```



Out[45]:

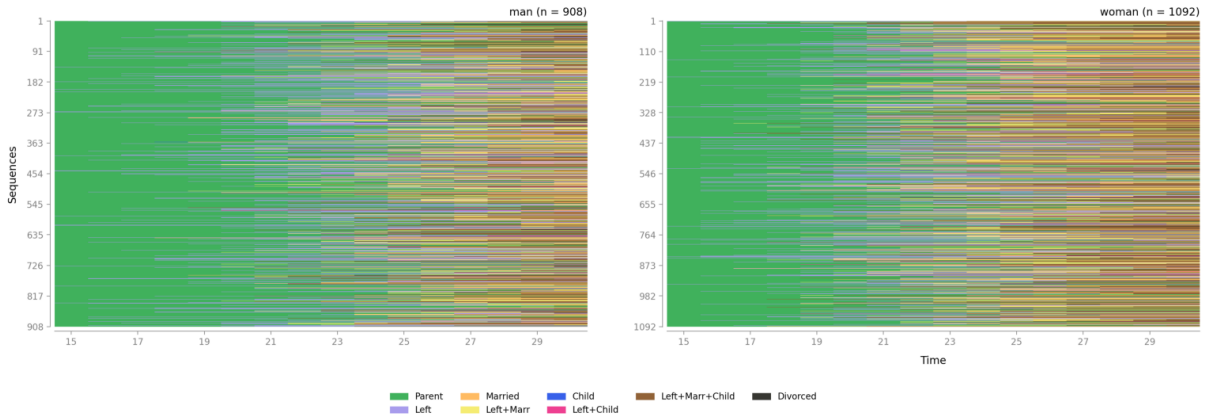
	ID	Transitions	Entropy	Turbulence	Complexity
0	722	0.266667	0.702820	0.343398	0.432919
1	37	0.266667	0.684140	0.330756	0.427127
2	1625	0.266667	0.658201	0.323178	0.418951
3	1954	0.266667	0.625000	0.306526	0.408248
4	379	0.200000	0.666667	0.304839	0.365148
5	1575	0.200000	0.659072	0.300220	0.363063
6	1563	0.200000	0.659072	0.300220	0.363063
7	671	0.200000	0.659072	0.300220	0.363063
8	1767	0.200000	0.659072	0.300220	0.363063

Paramètres de visualisaion

`sequenzo` propose plusieurs niveau de personnalisation des visuels comme le changement de la palette de couleurs ou la sépération d'une visualisation selon un groupe.

In [46]:
```
sequenzo.plot_sequence_index(
    biofam_seq,
    sort_by="unsorted",
    group_by_column="sex"
)
```

```
[>] Creating grouped plots by 'sex' with 2 categories
```



7. Obtention de typologies par l'analyse de séquences

L'une des application majeur de l'analyse de séquences est la construction de typologies : le regroupement des séquences partageant des propriétés similaires. L'identification de ces profils types permet ensuite de travailler avec des covariables afin d'expliquer la formation

de ces différents groupes.

La construction de ces typologies se fait en deux étapes : l'évaluation des écarts entre chaque paire de séquence qui fournit une matrice de (di)similarités, puis l'application d'un algorithme de classification (clustering) sur cette matrice afin de former des groupes d'individus similaires (ou peu dissimilaires).

Pour chacune de ces étapes, le package `sequenzo` propose plusieurs possibilités que nous allons illustrer sur le jeu de données `biofam` .

7.1 Calcul matrice de distances (dissimilarités)

La librairie `sequenzo` propose de nombreuses métrique pour calculer la dissimilarité entre les séquences :

In [47]:
```
start_time = time.time()
distance_matrix = sequenzo.get_distance_matrix(biofam_seq, method="OM", sm="constan
distance_matrix
time_taken = time.time() - start_time
print(str(time_taken)+" s")
```

```
[>] Processing 2000 sequences with 8 unique states.
  - Creating 9x9 substitution-cost matrix using 2 as constant value
[>] generated an indel of type number

[>] Identified 537 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
0.11284732818603516 s
```

In [48]:
```
distance_matrix.index
```

Out[48]:
```
Index([   0,    1,    2,    3,    4,    5,    6,    7,    8,    9,
       ...
       1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999],
      dtype='int64', length=2000)
```

7.2 Clustering

Une fois la matrice de dissimilarité calculée, la seconde étape consiste à regrouper les trajectoires similaires au moyen d'un algorithme de clustering. `sequenzo` propose directement plusieurs algorithmes de clustering...

In [49]:
```
start_time = time.time()
cluster = sequenzo.Cluster(
    matrix=distance_matrix,
    entity_ids=distance_matrix.index,
    clustering_method="ward"
)
```

```
cluster_results = sequenzo.ClusterResults(cluster)
members = cluster_results.get_cluster_memberships(num_clusters=2)
time_taken = time.time() - start_time
print(str(time_taken)+" s")
```

```
26.185077905654907 s
c:\Users\malnoeg\Documents\Stage_M2_SequenceAnalysis\venv-sequenzo\Lib\site-packages
\sequenzo\clustering\hierarchical_clustering.py:308: UserWarning:
[!] Ward linkage method detected with potentially non-Euclidean distance matrix!
   Ward clustering (both Ward D and Ward D2) assumes Euclidean distances for theoret
ical validity.

   Ward method variants:
   - 'ward_d' (classic): Uses squared Euclidean distances ÷ 2
   - 'ward_d2': Uses squared Euclidean distances

   For sequence distances (OM, LCS, etc.), consider using:
   - method='average' (UPGMA)
   - method='complete' (complete linkage)
   - method='single' (single linkage)

   Note: 'centroid' and 'median' methods may also produce inversions
   (non-monotonic dendrograms) with non-Euclidean distances.

   This warning is shown only once per session.
  self._warn_from_flags(int(result["warning_flags"]), self.clustering_method)
```

In [50]:
```
members
```

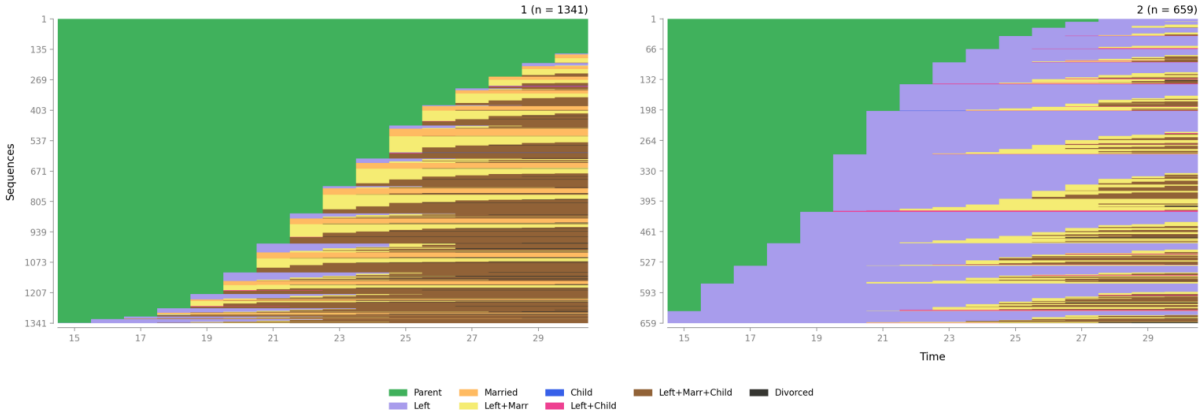
Out[50]:

	Entity ID	Cluster
0	0	1
1	1	2
2	2	2
3	3	2
4	4	2
...	...	...
1995	1995	1
1996	1996	2
1997	1997	1
1998	1998	1
1999	1999	1

2000 rows × 2 columns

In [51]:
```
sequenzo.plot_sequence_index(biofam_seq,
                    group_dataframe=members,
```

```
                                group_column_name="Cluster")
```



7.3 Cas du multichannel

L'analyse de séquences ne se limite pas qu'au cas unidimensionnel comme illustré précedemment et peut-être également intéressantes dans un contexte multidiomensionnel pour plusieurs individus reliés entre eux ou plusieurs aspects d'un même individu.

Le processus analytique dans le cadre multidimensionnel reste globalement le même : il s'agit toujours de définir des séquences à partir d'un certain alphabet d'états, de choisir le coût du passage d'un état à un autre, de calculer la dissimilarité entre les trajectoires et d'appliquer une algorithme de clustering. L'enjeu devient alors de choisir le moment auquel considérer l'aspect multidimensionnel. Quatre approches se distingues : extension de l'alphabet (EA, *Extended Alphabet*), association des matrices de coût (CAT, *Cost Additive Trick*), association des matrices de dissimilarité (DAT, *Distance Additive Trick*) et combinaison des clusters (CombT, *Combining Trajectories*) qui sont toutes implémentables en Python avec `sequenzo` .

Nous allons illustrer ces possibilités avec le jeu de données `extract_shp` .

Extended Alphabet

In [52]:
```
start_time = time.time()
# Lien vers les fichiers CSV préparés précédemment
csv_paths = [
    "C:/Users/malnoeg/Documents/Stage_M2_SequenceAnalysis/Protocole/Code/data_SHP_s
    "C:/Users/malnoeg/Documents/Stage_M2_SequenceAnalysis/Protocole/Code/data_SHP_s
    "C:/Users/malnoeg/Documents/Stage_M2_SequenceAnalysis/Protocole/Code/data_SHP_s
    "C:/Users/malnoeg/Documents/Stage_M2_SequenceAnalysis/Protocole/Code/data_SHP_s
    "C:/Users/malnoeg/Documents/Stage_M2_SequenceAnalysis/Protocole/Code/data_SHP_s
    "C:/Users/malnoeg/Documents/Stage_M2_SequenceAnalysis/Protocole/Code/data_SHP_s
]
time_cols_shp = ["1999","2000","2001","2002","2003","2004","2005","2006","2007","20
# Sequences pour EA (IDCD)
sequence_data = sequenzo.create_idcd_sequence_from_csvs(
    id_col="idhous",
    csv_paths=csv_paths,
```

```
    time_cols=time_cols_shp
)

distance_matrix = sequenzo.dissimilarity_measures.get_distance_matrix(
    seqdata=sequence_data,
    method="OM",
    sm="CONSTANT",
    indel="auto"
)

cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels = cluster_result.get_cluster_labels(num_clusters=4)

time_taken = time.time() - start_time
print(f"Temps d'exécution : {time_taken:.2f} s")
```

[IDCD] Observed Combined States Frequency Table:

State	Label	Frequency	Proportion (%)
4+2+2+2+2+7-8	4+2+2+2+2+7-8	197	0.95
4+2+2+2+2+9-10	4+2+2+2+2+9-10	181	0.88
4+1+1+2+2+7-8	4+1+1+2+2+7-8	147	0.71
4+1+1+1+2+7-8	4+1+1+1+2+7-8	135	0.65
4+2+1+1+2+7-8	4+2+1+1+2+7-8	134	0.65
2+2+2+2+2+9-10	2+2+2+2+2+9-10	120	0.58
1+1+1+1+2+7-8	1+1+1+1+2+7-8	117	0.57
2+1+1+2+2+7-8	2+1+1+2+2+7-8	103	0.50
3+2+2+2+2+7-8	3+2+2+2+2+7-8	100	0.48
2+4+2+2+2+7-8	2+4+2+2+2+7-8	99	0.48
4+2+1+2+2+7-8	4+2+1+2+2+7-8	99	0.48
2+2+1+1+2+7-8	2+2+1+1+2+7-8	98	0.47
2+2+2+2+2+7-8	2+2+2+2+2+7-8	97	0.47
1+1+1+2+2+7-8	1+1+1+2+2+7-8	96	0.47
2+1+1+1+2+7-8	2+1+1+1+2+7-8	96	0.47
2+1+2+2+2+7-8	2+1+2+2+2+7-8	93	0.45
1+2+1+1+2+7-8	1+2+1+1+2+7-8	85	0.41
2+2+2+4+2+7-8	2+2+2+4+2+7-8	85	0.41
1+2+1+2+2+7-8	1+2+1+2+2+7-8	79	0.38
6+1+2+2+2+9-10	6+1+2+2+2+9-10	79	0.38
4+1+1+2+2+9-10	4+1+1+2+2+9-10	78	0.38
1+1+2+2+2+7-8	1+1+2+2+2+7-8	77	0.37
5+2+2+2+2+7-8	5+2+2+2+2+7-8	74	0.36
6+2+1+2+2+7-8	6+2+1+2+2+7-8	74	0.36
2+2+1+2+2+7-8	2+2+1+2+2+7-8	73	0.35
3+1+2+2+2+9-10	3+1+2+2+2+9-10	72	0.35
3+4+2+2+2+9-10	3+4+2+2+2+9-10	71	0.34
3+2+2+2+2+9-10	3+2+2+2+2+9-10	71	0.34
2+1+2+2+2+9-10	2+1+2+2+2+9-10	69	0.33
3+2+2+4+2+7-8	3+2+2+4+2+7-8	67	0.32
1+4+2+2+2+9-10	1+4+2+2+2+9-10	66	0.32
1+1+1+2+2+4-6	1+1+1+2+2+4-6	65	0.31
2+7+2+2+2+7-8	2+7+2+2+2+7-8	65	0.31
4+2+2+2+3+9-10	4+2+2+2+3+9-10	65	0.31
4+2+2+4+2+7-8	4+2+2+4+2+7-8	64	0.31
4+2+2+2+2+4-6	4+2+2+2+2+4-6	63	0.31
5+6+2+2+2+9-10	5+6+2+2+2+9-10	63	0.31
2+2+1+1+2+9-10	2+2+1+1+2+9-10	62	0.30
3+2+1+1+2+7-8	3+2+1+1+2+7-8	62	0.30
1+2+2+2+2+7-8	1+2+2+2+2+7-8	60	0.29
3+1+1+1+2+7-8	3+1+1+1+2+7-8	59	0.29
3+2+1+2+2+7-8	3+2+1+2+2+7-8	59	0.29
5+2+2+4+2+7-8	5+2+2+4+2+7-8	58	0.28
3+4+2+2+2+7-8	3+4+2+2+2+7-8	57	0.28
2+8+2+2+2+7-8	2+8+2+2+2+7-8	56	0.27
2+1+2+1+2+7-8	2+1+2+1+2+7-8	53	0.26
3+1+2+2+2+7-8	3+1+2+2+2+7-8	53	0.26
3+7+2+2+2+7-8	3+7+2+2+2+7-8	53	0.26
3+7+2+2+2+9-10	3+7+2+2+2+9-10	53	0.26
4+2+1+2+2+9-10	4+2+1+2+2+9-10	51	0.25
4+2+1+1+2+9-10	4+2+1+1+2+9-10	51	0.25
4+1+1+1+2+9-10	4+1+1+1+2+9-10	50	0.24
4+2+1+1+2+4-6	4+2+1+1+2+4-6	50	0.24
2+6+2+2+2+7-8	2+6+2+2+2+7-8	49	0.24

4+4+2+2+2+9-10	4+4+2+2+2+9-10	49	0.24
6+1+1+1+2+4-6	6+1+1+1+2+4-6	49	0.24
1+1+2+2+2+9-10	1+1+2+2+2+9-10	49	0.24
1+1+1+1+2+4-6	1+1+1+1+2+4-6	48	0.23
4+2+2+4+2+9-10	4+2+2+4+2+9-10	48	0.23
4+2+2+1+2+7-8	4+2+2+1+2+7-8	47	0.23
4+1+1+4+2+7-8	4+1+1+4+2+7-8	47	0.23
4+2+1+2+2+4-6	4+2+1+2+2+4-6	47	0.23
6+2+1+2+2+4-6	6+2+1+2+2+4-6	47	0.23
1+1+1+2+2+9-10	1+1+1+2+2+9-10	46	0.22
2+1+1+1+2+4-6	2+1+1+1+2+4-6	46	0.22
3+4+2+3+2+7-8	3+4+2+3+2+7-8	46	0.22
5+1+1+1+2+7-8	5+1+1+1+2+7-8	46	0.22
1+2+1+2+2+9-10	1+2+1+2+2+9-10	45	0.22
1+4+2+2+2+7-8	1+4+2+2+2+7-8	44	0.21
1+2+1+1+2+4-6	1+2+1+1+2+4-6	44	0.21
2+1+2+5plus+2+7-8	2+1+2+5plus+2+7-8	44	0.21
3+1+1+1+2+4-6	3+1+1+1+2+4-6	44	0.21
4+1+2+2+2+7-8	4+1+2+2+2+7-8	44	0.21
2+1+1+2+2+4-6	2+1+1+2+2+4-6	43	0.21
2+2+1+1+2+4-6	2+2+1+1+2+4-6	43	0.21
4+2+1+5plus+2+7-8	4+2+1+5plus+2+7-8	43	0.21
6+1+1+2+2+7-8	6+1+1+2+2+7-8	43	0.21
3+7+2+4+2+7-8	3+7+2+4+2+7-8	42	0.20
4+4+2+4+2+7-8	4+4+2+4+2+7-8	42	0.20
5+6+2+4+2+7-8	5+6+2+4+2+7-8	42	0.20
2+8+2+2+2+9-10	2+8+2+2+2+9-10	41	0.20
2+6+2+3+2+7-8	2+6+2+3+2+7-8	41	0.20
4+2+1+3+2+9-10	4+2+1+3+2+9-10	41	0.20
5+1+1+1+2+9-10	5+1+1+1+2+9-10	41	0.20
2+2+2+3+2+7-8	2+2+2+3+2+7-8	41	0.20
2+7+2+5plus+2+7-8	2+7+2+5plus+2+7-8	40	0.19
3+3+1+1+2+7-8	3+3+1+1+2+7-8	40	0.19
4+2+2+2+3+7-8	4+2+2+2+3+7-8	40	0.19
1+1+2+1+2+7-8	1+1+2+1+2+7-8	39	0.19
2+8+2+3+2+7-8	2+8+2+3+2+7-8	39	0.19
2+1+1+2+2+9-10	2+1+1+2+2+9-10	39	0.19
2+6+2+2+2+9-10	2+6+2+2+2+9-10	39	0.19
3+2+2+3+2+7-8	3+2+2+3+2+7-8	39	0.19
3+4+2+2+3+9-10	3+4+2+2+3+9-10	39	0.19
5+8+2+2+2+9-10	5+8+2+2+2+9-10	39	0.19
5+2+2+2+2+9-10	5+2+2+2+2+9-10	39	0.19
2+1+2+1+2+9-10	2+1+2+1+2+9-10	38	0.18
5+1+1+2+2+7-8	5+1+1+2+2+7-8	38	0.18
1+1+1+4+2+4-6	1+1+1+4+2+4-6	37	0.18
2+7+2+4+2+7-8	2+7+2+4+2+7-8	37	0.18
2+7+2+2+2+4-6	2+7+2+2+2+4-6	37	0.18
6+1+1+1+2+9-10	6+1+1+1+2+9-10	37	0.18
4+2+2+3+2+7-8	4+2+2+3+2+7-8	37	0.18
4+2+1+1+3+7-8	4+2+1+1+3+7-8	37	0.18
6+8+2+5plus+2+7-8	6+8+2+5plus+2+7-8	37	0.18
5+6+2+2+2+7-8	5+6+2+2+2+7-8	37	0.18
1+4+2+5plus+2+7-8	1+4+2+5plus+2+7-8	36	0.17
1+2+2+2+2+9-10	1+2+2+2+2+9-10	36	0.17
1+1+1+3+2+7-8	1+1+1+3+2+7-8	36	0.17
2+8+2+4+2+7-8	2+8+2+4+2+7-8	36	0.17

3+4+2+1+2+7-8	3+4+2+1+2+7-8	36	0.17
4+1+1+3+2+7-8	4+1+1+3+2+7-8	36	0.17
4+2+1+4+2+7-8	4+2+1+4+2+7-8	36	0.17
4+3+2+2+2+7-8	4+3+2+2+2+7-8	36	0.17
6+2+1+1+2+7-8	6+2+1+1+2+7-8	36	0.17
2+6+1+4+2+7-8	2+6+1+4+2+7-8	35	0.17
4+1+1+1+2+4-6	4+1+1+1+2+4-6	35	0.17
4+1+1+2+2+4-6	4+1+1+2+2+4-6	35	0.17
2+4+2+3+2+7-8	2+4+2+3+2+7-8	35	0.17
1+2+2+4+2+7-8	1+2+2+4+2+7-8	34	0.16
1+2+1+3+2+4-6	1+2+1+3+2+4-6	34	0.16
1+2+2+4+2+9-10	1+2+2+4+2+9-10	34	0.16
2+1+2+4+2+7-8	2+1+2+4+2+7-8	34	0.16
2+8+2+5plus+2+7-8	2+8+2+5plus+2+7-8	34	0.16
1+4+2+4+2+7-8	1+4+2+4+2+7-8	33	0.16
2+4+2+2+2+9-10	2+4+2+2+2+9-10	33	0.16
2+8+2+5plus+2+9-10	2+8+2+5plus+2+9-10	33	0.16
2+2+1+2+2+4-6	2+2+1+2+2+4-6	33	0.16
3+4+1+1+2+7-8	3+4+1+1+2+7-8	33	0.16
4+2+2+5plus+2+7-8	4+2+2+5plus+2+7-8	33	0.16
4+1+2+1+2+7-8	4+1+2+1+2+7-8	33	0.16
5+1+1+2+2+9-10	5+1+1+2+2+9-10	33	0.16
1+1+1+1+2+9-10	1+1+1+1+2+9-10	32	0.16
2+2+2+1+2+7-8	2+2+2+1+2+7-8	32	0.16
2+2+2+5plus+2+7-8	2+2+2+5plus+2+7-8	32	0.16
3+1+1+1+2+9-10	3+1+1+1+2+9-10	32	0.16
3+3+2+2+2+9-10	3+3+2+2+2+9-10	32	0.16
6+1+1+2+2+9-10	6+1+1+2+2+9-10	32	0.16
5+1+1+1+2+4-6	5+1+1+1+2+4-6	32	0.16
7+4+2+2+2+9-10	7+4+2+2+2+9-10	32	0.16
1+2+2+1+2+7-8	1+2+2+1+2+7-8	31	0.15
2+2+2+4+2+9-10	2+2+2+4+2+9-10	31	0.15
3+2+1+2+2+9-10	3+2+1+2+2+9-10	31	0.15
4+1+2+2+2+9-10	4+1+2+2+2+9-10	31	0.15
5+2+1+2+2+7-8	5+2+1+2+2+7-8	31	0.15
6+1+1+1+2+7-8	6+1+1+1+2+7-8	31	0.15
1+2+1+3+2+7-8	1+2+1+3+2+7-8	30	0.15
2+1+1+4+2+7-8	2+1+1+4+2+7-8	30	0.15
2+6+2+4+2+7-8	2+6+2+4+2+7-8	30	0.15
2+1+2+2+2+4-6	2+1+2+2+2+4-6	30	0.15
4+1+2+4+2+7-8	4+1+2+4+2+7-8	30	0.15
4+4+2+4+2+4-6	4+4+2+4+2+4-6	30	0.15
6+2+2+2+2+9-10	6+2+2+2+2+9-10	30	0.15
1+7+2+4+2+7-8	1+7+2+4+2+7-8	29	0.14
2+1+1+1+2+9-10	2+1+1+1+2+9-10	29	0.14
2+2+2+4+3+7-8	2+2+2+4+3+7-8	29	0.14
2+1+1+4+2+4-6	2+1+1+4+2+4-6	29	0.14
3+7+1+2+2+7-8	3+7+1+2+2+7-8	29	0.14
4+4+2+2+2+7-8	4+4+2+2+2+7-8	29	0.14
4+1+1+3+2+9-10	4+1+1+3+2+9-10	29	0.14
4+1+1+2+3+7-8	4+1+1+2+3+7-8	29	0.14
5+8+2+2+2+7-8	5+8+2+2+2+7-8	29	0.14
2+2+2+1+2+9-10	2+2+2+1+2+9-10	28	0.14
3+3+1+1+2+4-6	3+3+1+1+2+4-6	28	0.14
3+2+2+1+2+7-8	3+2+2+1+2+7-8	28	0.14
5+6+1+2+2+7-8	5+6+1+2+2+7-8	28	0.14

1+8+2+2+2+9-10	1+8+2+2+2+9-10	28	0.14
2+2+2+2+3+9-10	2+2+2+2+3+9-10	27	0.13
2+1+2+5plus+2+9-10	2+1+2+5plus+2+9-10	27	0.13
3+1+2+1+2+7-8	3+1+2+1+2+7-8	27	0.13
4+2+2+3+2+9-10	4+2+2+3+2+9-10	27	0.13
4+4+2+1+2+9-10	4+4+2+1+2+9-10	27	0.13
6+1+2+2+2+7-8	6+1+2+2+2+7-8	27	0.13
1+7+2+2+2+7-8	1+7+2+2+2+7-8	26	0.13
2+4+2+2+2+4-6	2+4+2+2+2+4-6	26	0.13
2+2+1+3+2+7-8	2+2+1+3+2+7-8	26	0.13
2+8+2+5plus+2+4-6	2+8+2+5plus+2+4-6	26	0.13
2+2+2+2+2+4-6	2+2+2+2+2+4-6	26	0.13
3+2+1+3+2+7-8	3+2+1+3+2+7-8	26	0.13
5+2+1+1+2+4-6	5+2+1+1+2+4-6	26	0.13
6+2+1+2+2+9-10	6+2+1+2+2+9-10	26	0.13
5+1+2+1+2+9-10	5+1+2+1+2+9-10	26	0.13
1+2+1+5plus+2+4-6	1+2+1+5plus+2+4-6	25	0.12
1+1+1+4+2+7-8	1+1+1+4+2+7-8	25	0.12
1+2+2+1+2+4-6	1+2+2+1+2+4-6	25	0.12
2+7+2+4+3+7-8	2+7+2+4+3+7-8	25	0.12
2+6+2+2+3+7-8	2+6+2+2+3+7-8	25	0.12
3+1+2+3+2+9-10	3+1+2+3+2+9-10	25	0.12
3+4+2+3+2+9-10	3+4+2+3+2+9-10	25	0.12
4+4+2+4+2+9-10	4+4+2+4+2+9-10	25	0.12
4+2+2+1+2+9-10	4+2+2+1+2+9-10	25	0.12
1+2+2+3+2+9-10	1+2+2+3+2+9-10	24	0.12
2+8+2+2+2+4-6	2+8+2+2+2+4-6	24	0.12
3+7+2+4+2+9-10	3+7+2+4+2+9-10	24	0.12
3+2+2+5plus+2+7-8	3+2+2+5plus+2+7-8	24	0.12
3+2+2+4+2+4-6	3+2+2+4+2+4-6	24	0.12
3+2+2+4+3+7-8	3+2+2+4+3+7-8	24	0.12
4+1+1+4+2+9-10	4+1+1+4+2+9-10	24	0.12
4+1+1+3+2+4-6	4+1+1+3+2+4-6	24	0.12
4+3+1+2+2+7-8	4+3+1+2+2+7-8	24	0.12
2+7+2+2+3+7-8	2+7+2+2+3+7-8	23	0.11
3+1+1+2+2+9-10	3+1+1+2+2+9-10	23	0.11
3+2+1+4+2+7-8	3+2+1+4+2+7-8	23	0.11
3+1+2+4+2+7-8	3+1+2+4+2+7-8	23	0.11
5+2+2+5plus+2+7-8	5+2+2+5plus+2+7-8	23	0.11
5+8+2+4+2+7-8	5+8+2+4+2+7-8	23	0.11
5+1+2+2+2+7-8	5+1+2+2+2+7-8	23	0.11
6+1+1+2+2+4-6	6+1+1+2+2+4-6	23	0.11
6+2+1+1+2+4-6	6+2+1+1+2+4-6	23	0.11
5+7+2+2+2+9-10	5+7+2+2+2+9-10	23	0.11
6+4+2+4+2+7-8	6+4+2+4+2+7-8	23	0.11
1+1+2+2+3+7-8	1+1+2+2+3+7-8	22	0.11
1+2+1+1+2+9-10	1+2+1+1+2+9-10	22	0.11
2+4+1+2+2+7-8	2+4+1+2+2+7-8	22	0.11
2+2+1+4+2+7-8	2+2+1+4+2+7-8	22	0.11
2+6+2+5plus+2+7-8	2+6+2+5plus+2+7-8	22	0.11
3+2+2+2+2+4-6	3+2+2+2+2+4-6	22	0.11
3+1+2+1+2+9-10	3+1+2+1+2+9-10	22	0.11
3+7+2+5plus+2+9-10	3+7+2+5plus+2+9-10	22	0.11
3+4+2+4+2+7-8	3+4+2+4+2+7-8	22	0.11
3+2+2+3+2+9-10	3+2+2+3+2+9-10	22	0.11
4+1+2+4+2+9-10	4+1+2+4+2+9-10	22	0.11

5+1+1+2+2+4-6	5+1+1+2+2+4-6	22	0.11
7+2+1+4+2+4-6	7+2+1+4+2+4-6	22	0.11
1+2+1+5plus+2+7-8	1+2+1+5plus+2+7-8	22	0.11
3+4+1+1+2+9-10	3+4+1+1+2+9-10	22	0.11
2+1+2+1+3+7-8	2+1+2+1+3+7-8	21	0.10
2+2+1+2+2+9-10	2+2+1+2+2+9-10	21	0.10
3+7+2+5plus+2+7-8	3+7+2+5plus+2+7-8	21	0.10
4+3+2+2+2+9-10	4+3+2+2+2+9-10	21	0.10
5+2+2+2+2+4-6	5+2+2+2+2+4-6	21	0.10
6+8+2+2+2+9-10	6+8+2+2+2+9-10	21	0.10
1+2+2+5plus+2+7-8	1+2+2+5plus+2+7-8	21	0.10
1+1+1+2+3+7-8	1+1+1+2+3+7-8	20	0.10
1+2+2+1+2+9-10	1+2+2+1+2+9-10	20	0.10
1+1+2+4+3+7-8	1+1+2+4+3+7-8	20	0.10
1+4+2+2+3+7-8	1+4+2+2+3+7-8	20	0.10
2+2+1+4+2+4-6	2+2+1+4+2+4-6	20	0.10
2+7+2+2+2+9-10	2+7+2+2+2+9-10	20	0.10
3+1+2+5plus+2+7-8	3+1+2+5plus+2+7-8	20	0.10
3+2+2+5plus+2+9-10	3+2+2+5plus+2+9-10	20	0.10
4+1+1+1+3+7-8	4+1+1+1+3+7-8	20	0.10
4+2+2+5plus+2+9-10	4+2+2+5plus+2+9-10	20	0.10
4+3+1+1+2+7-8	4+3+1+1+2+7-8	20	0.10
5+6+2+4+3+9-10	5+6+2+4+3+9-10	20	0.10
5+1+2+2+2+9-10	5+1+2+2+2+9-10	20	0.10
5+2+2+4+3+7-8	5+2+2+4+3+7-8	20	0.10
6+7+2+2+2+9-10	6+7+2+2+2+9-10	20	0.10
7+1+1+1+2+7-8	7+1+1+1+2+7-8	20	0.10
1+3+2+1+2+9-10	1+3+2+1+2+9-10	20	0.10
4+4+2+5plus+2+9-10	4+4+2+5plus+2+9-10	20	0.10
6+2+2+3+2+9-10	6+2+2+3+2+9-10	20	0.10
1+6+2+1+2+7-8	1+6+2+1+2+7-8	19	0.09
1+2+1+4+2+4-6	1+2+1+4+2+4-6	19	0.09
2+1+1+1+3+7-8	2+1+1+1+3+7-8	19	0.09
2+1+2+1+2+4-6	2+1+2+1+2+4-6	19	0.09
2+1+1+1+1+4-6	2+1+1+1+1+4-6	19	0.09
2+2+1+2+3+7-8	2+2+1+2+3+7-8	19	0.09
2+2+1+1+2+2-3	2+2+1+1+2+2-3	19	0.09
3+4+2+2+3+7-8	3+4+2+2+3+7-8	19	0.09
4+1+2+5plus+2+7-8	4+1+2+5plus+2+7-8	19	0.09
4+4+2+1+2+7-8	4+4+2+1+2+7-8	19	0.09
5+2+2+3+2+7-8	5+2+2+3+2+7-8	19	0.09
5+8+2+5plus+2+7-8	5+8+2+5plus+2+7-8	19	0.09
6+2+2+2+2+7-8	6+2+2+2+2+7-8	19	0.09
1+1+2+4+2+7-8	1+1+2+4+2+7-8	19	0.09
3+8+1+1+2+7-8	3+8+1+1+2+7-8	19	0.09
1+4+2+2+2+4-6	1+4+2+2+2+4-6	18	0.09
1+7+1+1+2+7-8	1+7+1+1+2+7-8	18	0.09
1+2+2+4+2+4-6	1+2+2+4+2+4-6	18	0.09
1+2+2+3+2+7-8	1+2+2+3+2+7-8	18	0.09
1+4+2+4+3+7-8	1+4+2+4+3+7-8	18	0.09
1+4+2+4+3+9-10	1+4+2+4+3+9-10	18	0.09
2+1+1+3+2+7-8	2+1+1+3+2+7-8	18	0.09
2+7+2+3+2+7-8	2+7+2+3+2+7-8	18	0.09
2+2+1+4+2+9-10	2+2+1+4+2+9-10	18	0.09
2+1+2+5plus+3+9-10	2+1+2+5plus+3+9-10	18	0.09
3+2+1+1+2+9-10	3+2+1+1+2+9-10	18	0.09

2+1+2+3+2+7-8	2+1+2+3+2+7-8	18	0.09
3+4+2+4+2+9-10	3+4+2+4+2+9-10	18	0.09
3+2+2+4+2+9-10	3+2+2+4+2+9-10	18	0.09
4+2+1+2+3+9-10	4+2+1+2+3+9-10	18	0.09
4+3+1+1+2+9-10	4+3+1+1+2+9-10	18	0.09
4+7+2+5plus+2+9-10	4+7+2+5plus+2+9-10	18	0.09
6+2+1+3+2+7-8	6+2+1+3+2+7-8	18	0.09
7+1+2+2+2+9-10	7+1+2+2+2+9-10	18	0.09
7+4+2+2+3+9-10	7+4+2+2+3+9-10	18	0.09
1+1+1+1+3+7-8	1+1+1+1+3+7-8	17	0.08
1+2+1+2+2+4-6	1+2+1+2+2+4-6	17	0.08
1+1+2+2+3+9-10	1+1+2+2+3+9-10	17	0.08
1+2+1+4+2+7-8	1+2+1+4+2+7-8	17	0.08
2+8+2+2+3+7-8	2+8+2+2+3+7-8	17	0.08
2+1+1+1+3+9-10	2+1+1+1+3+9-10	17	0.08
2+2+2+2+3+7-8	2+2+2+2+3+7-8	17	0.08
2+1+2+2+3+9-10	2+1+2+2+3+9-10	17	0.08
2+7+2+2+3+4-6	2+7+2+2+3+4-6	17	0.08
2+1+2+4+2+9-10	2+1+2+4+2+9-10	17	0.08
3+7+2+3+2+7-8	3+7+2+3+2+7-8	17	0.08
3+7+2+1+2+9-10	3+7+2+1+2+9-10	17	0.08
6+1+1+3+2+7-8	6+1+1+3+2+7-8	17	0.08
4+2+1+3+2+7-8	4+2+1+3+2+7-8	17	0.08
4+2+1+2+3+7-8	4+2+1+2+3+7-8	17	0.08
5+2+1+1+2+7-8	5+2+1+1+2+7-8	17	0.08
5+6+2+4+2+9-10	5+6+2+4+2+9-10	17	0.08
5+2+1+2+2+9-10	5+2+1+2+2+9-10	17	0.08
5+8+2+4+2+9-10	5+8+2+4+2+9-10	17	0.08
7+2+1+1+2+7-8	7+2+1+1+2+7-8	17	0.08
7+4+2+2+2+7-8	7+4+2+2+2+7-8	17	0.08
3+4+2+4+2+4-6	3+4+2+4+2+4-6	17	0.08
3+7+1+2+2+9-10	3+7+1+2+2+9-10	17	0.08
3+1+2+4+2+9-10	3+1+2+4+2+9-10	17	0.08
4+4+1+1+2+7-8	4+4+1+1+2+7-8	17	0.08
6+6+2+4+2+7-8	6+6+2+4+2+7-8	17	0.08
1+1+1+3+2+4-6	1+1+1+3+2+4-6	16	0.08
1+2+2+2+3+7-8	1+2+2+2+3+7-8	16	0.08
1+4+1+2+2+4-6	1+4+1+2+2+4-6	16	0.08
1+4+2+2+3+9-10	1+4+2+2+3+9-10	16	0.08
1+5+2+2+2+7-8	1+5+2+2+2+7-8	16	0.08
2+6+2+3+3+7-8	2+6+2+3+3+7-8	16	0.08
2+8+2+3+2+4-6	2+8+2+3+2+4-6	16	0.08
2+7+2+4+2+4-6	2+7+2+4+2+4-6	16	0.08
3+1+2+3+2+7-8	3+1+2+3+2+7-8	16	0.08
3+7+2+3+2+9-10	3+7+2+3+2+9-10	16	0.08
4+1+2+4+3+7-8	4+1+2+4+3+7-8	16	0.08
6+7+2+4+2+7-8	6+7+2+4+2+7-8	16	0.08
4+6+2+4+2+7-8	4+6+2+4+2+7-8	16	0.08
4+4+2+3+2+9-10	4+4+2+3+2+9-10	16	0.08
4+2+2+3+3+9-10	4+2+2+3+3+9-10	16	0.08
5+2+2+2+3+7-8	5+2+2+2+3+7-8	16	0.08
5+2+2+4+2+9-10	5+2+2+4+2+9-10	16	0.08
6+7+2+2+2+7-8	6+7+2+2+2+7-8	16	0.08
6+2+1+5plus+2+4-6	6+2+1+5plus+2+4-6	16	0.08
6+1+2+2+3+9-10	6+1+2+2+3+9-10	16	0.08
6+2+2+4+2+7-8	6+2+2+4+2+7-8	16	0.08

6+8+2+4+2+9-10	6+8+2+4+2+9-10	16	0.08
3+4+2+3+2+4-6	3+4+2+3+2+4-6	16	0.08
3+6+1+2+2+9-10	3+6+1+2+2+9-10	16	0.08
5+6+2+5plus+2+7-8	5+6+2+5plus+2+7-8	16	0.08
1+1+2+2+2+4-6	1+1+2+2+2+4-6	15	0.07
1+3+2+4+2+7-8	1+3+2+4+2+7-8	15	0.07
1+2+1+1+2+2-3	1+2+1+1+2+2-3	15	0.07
2+7+2+4+2+9-10	2+7+2+4+2+9-10	15	0.07
2+1+1+5plus+2+7-8	2+1+1+5plus+2+7-8	15	0.07
2+8+2+3+2+9-10	2+8+2+3+2+9-10	15	0.07
2+1+2+2+3+7-8	2+1+2+2+3+7-8	15	0.07
2+8+2+4+2+4-6	2+8+2+4+2+4-6	15	0.07
2+7+2+3+3+4-6	2+7+2+3+3+4-6	15	0.07
2+2+2+1+2+4-6	2+2+2+1+2+4-6	15	0.07
3+1+1+2+2+7-8	3+1+1+2+2+7-8	15	0.07
3+4+2+1+2+9-10	3+4+2+1+2+9-10	15	0.07
3+7+2+1+2+7-8	3+7+2+1+2+7-8	15	0.07
4+2+2+1+2+4-6	4+2+2+1+2+4-6	15	0.07
4+2+2+4+3+7-8	4+2+2+4+3+7-8	15	0.07
4+1+2+2+2+4-6	4+1+2+2+2+4-6	15	0.07
4+3+1+3+2+4-6	4+3+1+3+2+4-6	15	0.07
4+2+2+4+3+9-10	4+2+2+4+3+9-10	15	0.07
5+1+1+1+1+7-8	5+1+1+1+1+7-8	15	0.07
5+4+2+3+2+7-8	5+4+2+3+2+7-8	15	0.07
6+1+1+5plus+2+7-8	6+1+1+5plus+2+7-8	15	0.07
6+2+2+4+2+9-10	6+2+2+4+2+9-10	15	0.07
6+7+2+1+2+9-10	6+7+2+1+2+9-10	15	0.07
1+1+2+4+2+9-10	1+1+2+4+2+9-10	15	0.07
2+5+2+1+2+9-10	2+5+2+1+2+9-10	15	0.07
4+2+1+4+2+4-6	4+2+1+4+2+4-6	15	0.07
4+4+2+5plus+2+7-8	4+4+2+5plus+2+7-8	15	0.07
1+3+2+5plus+2+9-10	1+3+2+5plus+2+9-10	15	0.07
1+6+1+1+2+7-8	1+6+1+1+2+7-8	14	0.07
1+6+2+4+2+7-8	1+6+2+4+2+7-8	14	0.07
1+1+2+1+2+9-10	1+1+2+1+2+9-10	14	0.07
1+3+1+2+2+7-8	1+3+1+2+2+7-8	14	0.07
1+1+1+1+2+2-3	1+1+1+1+2+2-3	14	0.07
2+8+2+2+3+9-10	2+8+2+2+3+9-10	14	0.07
1+6+2+1+2+4-6	1+6+2+1+2+4-6	14	0.07
2+4+2+2+3+7-8	2+4+2+2+3+7-8	14	0.07
2+2+1+1+3+7-8	2+2+1+1+3+7-8	14	0.07
2+8+2+3+3+7-8	2+8+2+3+3+7-8	14	0.07
2+8+2+1+2+7-8	2+8+2+1+2+7-8	14	0.07
2+6+2+5plus+2+9-10	2+6+2+5plus+2+9-10	14	0.07
2+2+2+3+2+4-6	2+2+2+3+2+4-6	14	0.07
2+6+1+2+2+7-8	2+6+1+2+2+7-8	14	0.07
3+1+1+2+2+4-6	3+1+1+2+2+4-6	14	0.07
3+1+2+2+2+4-6	3+1+2+2+2+4-6	14	0.07
3+2+2+5plus+3+9-10	3+2+2+5plus+3+9-10	14	0.07
7+4+2+4+2+9-10	7+4+2+4+2+9-10	14	0.07
5+5+2+2+2+9-10	5+5+2+2+2+9-10	14	0.07
5+5+2+1+2+4-6	5+5+2+1+2+4-6	14	0.07
5+8+1+2+2+7-8	5+8+1+2+2+7-8	14	0.07
5+6+2+3+2+9-10	5+6+2+3+2+9-10	14	0.07
6+1+2+1+2+7-8	6+1+2+1+2+7-8	14	0.07
1+1+2+3+2+9-10	1+1+2+3+2+9-10	14	0.07

1+7+2+2+2+9-10	1+7+2+2+2+9-10	14	0.07
2+4+2+1+2+4-6	2+4+2+1+2+4-6	14	0.07
2+2+1+5plus+2+7-8	2+2+1+5plus+2+7-8	14	0.07
2+6+2+5plus+2+4-6	2+6+2+5plus+2+4-6	14	0.07
6+7+2+5plus+2+9-10	6+7+2+5plus+2+9-10	14	0.07
3+3+2+4+2+7-8	3+3+2+4+2+7-8	14	0.07
1+4+1+4+2+4-6	1+4+1+4+2+4-6	14	0.07
6+8+2+3+2+9-10	6+8+2+3+2+9-10	14	0.07
1+4+2+3+2+7-8	1+4+2+3+2+7-8	13	0.06
1+2+2+2+2+4-6	1+2+2+2+2+4-6	13	0.06
1+7+1+3+2+7-8	1+7+1+3+2+7-8	13	0.06
1+1+1+1+3+4-6	1+1+1+1+3+4-6	13	0.06
2+4+1+2+2+4-6	2+4+1+2+2+4-6	13	0.06
2+8+2+5plus+3+7-8	2+8+2+5plus+3+7-8	13	0.06
2+2+2+4+3+9-10	2+2+2+4+3+9-10	13	0.06
2+6+2+1+2+7-8	2+6+2+1+2+7-8	13	0.06
2+1+2+1+3+9-10	2+1+2+1+3+9-10	13	0.06
2+7+2+3+2+4-6	2+7+2+3+2+4-6	13	0.06
2+6+2+2+2+4-6	2+6+2+2+2+4-6	13	0.06
2+8+2+1+2+4-6	2+8+2+1+2+4-6	13	0.06
2+1+2+4+3+7-8	2+1+2+4+3+7-8	13	0.06
3+2+1+2+2+4-6	3+2+1+2+2+4-6	13	0.06
3+1+1+1+3+7-8	3+1+1+1+3+7-8	13	0.06
3+6+2+2+2+9-10	3+6+2+2+2+9-10	13	0.06
3+6+1+1+2+9-10	3+6+1+1+2+9-10	13	0.06
3+7+2+4+2+4-6	3+7+2+4+2+4-6	13	0.06
4+2+1+5plus+2+4-6	4+2+1+5plus+2+4-6	13	0.06
4+2+2+4+2+4-6	4+2+2+4+2+4-6	13	0.06
4+3+1+2+2+9-10	4+3+1+2+2+9-10	13	0.06
4+4+2+3+2+7-8	4+4+2+3+2+7-8	13	0.06
5+6+2+4+3+7-8	5+6+2+4+3+7-8	13	0.06
5+1+1+2+3+7-8	5+1+1+2+3+7-8	13	0.06
5+2+2+5plus+3+7-8	5+2+2+5plus+3+7-8	13	0.06
6+2+1+5plus+2+7-8	6+2+1+5plus+2+7-8	13	0.06
6+9+2+4+2+7-8	6+9+2+4+2+7-8	13	0.06
6+6+1+5plus+2+9-10	6+6+1+5plus+2+9-10	13	0.06
6+2+2+1+2+7-8	6+2+2+1+2+7-8	13	0.06
7+1+2+1+2+7-8	7+1+2+1+2+7-8	13	0.06
1+2+2+5plus+3+7-8	1+2+2+5plus+3+7-8	13	0.06
2+6+1+1+2+7-8	2+6+1+1+2+7-8	13	0.06
6+8+2+4+2+4-6	6+8+2+4+2+4-6	13	0.06
4+3+1+2+2+4-6	4+3+1+2+2+4-6	13	0.06
4+7+1+2+2+7-8	4+7+1+2+2+7-8	13	0.06
5+8+1+1+2+7-8	5+8+1+1+2+7-8	13	0.06
5+6+2+1+2+9-10	5+6+2+1+2+9-10	13	0.06
2+9+2+1+2+4-6	2+9+2+1+2+4-6	13	0.06
5+2+2+4+2+4-6	5+2+2+4+2+4-6	13	0.06
5+7+2+5plus+2+7-8	5+7+2+5plus+2+7-8	13	0.06
5+2+2+1+2+9-10	5+2+2+1+2+9-10	13	0.06
1+7+1+2+2+7-8	1+7+1+2+2+7-8	12	0.06
1+6+2+4+3+7-8	1+6+2+4+3+7-8	12	0.06
1+1+2+4+3+4-6	1+1+2+4+3+4-6	12	0.06
2+4+2+4+3+7-8	2+4+2+4+3+7-8	12	0.06
1+3+2+2+2+7-8	1+3+2+2+2+7-8	12	0.06
2+1+1+2+3+7-8	2+1+1+2+3+7-8	12	0.06
2+1+1+4+2+9-10	2+1+1+4+2+9-10	12	0.06

2+8+1+2+3+7-8	2+8+1+2+3+7-8	12	0.06
2+8+2+4+3+7-8	2+8+2+4+3+7-8	12	0.06
2+2+2+5plus+2+9-10	2+2+2+5plus+2+9-10	12	0.06
3+4+2+2+2+4-6	3+4+2+2+2+4-6	12	0.06
3+2+2+2+3+9-10	3+2+2+2+3+9-10	12	0.06
3+2+2+2+3+7-8	3+2+2+2+3+7-8	12	0.06
4+2+2+3+3+7-8	4+2+2+3+3+7-8	12	0.06
7+4+2+4+2+7-8	7+4+2+4+2+7-8	12	0.06
4+1+2+4+2+4-6	4+1+2+4+2+4-6	12	0.06
6+7+2+3+2+7-8	6+7+2+3+2+7-8	12	0.06
4+4+2+4+3+7-8	4+4+2+4+3+7-8	12	0.06
5+1+1+4+2+7-8	5+1+1+4+2+7-8	12	0.06
5+6+2+2+3+7-8	5+6+2+2+3+7-8	12	0.06
6+6+2+4+2+9-10	6+6+2+4+2+9-10	12	0.06
7+2+2+2+2+9-10	7+2+2+2+2+9-10	12	0.06
7+6+2+2+2+9-10	7+6+2+2+2+9-10	12	0.06
1+3+2+1+2+7-8	1+3+2+1+2+7-8	12	0.06
2+6+1+2+2+9-10	2+6+1+2+2+9-10	12	0.06
2+6+1+2+2+4-6	2+6+1+2+2+4-6	12	0.06
2+4+1+1+2+7-8	2+4+1+1+2+7-8	12	0.06
2+4+2+3+2+9-10	2+4+2+3+2+9-10	12	0.06
3+4+1+3+2+7-8	3+4+1+3+2+7-8	12	0.06
5+4+2+5plus+2+7-8	5+4+2+5plus+2+7-8	12	0.06
6+6+2+2+2+9-10	6+6+2+2+2+9-10	12	0.06
5+4+1+1+2+9-10	5+4+1+1+2+9-10	12	0.06
6+4+2+4+2+9-10	6+4+2+4+2+9-10	12	0.06
1+2+1+2+3+9-10	1+2+1+2+3+9-10	11	0.05
1+7+1+2+2+9-10	1+7+1+2+2+9-10	11	0.05
1+7+2+3+2+4-6	1+7+2+3+2+4-6	11	0.05
1+4+2+3+2+4-6	1+4+2+3+2+4-6	11	0.05
1+6+2+2+2+7-8	1+6+2+2+2+7-8	11	0.05
2+7+1+1+2+4-6	2+7+1+1+2+4-6	11	0.05
2+1+2+1+3+4-6	2+1+2+1+3+4-6	11	0.05
2+6+2+2+3+9-10	2+6+2+2+3+9-10	11	0.05
1+8+2+4+2+7-8	1+8+2+4+2+7-8	11	0.05
2+7+2+5plus+2+4-6	2+7+2+5plus+2+4-6	11	0.05
2+1+1+3+2+4-6	2+1+1+3+2+4-6	11	0.05
2+7+2+3+2+9-10	2+7+2+3+2+9-10	11	0.05
3+7+2+3+2+4-6	3+7+2+3+2+4-6	11	0.05
3+6+2+2+2+7-8	3+6+2+2+2+7-8	11	0.05
3+8+2+2+2+7-8	3+8+2+2+2+7-8	11	0.05
4+2+1+1+3+9-10	4+2+1+1+3+9-10	11	0.05
4+2+1+5plus+2+9-10	4+2+1+5plus+2+9-10	11	0.05
4+1+2+2+3+7-8	4+1+2+2+3+7-8	11	0.05
4+1+2+5plus+2+9-10	4+1+2+5plus+2+9-10	11	0.05
4+1+1+5plus+2+7-8	4+1+1+5plus+2+7-8	11	0.05
4+2+1+3+2+4-6	4+2+1+3+2+4-6	11	0.05
5+8+2+2+3+7-8	5+8+2+2+3+7-8	11	0.05
5+1+1+5plus+2+4-6	5+1+1+5plus+2+4-6	11	0.05
5+6+2+3+2+7-8	5+6+2+3+2+7-8	11	0.05
5+1+2+4+2+9-10	5+1+2+4+2+9-10	11	0.05
6+2+1+2+3+7-8	6+2+1+2+3+7-8	11	0.05
6+2+1+3+2+9-10	6+2+1+3+2+9-10	11	0.05
6+8+2+5plus+2+4-6	6+8+2+5plus+2+4-6	11	0.05
3+2+1+1+3+7-8	3+2+1+1+3+7-8	11	0.05
6+2+2+4+2+4-6	6+2+2+4+2+4-6	11	0.05

6+8+2+2+2+7-8	6+8+2+2+2+7-8	11	0.05
7+4+2+5plus+2+7-8	7+4+2+5plus+2+7-8	11	0.05
7+3+1+5plus+2+7-8	7+3+1+5plus+2+7-8	11	0.05
7+2+1+1+2+4-6	7+2+1+1+2+4-6	11	0.05
7+2+1+4+2+7-8	7+2+1+4+2+7-8	11	0.05
1+4+1+1+2+7-8	1+4+1+1+2+7-8	11	0.05
3+6+1+2+2+7-8	3+6+1+2+2+7-8	11	0.05
5+4+1+4+2+7-8	5+4+1+4+2+7-8	11	0.05
5+7+2+2+2+7-8	5+7+2+2+2+7-8	11	0.05
7+2+1+2+2+4-6	7+2+1+2+2+4-6	11	0.05
3+8+2+1+2+9-10	3+8+2+1+2+9-10	11	0.05
1+1+1+5plus+2+4-6	1+1+1+5plus+2+4-6	10	0.05
1+1+1+5plus+2+2-3	1+1+1+5plus+2+2-3	10	0.05
1+4+2+5plus+3+7-8	1+4+2+5plus+3+7-8	10	0.05
1+4+2+5plus+2+9-10	1+4+2+5plus+2+9-10	10	0.05
1+4+2+3+2+9-10	1+4+2+3+2+9-10	10	0.05
1+2+2+4+3+7-8	1+2+2+4+3+7-8	10	0.05
1+4+2+4+2+9-10	1+4+2+4+2+9-10	10	0.05
1+7+2+1+2+7-8	1+7+2+1+2+7-8	10	0.05
1+1+1+2+3+4-6	1+1+1+2+3+4-6	10	0.05
2+1+1+1+3+4-6	2+1+1+1+3+4-6	10	0.05
2+6+2+3+2+9-10	2+6+2+3+2+9-10	10	0.05
2+6+2+4+2+9-10	2+6+2+4+2+9-10	10	0.05
2+6+2+3+2+4-6	2+6+2+3+2+4-6	10	0.05
2+1+2+5plus+3+7-8	2+1+2+5plus+3+7-8	10	0.05
2+8+2+3+3+4-6	2+8+2+3+3+4-6	10	0.05
2+8+1+1+2+9-10	2+8+1+1+2+9-10	10	0.05
2+7+1+1+2+7-8	2+7+1+1+2+7-8	10	0.05
2+1+2+3+2+4-6	2+1+2+3+2+4-6	10	0.05
2+7+2+1+2+7-8	2+7+2+1+2+7-8	10	0.05
2+7+1+2+2+7-8	2+7+1+2+2+7-8	10	0.05
3+2+1+4+2+4-6	3+2+1+4+2+4-6	10	0.05
3+4+2+5plus+2+7-8	3+4+2+5plus+2+7-8	10	0.05
3+6+2+2+1+9-10	3+6+2+2+1+9-10	10	0.05
3+1+2+2+3+9-10	3+1+2+2+3+9-10	10	0.05
3+8+2+2+2+4-6	3+8+2+2+2+4-6	10	0.05
3+2+1+1+2+4-6	3+2+1+1+2+4-6	10	0.05
6+7+2+3+2+9-10	6+7+2+3+2+9-10	10	0.05
4+6+2+4+2+4-6	4+6+2+4+2+4-6	10	0.05
4+3+1+4+2+4-6	4+3+1+4+2+4-6	10	0.05
4+2+2+5plus+2+4-6	4+2+2+5plus+2+4-6	10	0.05
4+7+2+5plus+2+7-8	4+7+2+5plus+2+7-8	10	0.05
4+3+2+4+2+7-8	4+3+2+4+2+7-8	10	0.05
5+5+1+2+2+7-8	5+5+1+2+2+7-8	10	0.05
5+5+2+2+2+7-8	5+5+2+2+2+7-8	10	0.05
5+1+2+5plus+2+7-8	5+1+2+5plus+2+7-8	10	0.05
6+1+1+1+3+4-6	6+1+1+1+3+4-6	10	0.05
6+5+2+4+2+7-8	6+5+2+4+2+7-8	10	0.05
6+2+2+4+3+9-10	6+2+2+4+3+9-10	10	0.05
7+1+2+3+2+7-8	7+1+2+3+2+7-8	10	0.05
1+1+1+4+2+9-10	1+1+1+4+2+9-10	10	0.05
1+5+2+2+2+4-6	1+5+2+2+2+4-6	10	0.05
1+1+2+3+2+7-8	1+1+2+3+2+7-8	10	0.05
2+6+2+1+2+4-6	2+6+2+1+2+4-6	10	0.05
2+2+1+1+3+9-10	2+2+1+1+3+9-10	10	0.05
2+2+1+5plus+2+4-6	2+2+1+5plus+2+4-6	10	0.05

6+8+2+4+2+7-8	6+8+2+4+2+7-8	10	0.05
3+7+2+5plus+2+4-6	3+7+2+5plus+2+4-6	10	0.05
4+4+1+2+2+9-10	4+4+1+2+2+9-10	10	0.05
5+2+2+4+3+4-6	5+2+2+4+3+4-6	10	0.05
6+3+1+4+2+9-10	6+3+1+4+2+9-10	10	0.05
5+7+2+1+2+7-8	5+7+2+1+2+7-8	10	0.05
1+2+1+1+3+4-6	1+2+1+1+3+4-6	9	0.04
1+2+1+1+3+7-8	1+2+1+1+3+7-8	9	0.04
1+4+2+4+2+4-6	1+4+2+4+2+4-6	9	0.04
1+1+1+2+3+9-10	1+1+1+2+3+9-10	9	0.04
1+3+1+2+2+4-6	1+3+1+2+2+4-6	9	0.04
1+7+2+3+2+7-8	1+7+2+3+2+7-8	9	0.04
1+2+2+1+2+2-3	1+2+2+1+2+2-3	9	0.04
1+2+2+2+3+9-10	1+2+2+2+3+9-10	9	0.04
1+6+2+5plus+2+7-8	1+6+2+5plus+2+7-8	9	0.04
2+7+1+3+2+4-6	2+7+1+3+2+4-6	9	0.04
2+2+2+2+3+4-6	2+2+2+2+3+4-6	9	0.04
2+2+1+3+2+4-6	2+2+1+3+2+4-6	9	0.04
2+6+2+3+3+9-10	2+6+2+3+3+9-10	9	0.04
2+6+1+4+2+9-10	2+6+1+4+2+9-10	9	0.04
2+4+2+1+2+7-8	2+4+2+1+2+7-8	9	0.04
2+8+1+1+2+7-8	2+8+1+1+2+7-8	9	0.04
2+2+2+3+3+7-8	2+2+2+3+3+7-8	9	0.04
3+2+1+3+2+4-6	3+2+1+3+2+4-6	9	0.04
3+4+2+5plus+2+9-10	3+4+2+5plus+2+9-10	9	0.04
3+7+2+2+2+4-6	3+7+2+2+2+4-6	9	0.04
3+1+1+1+3+9-10	3+1+1+1+3+9-10	9	0.04
3+2+1+1+3+9-10	3+2+1+1+3+9-10	9	0.04
3+8+1+2+2+9-10	3+8+1+2+2+9-10	9	0.04
4+2+2+2+3+4-6	4+2+2+2+3+4-6	9	0.04
4+1+1+2+3+9-10	4+1+1+2+3+9-10	9	0.04
4+1+1+3+3+4-6	4+1+1+3+3+4-6	9	0.04
4+1+1+2+2+2-3	4+1+1+2+2+2-3	9	0.04
4+7+2+4+2+7-8	4+7+2+4+2+7-8	9	0.04
4+4+2+2+2+4-6	4+4+2+2+2+4-6	9	0.04
5+8+2+2+3+9-10	5+8+2+2+3+9-10	9	0.04
5+6+1+3+2+7-8	5+6+1+3+2+7-8	9	0.04
5+8+1+4+2+7-8	5+8+1+4+2+7-8	9	0.04
5+8+2+4+2+4-6	5+8+2+4+2+4-6	9	0.04
5+4+2+3+2+9-10	5+4+2+3+2+9-10	9	0.04
5+8+2+3+2+7-8	5+8+2+3+2+7-8	9	0.04
5+8+2+3+2+9-10	5+8+2+3+2+9-10	9	0.04
5+1+1+3+2+7-8	5+1+1+3+2+7-8	9	0.04
6+6+2+1+2+9-10	6+6+2+1+2+9-10	9	0.04
6+1+2+1+2+9-10	6+1+2+1+2+9-10	9	0.04
6+1+2+2+3+7-8	6+1+2+2+3+7-8	9	0.04
6+6+2+5plus+2+9-10	6+6+2+5plus+2+9-10	9	0.04
7+8+2+2+2+7-8	7+8+2+2+2+7-8	9	0.04
7+1+2+2+2+7-8	7+1+2+2+2+7-8	9	0.04
7+4+2+3+2+4-6	7+4+2+3+2+4-6	9	0.04
7+2+2+4+2+7-8	7+2+2+4+2+7-8	9	0.04
7+6+2+2+2+7-8	7+6+2+2+2+7-8	9	0.04
7+2+2+3+2+7-8	7+2+2+3+2+7-8	9	0.04
1+1+1+2+1+4-6	1+1+1+2+1+4-6	9	0.04
1+4+2+5plus+2+4-6	1+4+2+5plus+2+4-6	9	0.04
1+3+2+1+2+4-6	1+3+2+1+2+4-6	9	0.04

1+3+2+5plus+3+7-8	1+3+2+5plus+3+7-8	9	0.04
1+7+2+5plus+2+4-6	1+7+2+5plus+2+4-6	9	0.04
1+3+1+4+2+9-10	1+3+1+4+2+9-10	9	0.04
2+4+2+5plus+2+9-10	2+4+2+5plus+2+9-10	9	0.04
2+1+2+3+3+7-8	2+1+2+3+3+7-8	9	0.04
2+2+2+4+2+4-6	2+2+2+4+2+4-6	9	0.04
2+8+1+5plus+2+4-6	2+8+1+5plus+2+4-6	9	0.04
3+4+2+3+3+7-8	3+4+2+3+3+7-8	9	0.04
3+5+2+5plus+2+7-8	3+5+2+5plus+2+7-8	9	0.04
4+3+1+4+2+7-8	4+3+1+4+2+7-8	9	0.04
4+1+2+3+2+4-6	4+1+2+3+2+4-6	9	0.04
4+1+1+2+1+2-3	4+1+1+2+1+2-3	9	0.04
5+6+1+2+2+9-10	5+6+1+2+2+9-10	9	0.04
5+6+2+2+3+9-10	5+6+2+2+3+9-10	9	0.04
6+6+2+2+3+9-10	6+6+2+2+3+9-10	9	0.04
6+1+1+4+2+7-8	6+1+1+4+2+7-8	9	0.04
7+3+2+3+2+4-6	7+3+2+3+2+4-6	9	0.04
2+9+2+5plus+2+7-8	2+9+2+5plus+2+7-8	9	0.04
2+4+1+4+2+9-10	2+4+1+4+2+9-10	9	0.04
1+6+2+2+2+4-6	1+6+2+2+2+4-6	9	0.04
4+6+2+4+2+9-10	4+6+2+4+2+9-10	9	0.04
5+2+2+3+2+9-10	5+2+2+3+2+9-10	9	0.04
5+4+2+2+2+9-10	5+4+2+2+2+9-10	9	0.04
1+2+1+2+3+7-8	1+2+1+2+3+7-8	8	0.04
1+1+1+1+1+7-8	1+1+1+1+1+7-8	8	0.04
1+7+2+2+3+7-8	1+7+2+2+3+7-8	8	0.04
1+6+1+1+2+9-10	1+6+1+1+2+9-10	8	0.04
1+1+2+4+2+4-6	1+1+2+4+2+4-6	8	0.04
1+1+2+3+2+4-6	1+1+2+3+2+4-6	8	0.04
1+6+1+1+3+7-8	1+6+1+1+3+7-8	8	0.04
2+6+2+4+3+9-10	2+6+2+4+3+9-10	8	0.04
2+1+1+5plus+2+4-6	2+1+1+5plus+2+4-6	8	0.04
2+7+1+4+2+4-6	2+7+1+4+2+4-6	8	0.04
1+8+2+4+3+7-8	1+8+2+4+3+7-8	8	0.04
1+8+2+4+2+9-10	1+8+2+4+2+9-10	8	0.04
2+6+1+3+2+7-8	2+6+1+3+2+7-8	8	0.04
2+2+2+3+2+9-10	2+2+2+3+2+9-10	8	0.04
2+6+1+3+2+4-6	2+6+1+3+2+4-6	8	0.04
2+7+2+4+3+4-6	2+7+2+4+3+4-6	8	0.04
2+8+2+1+3+9-10	2+8+2+1+3+9-10	8	0.04
1+8+1+1+2+9-10	1+8+1+1+2+9-10	8	0.04
1+8+1+1+2+4-6	1+8+1+1+2+4-6	8	0.04
2+8+1+2+2+4-6	2+8+1+2+2+4-6	8	0.04
2+6+2+5plus+3+7-8	2+6+2+5plus+3+7-8	8	0.04
3+3+1+2+2+7-8	3+3+1+2+2+7-8	8	0.04
3+2+2+1+2+9-10	3+2+2+1+2+9-10	8	0.04
7+4+2+4+3+9-10	7+4+2+4+3+9-10	8	0.04
4+4+2+1+3+7-8	4+4+2+1+3+7-8	8	0.04
4+1+2+2+3+9-10	4+1+2+2+3+9-10	8	0.04
4+1+1+1+3+9-10	4+1+1+1+3+9-10	8	0.04
4+2+2+5plus+3+7-8	4+2+2+5plus+3+7-8	8	0.04
4+1+2+1+2+9-10	4+1+2+1+2+9-10	8	0.04
4+1+1+5plus+2+9-10	4+1+1+5plus+2+9-10	8	0.04
4+3+1+1+2+4-6	4+3+1+1+2+4-6	8	0.04
4+2+1+1+3+4-6	4+2+1+1+3+4-6	8	0.04
5+8+2+4+3+7-8	5+8+2+4+3+7-8	8	0.04

5+8+1+2+2+9-10	5+8+1+2+2+9-10	8	0.04
5+4+2+3+2+4-6	5+4+2+3+2+4-6	8	0.04
5+2+2+5plus+2+9-10	5+2+2+5plus+2+9-10	8	0.04
5+1+1+1+3+9-10	5+1+1+1+3+9-10	8	0.04
5+4+2+4+3+7-8	5+4+2+4+3+7-8	8	0.04
5+1+2+5plus+2+9-10	5+1+2+5plus+2+9-10	8	0.04
6+2+1+1+2+9-10	6+2+1+1+2+9-10	8	0.04
6+9+2+4+2+4-6	6+9+2+4+2+4-6	8	0.04
6+2+2+1+2+9-10	6+2+2+1+2+9-10	8	0.04
7+2+2+2+2+7-8	7+2+2+2+2+7-8	8	0.04
7+4+2+1+2+7-8	7+4+2+1+2+7-8	8	0.04
7+2+2+2+3+9-10	7+2+2+2+3+9-10	8	0.04
7+3+1+4+2+9-10	7+3+1+4+2+9-10	8	0.04
2+4+2+5plus+2+7-8	2+4+2+5plus+2+7-8	8	0.04
2+1+2+3+2+9-10	2+1+2+3+2+9-10	8	0.04
2+8+1+4+2+7-8	2+8+1+4+2+7-8	8	0.04
2+7+1+3+2+9-10	2+7+1+3+2+9-10	8	0.04
4+4+1+2+2+7-8	4+4+1+2+2+7-8	8	0.04
4+3+2+2+3+7-8	4+3+2+2+3+7-8	8	0.04
4+1+1+1+1+7-8	4+1+1+1+1+7-8	8	0.04
4+7+2+2+2+4-6	4+7+2+2+2+4-6	8	0.04
5+1+2+1+2+7-8	5+1+2+1+2+7-8	8	0.04
6+2+1+4+2+9-10	6+2+1+4+2+9-10	8	0.04
6+6+2+2+3+7-8	6+6+2+2+3+7-8	8	0.04
2+9+2+1+2+7-8	2+9+2+1+2+7-8	8	0.04
3+4+2+3+3+9-10	3+4+2+3+3+9-10	8	0.04
3+2+2+1+2+4-6	3+2+2+1+2+4-6	8	0.04
1+3+2+5plus+2+7-8	1+3+2+5plus+2+7-8	8	0.04
1+4+2+3+3+7-8	1+4+2+3+3+7-8	8	0.04
1+3+2+5plus+2+4-6	1+3+2+5plus+2+4-6	8	0.04
4+3+2+1+2+7-8	4+3+2+1+2+7-8	8	0.04
6+8+3+1+2+9-10	6+8+3+1+2+9-10	8	0.04
7+6+2+1+2+7-8	7+6+2+1+2+7-8	8	0.04
1+1+2+2+3+4-6	1+1+2+2+3+4-6	7	0.03
1+3+2+4+2+4-6	1+3+2+4+2+4-6	7	0.03
1+8+2+1+2+9-10	1+8+2+1+2+9-10	7	0.03
1+7+2+2+3+9-10	1+7+2+2+3+9-10	7	0.03
1+2+2+4+3+4-6	1+2+2+4+3+4-6	7	0.03
1+1+2+1+3+7-8	1+1+2+1+3+7-8	7	0.03
1+1+1+3+3+7-8	1+1+1+3+3+7-8	7	0.03
1+3+1+3+2+7-8	1+3+1+3+2+7-8	7	0.03
1+3+1+3+2+4-6	1+3+1+3+2+4-6	7	0.03
1+7+1+3+2+9-10	1+7+1+3+2+9-10	7	0.03
1+1+2+3+3+7-8	1+1+2+3+3+7-8	7	0.03
1+1+1+2+1+7-8	1+1+1+2+1+7-8	7	0.03
1+6+1+1+2+4-6	1+6+1+1+2+4-6	7	0.03
1+6+1+2+2+4-6	1+6+1+2+2+4-6	7	0.03
1+1+2+5plus+2+9-10	1+1+2+5plus+2+9-10	7	0.03
1+4+2+4+3+4-6	1+4+2+4+3+4-6	7	0.03
1+3+2+4+2+9-10	1+3+2+4+2+9-10	7	0.03
1+6+2+5plus+2+9-10	1+6+2+5plus+2+9-10	7	0.03
2+4+2+2+3+9-10	2+4+2+2+3+9-10	7	0.03
2+2+1+1+3+4-6	2+2+1+1+3+4-6	7	0.03
2+1+1+5plus+3+7-8	2+1+1+5plus+3+7-8	7	0.03
2+6+2+2+3+4-6	2+6+2+2+3+4-6	7	0.03
2+6+2+4+3+7-8	2+6+2+4+3+7-8	7	0.03

2+5+2+5plus+2+7-8	2+5+2+5plus+2+7-8	7	0.03
2+7+1+3+2+7-8	2+7+1+3+2+7-8	7	0.03
2+9+1+4+2+4-6	2+9+1+4+2+4-6	7	0.03
2+6+1+4+2+4-6	2+6+1+4+2+4-6	7	0.03
2+7+2+1+2+9-10	2+7+2+1+2+9-10	7	0.03
3+1+1+3+2+7-8	3+1+1+3+2+7-8	7	0.03
2+2+1+1+2+0-1	2+2+1+1+2+0-1	7	0.03
3+6+2+2+3+7-8	3+6+2+2+3+7-8	7	0.03
3+1+2+4+3+7-8	3+1+2+4+3+7-8	7	0.03
3+7+1+4+2+4-6	3+7+1+4+2+4-6	7	0.03
3+1+2+5plus+2+9-10	3+1+2+5plus+2+9-10	7	0.03
3+2+2+4+3+4-6	3+2+2+4+3+4-6	7	0.03
3+4+1+1+2+4-6	3+4+1+1+2+4-6	7	0.03
4+4+2+4+3+9-10	4+4+2+4+3+9-10	7	0.03
4+1+1+4+1+7-8	4+1+1+4+1+7-8	7	0.03
4+6+2+3+2+7-8	4+6+2+3+2+7-8	7	0.03
4+2+1+4+2+9-10	4+2+1+4+2+9-10	7	0.03
4+3+1+3+2+7-8	4+3+1+3+2+7-8	7	0.03
4+1+2+5plus+3+9-10	4+1+2+5plus+3+9-10	7	0.03
4+4+2+1+3+9-10	4+4+2+1+3+9-10	7	0.03
4+1+1+3+3+7-8	4+1+1+3+3+7-8	7	0.03
5+2+2+4+3+9-10	5+2+2+4+3+9-10	7	0.03
5+1+1+2+3+9-10	5+1+1+2+3+9-10	7	0.03
5+2+1+3+2+7-8	5+2+1+3+2+7-8	7	0.03
5+1+1+3+2+9-10	5+1+1+3+2+9-10	7	0.03
5+4+2+4+2+7-8	5+4+2+4+2+7-8	7	0.03
5+2+1+2+3+7-8	5+2+1+2+3+7-8	7	0.03
5+2+1+2+2+4-6	5+2+1+2+2+4-6	7	0.03
5+1+2+4+3+9-10	5+1+2+4+3+9-10	7	0.03
6+2+1+2+3+9-10	6+2+1+2+3+9-10	7	0.03
6+1+1+1+2+2-3	6+1+1+1+2+2-3	7	0.03
6+1+2+4+2+7-8	6+1+2+4+2+7-8	7	0.03
6+1+2+3+2+7-8	6+1+2+3+2+7-8	7	0.03
6+1+1+4+3+4-6	6+1+1+4+3+4-6	7	0.03
6+8+2+5plus+3+7-8	6+8+2+5plus+3+7-8	7	0.03
6+8+2+2+3+9-10	6+8+2+2+3+9-10	7	0.03
1+2+2+1+3+7-8	1+2+2+1+3+7-8	7	0.03
1+1+1+5plus+2+7-8	1+1+1+5plus+2+7-8	7	0.03
1+1+2+1+2+4-6	1+1+2+1+2+4-6	7	0.03
2+4+1+3+2+7-8	2+4+1+3+2+7-8	7	0.03
2+7+2+2+2+2-3	2+7+2+2+2+2-3	7	0.03
2+4+1+1+2+4-6	2+4+1+1+2+4-6	7	0.03
3+1+1+1+1+4-6	3+1+1+1+1+4-6	7	0.03
3+7+2+4+3+9-10	3+7+2+4+3+9-10	7	0.03
4+4+1+1+2+9-10	4+4+1+1+2+9-10	7	0.03
4+4+2+5plus+2+4-6	4+4+2+5plus+2+4-6	7	0.03
4+6+2+4+3+9-10	4+6+2+4+3+9-10	7	0.03
4+3+2+2+3+9-10	4+3+2+2+3+9-10	7	0.03
4+2+1+4+3+4-6	4+2+1+4+3+4-6	7	0.03
5+4+1+3+2+7-8	5+4+1+3+2+7-8	7	0.03
5+9+2+5plus+3+4-6	5+9+2+5plus+3+4-6	7	0.03
6+3+2+2+3+9-10	6+3+2+2+3+9-10	7	0.03
6+7+2+4+2+9-10	6+7+2+4+2+9-10	7	0.03
6+2+1+4+2+7-8	6+2+1+4+2+7-8	7	0.03
6+1+1+4+2+4-6	6+1+1+4+2+4-6	7	0.03
6+2+2+3+2+7-8	6+2+2+3+2+7-8	7	0.03

6+6+2+4+2+4-6	6+6+2+4+2+4-6	7	0.03
6+1+2+3+1+7-8	6+1+2+3+1+7-8	7	0.03
2+8+3+2+2+9-10	2+8+3+2+2+9-10	7	0.03
2+9+2+5plus+2+4-6	2+9+2+5plus+2+4-6	7	0.03
2+7+3+1+2+9-10	2+7+3+1+2+9-10	7	0.03
3+3+1+5plus+2+7-8	3+3+1+5plus+2+7-8	7	0.03
3+3+2+5plus+3+4-6	3+3+2+5plus+3+4-6	7	0.03
4+3+2+5plus+2+7-8	4+3+2+5plus+2+7-8	7	0.03
1+2+2+5plus+1+4-6	1+2+2+5plus+1+4-6	7	0.03
4+3+2+1+2+9-10	4+3+2+1+2+9-10	7	0.03
5+5+2+5plus+2+7-8	5+5+2+5plus+2+7-8	7	0.03
5+5+2+2+3+7-8	5+5+2+2+3+7-8	7	0.03
6+4+2+2+2+4-6	6+4+2+2+2+4-6	7	0.03
6+6+1+5plus+2+7-8	6+6+1+5plus+2+7-8	7	0.03
6+2+2+5plus+2+7-8	6+2+2+5plus+2+7-8	7	0.03
6+2+1+1+2+0-1	6+2+1+1+2+0-1	7	0.03
1+3+2+4+3+7-8	1+3+2+4+3+7-8	6	0.03
1+7+2+4+2+4-6	1+7+2+4+2+4-6	6	0.03
1+1+1+3+3+4-6	1+1+1+3+3+4-6	6	0.03
1+8+1+2+2+7-8	1+8+1+2+2+7-8	6	0.03
1+7+2+1+2+9-10	1+7+2+1+2+9-10	6	0.03
1+6+2+1+3+7-8	1+6+2+1+3+7-8	6	0.03
1+4+2+1+3+7-8	1+4+2+1+3+7-8	6	0.03
2+4+1+4+3+7-8	2+4+1+4+3+7-8	6	0.03
2+4+1+3+2+4-6	2+4+1+3+2+4-6	6	0.03
2+6+1+2+3+7-8	2+6+1+2+3+7-8	6	0.03
2+7+2+2+3+9-10	2+7+2+2+3+9-10	6	0.03
2+1+1+2+1+7-8	2+1+1+2+1+7-8	6	0.03
2+1+1+2+3+4-6	2+1+1+2+3+4-6	6	0.03
2+2+1+3+2+9-10	2+2+1+3+2+9-10	6	0.03
2+8+1+2+2+7-8	2+8+1+2+2+7-8	6	0.03
2+2+1+2+1+7-8	2+2+1+2+1+7-8	6	0.03
2+6+2+1+3+4-6	2+6+2+1+3+4-6	6	0.03
2+7+2+5plus+3+7-8	2+7+2+5plus+3+7-8	6	0.03
2+7+2+3+2+2-3	2+7+2+3+2+2-3	6	0.03
2+1+2+3+3+9-10	2+1+2+3+3+9-10	6	0.03
1+8+1+1+2+7-8	1+8+1+1+2+7-8	6	0.03
3+1+1+4+2+7-8	3+1+1+4+2+7-8	6	0.03
3+2+2+3+3+9-10	3+2+2+3+3+9-10	6	0.03
6+6+1+1+2+7-8	6+6+1+1+2+7-8	6	0.03
3+4+2+2+3+4-6	3+4+2+2+3+4-6	6	0.03
3+1+1+3+2+4-6	3+1+1+3+2+4-6	6	0.03
3+1+1+5plus+2+7-8	3+1+1+5plus+2+7-8	6	0.03
3+1+1+1+3+4-6	3+1+1+1+3+4-6	6	0.03
3+2+2+3+2+4-6	3+2+2+3+2+4-6	6	0.03
4+1+1+1+3+4-6	4+1+1+1+3+4-6	6	0.03
4+1+2+4+3+9-10	4+1+2+4+3+9-10	6	0.03
4+1+1+4+3+7-8	4+1+1+4+3+7-8	6	0.03
4+1+2+5plus+2+4-6	4+1+2+5plus+2+4-6	6	0.03
6+1+1+3+2+9-10	6+1+1+3+2+9-10	6	0.03
4+7+2+2+2+7-8	4+7+2+2+2+7-8	6	0.03
4+4+2+3+3+7-8	4+4+2+3+3+7-8	6	0.03
4+2+1+2+3+4-6	4+2+1+2+3+4-6	6	0.03
4+4+2+2+3+9-10	4+4+2+2+3+9-10	6	0.03
5+6+1+4+2+9-10	5+6+1+4+2+9-10	6	0.03
5+4+1+2+2+7-8	5+4+1+2+2+7-8	6	0.03

5+6+2+3+3+9-10	5+6+2+3+3+9-10	6	0.03
5+8+1+4+2+4-6	5+8+1+4+2+4-6	6	0.03
5+1+2+4+2+7-8	5+1+2+4+2+7-8	6	0.03
5+1+2+3+2+9-10	5+1+2+3+2+9-10	6	0.03
5+5+1+2+2+4-6	5+5+1+2+2+4-6	6	0.03
5+1+1+3+2+4-6	5+1+1+3+2+4-6	6	0.03
5+6+2+4+2+4-6	5+6+2+4+2+4-6	6	0.03
5+4+2+4+3+9-10	5+4+2+4+3+9-10	6	0.03
5+4+2+4+2+9-10	5+4+2+4+2+9-10	6	0.03
5+1+2+5plus+3+7-8	5+1+2+5plus+3+7-8	6	0.03
6+1+2+4+2+4-6	6+1+2+4+2+4-6	6	0.03
6+1+1+2+3+7-8	6+1+1+2+3+7-8	6	0.03
6+1+1+1+3+7-8	6+1+1+1+3+7-8	6	0.03
6+4+1+2+2+7-8	6+4+1+2+2+7-8	6	0.03
6+7+2+1+2+7-8	6+7+2+1+2+7-8	6	0.03
7+1+2+5plus+2+7-8	7+1+2+5plus+2+7-8	6	0.03
7+4+2+3+2+7-8	7+4+2+3+2+7-8	6	0.03
7+3+1+5plus+2+4-6	7+3+1+5plus+2+4-6	6	0.03
7+4+2+1+2+4-6	7+4+2+1+2+4-6	6	0.03
1+1+1+4+3+4-6	1+1+1+4+3+4-6	6	0.03
1+2+1+3+3+4-6	1+2+1+3+3+4-6	6	0.03
1+6+2+2+2+9-10	1+6+2+2+2+9-10	6	0.03
2+3+2+2+2+4-6	2+3+2+2+2+4-6	6	0.03
2+8+2+3+3+9-10	2+8+2+3+3+9-10	6	0.03
2+8+1+4+2+4-6	2+8+1+4+2+4-6	6	0.03
3+2+1+3+2+9-10	3+2+1+3+2+9-10	6	0.03
3+7+2+4+3+7-8	3+7+2+4+3+7-8	6	0.03
3+4+1+4+1+7-8	3+4+1+4+1+7-8	6	0.03
4+4+1+1+2+4-6	4+4+1+1+2+4-6	6	0.03
4+2+1+4+3+7-8	4+2+1+4+3+7-8	6	0.03
4+1+1+1+2+2-3	4+1+1+1+2+2-3	6	0.03
2+6+1+1+2+9-10	2+6+1+1+2+9-10	6	0.03
4+2+2+5plus+3+9-10	4+2+2+5plus+3+9-10	6	0.03
5+1+2+1+2+4-6	5+1+2+1+2+4-6	6	0.03
5+4+1+3+2+9-10	5+4+1+3+2+9-10	6	0.03
5+8+2+5plus+2+4-6	5+8+2+5plus+2+4-6	6	0.03
6+6+2+5plus+2+7-8	6+6+2+5plus+2+7-8	6	0.03
6+3+1+4+2+7-8	6+3+1+4+2+7-8	6	0.03
6+1+1+4+3+7-8	6+1+1+4+3+7-8	6	0.03
6+8+2+5plus+2+9-10	6+8+2+5plus+2+9-10	6	0.03
6+2+1+3+2+4-6	6+2+1+3+2+4-6	6	0.03
6+2+2+4+3+7-8	6+2+2+4+3+7-8	6	0.03
7+2+1+3+2+7-8	7+2+1+3+2+7-8	6	0.03
2+9+3+1+2+4-6	2+9+3+1+2+4-6	6	0.03
3+3+2+3+2+7-8	3+3+2+3+2+7-8	6	0.03
3+3+1+5plus+2+9-10	3+3+1+5plus+2+9-10	6	0.03
3+3+2+5plus+2+7-8	3+3+2+5plus+2+7-8	6	0.03
4+3+1+5plus+2+4-6	4+3+1+5plus+2+4-6	6	0.03
1+3+1+4+2+7-8	1+3+1+4+2+7-8	6	0.03
1+8+2+2+1+0-1	1+8+2+2+1+0-1	6	0.03
2+6+1+5plus+2+9-10	2+6+1+5plus+2+9-10	6	0.03
5+7+2+1+2+4-6	5+7+2+1+2+4-6	6	0.03
5+5+2+5plus+2+4-6	5+5+2+5plus+2+4-6	6	0.03
6+8+3+1+2+7-8	6+8+3+1+2+7-8	6	0.03
6+4+2+3+2+7-8	6+4+2+3+2+7-8	6	0.03
6+4+2+2+2+7-8	6+4+2+2+2+7-8	6	0.03

7+6+2+1+2+4-6	7+6+2+1+2+4-6	6	0.03
1+1+1+4+2+2-3	1+1+1+4+2+2-3	5	0.02
1+6+2+1+2+9-10	1+6+2+1+2+9-10	5	0.02
1+4+2+5plus+3+9-10	1+4+2+5plus+3+9-10	5	0.02
1+7+1+2+2+4-6	1+7+1+2+2+4-6	5	0.02
1+6+2+4+3+9-10	1+6+2+4+3+9-10	5	0.02
1+2+1+1+3+2-3	1+2+1+1+3+2-3	5	0.02
1+2+1+2+3+4-6	1+2+1+2+3+4-6	5	0.02
1+6+1+1+3+9-10	1+6+1+1+3+9-10	5	0.02
1+3+1+3+3+7-8	1+3+1+3+3+7-8	5	0.02
1+2+2+3+3+9-10	1+2+2+3+3+9-10	5	0.02
1+1+2+4+3+9-10	1+1+2+4+3+9-10	5	0.02
1+1+1+1+1+9-10	1+1+1+1+1+9-10	5	0.02
1+2+2+4+3+9-10	1+2+2+4+3+9-10	5	0.02
1+1+2+3+3+9-10	1+1+2+3+3+9-10	5	0.02
1+2+1+1+1+7-8	1+2+1+1+1+7-8	5	0.02
2+4+2+4+3+4-6	2+4+2+4+3+4-6	5	0.02
2+4+2+4+2+9-10	2+4+2+4+2+9-10	5	0.02
1+7+2+2+2+4-6	1+7+2+2+2+4-6	5	0.02
1+1+1+1+1+4-6	1+1+1+1+1+4-6	5	0.02
2+4+1+2+3+7-8	2+4+1+2+3+7-8	5	0.02
2+2+2+1+3+7-8	2+2+2+1+3+7-8	5	0.02
2+2+2+1+3+4-6	2+2+2+1+3+4-6	5	0.02
2+1+2+5plus+2+4-6	2+1+2+5plus+2+4-6	5	0.02
2+8+2+4+3+4-6	2+8+2+4+3+4-6	5	0.02
2+6+2+4+2+4-6	2+6+2+4+2+4-6	5	0.02
2+9+1+3+2+4-6	2+9+1+3+2+4-6	5	0.02
2+8+1+1+2+4-6	2+8+1+1+2+4-6	5	0.02
2+1+2+2+3+4-6	2+1+2+2+3+4-6	5	0.02
2+1+1+2+1+9-10	2+1+1+2+1+9-10	5	0.02
2+7+1+2+2+9-10	2+7+1+2+2+9-10	5	0.02
3+2+2+4+1+7-8	3+2+2+4+1+7-8	5	0.02
3+6+2+2+3+9-10	3+6+2+2+3+9-10	5	0.02
3+3+1+2+2+9-10	3+3+1+2+2+9-10	5	0.02
3+3+1+1+3+4-6	3+3+1+1+3+4-6	5	0.02
3+7+1+1+2+7-8	3+7+1+1+2+7-8	5	0.02
3+1+1+1+2+2-3	3+1+1+1+2+2-3	5	0.02
3+1+2+2+3+7-8	3+1+2+2+3+7-8	5	0.02
3+6+1+1+2+7-8	3+6+1+1+2+7-8	5	0.02
3+8+1+2+2+7-8	3+8+1+2+2+7-8	5	0.02
3+3+2+2+3+9-10	3+3+2+2+3+9-10	5	0.02
3+3+2+2+2+7-8	3+3+2+2+2+7-8	5	0.02
3+4+2+4+3+7-8	3+4+2+4+3+7-8	5	0.02
3+1+1+4+2+4-6	3+1+1+4+2+4-6	5	0.02
3+2+1+3+3+7-8	3+2+1+3+3+7-8	5	0.02
1+2+2+5plus+3+9-10	1+2+2+5plus+3+9-10	5	0.02
4+1+1+4+2+4-6	4+1+1+4+2+4-6	5	0.02
4+1+1+3+2+2-3	4+1+1+3+2+2-3	5	0.02
4+1+1+4+3+9-10	4+1+1+4+3+9-10	5	0.02
4+1+2+3+2+7-8	4+1+2+3+2+7-8	5	0.02
4+4+1+4+2+7-8	4+4+1+4+2+7-8	5	0.02
6+7+2+3+3+9-10	6+7+2+3+3+9-10	5	0.02
4+6+2+4+3+7-8	4+6+2+4+3+7-8	5	0.02
4+3+2+4+2+9-10	4+3+2+4+2+9-10	5	0.02
4+2+1+2+1+7-8	4+2+1+2+1+7-8	5	0.02
5+1+1+5plus+2+9-10	5+1+1+5plus+2+9-10	5	0.02

5+8+1+3+2+7-8	5+8+1+3+2+7-8	5	0.02
5+5+1+3+2+4-6	5+5+1+3+2+4-6	5	0.02
5+8+2+4+3+9-10	5+8+2+4+3+9-10	5	0.02
2+3+1+3+2+9-10	2+3+1+3+2+9-10	5	0.02
5+1+1+4+2+4-6	5+1+1+4+2+4-6	5	0.02
5+5+1+4+2+7-8	5+5+1+4+2+7-8	5	0.02
5+4+2+2+2+7-8	5+4+2+2+2+7-8	5	0.02
5+2+1+2+3+9-10	5+2+1+2+3+9-10	5	0.02
6+2+1+2+1+7-8	6+2+1+2+1+7-8	5	0.02
6+2+1+1+3+7-8	6+2+1+1+3+7-8	5	0.02
3+2+2+4+3+9-10	3+2+2+4+3+9-10	5	0.02
6+6+2+1+2+7-8	6+6+2+1+2+7-8	5	0.02
6+1+2+1+2+4-6	6+1+2+1+2+4-6	5	0.02
6+5+2+3+2+4-6	6+5+2+3+2+4-6	5	0.02
6+2+1+2+1+9-10	6+2+1+2+1+9-10	5	0.02
6+6+1+5plus+3+9-10	6+6+1+5plus+3+9-10	5	0.02
7+1+2+3+3+7-8	7+1+2+3+3+7-8	5	0.02
7+4+2+4+2+4-6	7+4+2+4+2+4-6	5	0.02
7+4+2+5plus+3+7-8	7+4+2+5plus+3+7-8	5	0.02
7+8+2+1+2+7-8	7+8+2+1+2+7-8	5	0.02
7+4+2+3+2+9-10	7+4+2+3+2+9-10	5	0.02
7+2+2+5plus+2+7-8	7+2+2+5plus+2+7-8	5	0.02
1+1+1+4+3+7-8	1+1+1+4+3+7-8	5	0.02
1+2+1+3+2+9-10	1+2+1+3+2+9-10	5	0.02
1+3+1+1+2+7-8	1+3+1+1+2+7-8	5	0.02
1+4+1+1+2+4-6	1+4+1+1+2+4-6	5	0.02
1+4+1+1+2+9-10	1+4+1+1+2+9-10	5	0.02
1+7+2+5plus+2+7-8	1+7+2+5plus+2+7-8	5	0.02
1+1+1+1+2+0-1	1+1+1+1+2+0-1	5	0.02
1+2+1+1+2+0-1	1+2+1+1+2+0-1	5	0.02
1+3+1+2+3+7-8	1+3+1+2+3+7-8	5	0.02
1+3+1+3+1+4-6	1+3+1+3+1+4-6	5	0.02
2+6+1+3+2+9-10	2+6+1+3+2+9-10	5	0.02
2+6+3+2+2+9-10	2+6+3+2+2+9-10	5	0.02
2+6+3+2+2+4-6	2+6+3+2+2+4-6	5	0.02
2+7+2+5plus+2+9-10	2+7+2+5plus+2+9-10	5	0.02
2+8+1+3+2+7-8	2+8+1+3+2+7-8	5	0.02
6+6+1+1+1+7-8	6+6+1+1+1+7-8	5	0.02
2+2+1+1+1+9-10	2+2+1+1+1+9-10	5	0.02
2+4+2+3+2+4-6	2+4+2+3+2+4-6	5	0.02
3+4+2+1+2+4-6	3+4+2+1+2+4-6	5	0.02
3+1+1+5plus+2+4-6	3+1+1+5plus+2+4-6	5	0.02
3+4+1+3+2+9-10	3+4+1+3+2+9-10	5	0.02
4+1+1+2+1+0-1	4+1+1+2+1+0-1	5	0.02
4+2+2+2+1+9-10	4+2+2+2+1+9-10	5	0.02
4+4+1+2+2+4-6	4+4+1+2+2+4-6	5	0.02
5+2+1+4+2+4-6	5+2+1+4+2+4-6	5	0.02
5+6+1+2+2+4-6	5+6+1+2+2+4-6	5	0.02
5+8+2+5plus+2+9-10	5+8+2+5plus+2+9-10	5	0.02
5+7+2+5plus+2+9-10	5+7+2+5plus+2+9-10	5	0.02
5+9+2+5plus+3+7-8	5+9+2+5plus+3+7-8	5	0.02
5+4+2+5plus+2+4-6	5+4+2+5plus+2+4-6	5	0.02
5+5+1+1+2+9-10	5+5+1+1+2+9-10	5	0.02
6+6+1+4+3+9-10	6+6+1+4+3+9-10	5	0.02
5+6+2+5plus+3+7-8	5+6+2+5plus+3+7-8	5	0.02
5+6+2+5plus+2+9-10	5+6+2+5plus+2+9-10	5	0.02

6+6+2+2+2+7-8	6+6+2+2+2+7-8	5	0.02
6+3+2+2+2+9-10	6+3+2+2+2+9-10	5	0.02
6+1+1+4+2+9-10	6+1+1+4+2+9-10	5	0.02
6+2+1+1+2+2-3	6+2+1+1+2+2-3	5	0.02
6+4+2+5plus+2+7-8	6+4+2+5plus+2+7-8	5	0.02
2+9+3+1+2+7-8	2+9+3+1+2+7-8	5	0.02
2+8+1+4+1+7-8	2+8+1+4+1+7-8	5	0.02
3+8+2+1+3+9-10	3+8+2+1+3+9-10	5	0.02
3+7+2+5plus+3+9-10	3+7+2+5plus+3+9-10	5	0.02
3+6+2+4+2+7-8	3+6+2+4+2+7-8	5	0.02
4+3+1+5plus+2+7-8	4+3+1+5plus+2+7-8	5	0.02
1+3+1+4+2+4-6	1+3+1+4+2+4-6	5	0.02
1+3+1+1+3+7-8	1+3+1+1+3+7-8	5	0.02
1+5+1+4+2+7-8	1+5+1+4+2+7-8	5	0.02
2+6+3+1+2+4-6	2+6+3+1+2+4-6	5	0.02
2+6+1+1+2+4-6	2+6+1+1+2+4-6	5	0.02
4+2+2+5plus+2+2-3	4+2+2+5plus+2+2-3	5	0.02
5+2+2+3+2+4-6	5+2+2+3+2+4-6	5	0.02
5+4+2+5plus+2+9-10	5+4+2+5plus+2+9-10	5	0.02
6+4+2+4+3+9-10	6+4+2+4+3+9-10	5	0.02
6+6+1+5plus+2+4-6	6+6+1+5plus+2+4-6	5	0.02
5+7+1+2+2+7-8	5+7+1+2+2+7-8	5	0.02
7+2+2+4+3+4-6	7+2+2+4+3+4-6	5	0.02
1+6+2+4+3+4-6	1+6+2+4+3+4-6	4	0.02
1+2+1+1+3+9-10	1+2+1+1+3+9-10	4	0.02
1+1+1+3+2+9-10	1+1+1+3+2+9-10	4	0.02
1+3+1+3+2+9-10	1+3+1+3+2+9-10	4	0.02
1+7+1+1+2+4-6	1+7+1+1+2+4-6	4	0.02
1+1+2+5plus+3+7-8	1+1+2+5plus+3+7-8	4	0.02
2+8+1+2+3+9-10	2+8+1+2+3+9-10	4	0.02
1+4+2+3+3+9-10	1+4+2+3+3+9-10	4	0.02
2+4+2+4+2+7-8	2+4+2+4+2+7-8	4	0.02
1+8+1+2+3+7-8	1+8+1+2+3+7-8	4	0.02
1+8+1+2+2+9-10	1+8+1+2+2+9-10	4	0.02
1+7+2+1+2+4-6	1+7+2+1+2+4-6	4	0.02
2+7+2+4+3+9-10	2+7+2+4+3+9-10	4	0.02
2+1+2+4+2+4-6	2+1+2+4+2+4-6	4	0.02
2+2+1+3+3+7-8	2+2+1+3+3+7-8	4	0.02
2+7+1+1+3+4-6	2+7+1+1+3+4-6	4	0.02
2+1+1+3+2+9-10	2+1+1+3+2+9-10	4	0.02
2+2+1+1+1+4-6	2+2+1+1+1+4-6	4	0.02
2+4+1+4+2+7-8	2+4+1+4+2+7-8	4	0.02
1+8+2+4+2+2-3	1+8+2+4+2+2-3	4	0.02
2+8+2+4+2+9-10	2+8+2+4+2+9-10	4	0.02
2+2+2+3+3+9-10	2+2+2+3+3+9-10	4	0.02
2+7+1+2+3+7-8	2+7+1+2+3+7-8	4	0.02
2+2+1+2+3+4-6	2+2+1+2+3+4-6	4	0.02
2+8+2+1+2+9-10	2+8+2+1+2+9-10	4	0.02
2+1+2+4+3+9-10	2+1+2+4+3+9-10	4	0.02
2+4+1+1+3+7-8	2+4+1+1+3+7-8	4	0.02
2+9+2+4+2+4-6	2+9+2+4+2+4-6	4	0.02
2+1+1+2+1+4-6	2+1+1+2+1+4-6	4	0.02
2+1+1+1+1+7-8	2+1+1+1+1+7-8	4	0.02
2+1+2+3+3+4-6	2+1+2+3+3+4-6	4	0.02
3+1+1+3+2+9-10	3+1+1+3+2+9-10	4	0.02
2+4+1+2+2+9-10	2+4+1+2+2+9-10	4	0.02

2+8+3+1+2+9-10	2+8+3+1+2+9-10	4	0.02
3+2+2+3+3+7-8	3+2+2+3+3+7-8	4	0.02
3+4+2+3+3+4-6	3+4+2+3+3+4-6	4	0.02
3+3+1+2+2+4-6	3+3+1+2+2+4-6	4	0.02
3+1+2+1+3+7-8	3+1+2+1+3+7-8	4	0.02
3+1+2+1+2+4-6	3+1+2+1+2+4-6	4	0.02
3+2+1+2+3+9-10	3+2+1+2+3+9-10	4	0.02
3+7+2+1+3+7-8	3+7+2+1+3+7-8	4	0.02
3+7+2+2+3+7-8	3+7+2+2+3+7-8	4	0.02
3+6+2+3+2+4-6	3+6+2+3+2+4-6	4	0.02
3+6+1+4+2+4-6	3+6+1+4+2+4-6	4	0.02
3+6+1+1+3+7-8	3+6+1+1+3+7-8	4	0.02
3+8+1+2+3+9-10	3+8+1+2+3+9-10	4	0.02
3+2+2+4+2+2-3	3+2+2+4+2+2-3	4	0.02
4+2+2+3+3+4-6	4+2+2+3+3+4-6	4	0.02
4+2+2+3+2+4-6	4+2+2+3+2+4-6	4	0.02
4+1+1+2+1+9-10	4+1+1+2+1+9-10	4	0.02
1+1+1+2+2+2-3	1+1+1+2+2+2-3	4	0.02
4+1+1+2+3+4-6	4+1+1+2+3+4-6	4	0.02
4+3+1+2+3+7-8	4+3+1+2+3+7-8	4	0.02
4+1+2+2+3+4-6	4+1+2+2+3+4-6	4	0.02
4+6+2+2+2+7-8	4+6+2+2+2+7-8	4	0.02
4+7+1+2+2+9-10	4+7+1+2+2+9-10	4	0.02
4+7+2+1+3+7-8	4+7+2+1+3+7-8	4	0.02
4+7+2+3+2+7-8	4+7+2+3+2+7-8	4	0.02
4+3+2+4+2+4-6	4+3+2+4+2+4-6	4	0.02
4+1+1+5plus+2+4-6	4+1+1+5plus+2+4-6	4	0.02
4+3+1+1+3+7-8	4+3+1+1+3+7-8	4	0.02
5+6+2+4+3+4-6	5+6+2+4+3+4-6	4	0.02
5+1+1+1+3+7-8	5+1+1+1+3+7-8	4	0.02
5+2+1+4+2+9-10	5+2+1+4+2+9-10	4	0.02
5+2+2+5plus+2+4-6	5+2+2+5plus+2+4-6	4	0.02
5+1+2+3+2+7-8	5+1+2+3+2+7-8	4	0.02
5+4+1+1+2+4-6	5+4+1+1+2+4-6	4	0.02
5+5+1+5plus+2+7-8	5+5+1+5plus+2+7-8	4	0.02
5+6+1+2+3+7-8	5+6+1+2+3+7-8	4	0.02
5+6+2+3+3+7-8	5+6+2+3+3+7-8	4	0.02
5+2+2+2+3+9-10	5+2+2+2+3+9-10	4	0.02
5+1+2+5plus+3+9-10	5+1+2+5plus+3+9-10	4	0.02
6+1+1+5plus+2+9-10	6+1+1+5plus+2+9-10	4	0.02
6+1+2+3+2+9-10	6+1+2+3+2+9-10	4	0.02
6+1+1+2+3+9-10	6+1+1+2+3+9-10	4	0.02
6+2+1+2+3+4-6	6+2+1+2+3+4-6	4	0.02
6+5+1+4+2+7-8	6+5+1+4+2+7-8	4	0.02
6+5+2+3+2+7-8	6+5+2+3+2+7-8	4	0.02
6+8+2+5plus+3+4-6	6+8+2+5plus+3+4-6	4	0.02
6+2+1+1+3+4-6	6+2+1+1+3+4-6	4	0.02
6+3+2+1+3+7-8	6+3+2+1+3+7-8	4	0.02
6+3+2+2+3+7-8	6+3+2+2+3+7-8	4	0.02
6+6+2+3+2+7-8	6+6+2+3+2+7-8	4	0.02
7+8+2+1+2+9-10	7+8+2+1+2+9-10	4	0.02
7+4+2+3+3+9-10	7+4+2+3+3+9-10	4	0.02
7+2+2+3+2+9-10	7+2+2+3+2+9-10	4	0.02
7+1+2+1+2+4-6	7+1+2+1+2+4-6	4	0.02
7+4+2+1+2+0-1	7+4+2+1+2+0-1	4	0.02
7+3+1+4+3+9-10	7+3+1+4+3+9-10	4	0.02

7+2+2+1+2+7-8	7+2+2+1+2+7-8	4	0.02
7+2+2+1+2+9-10	7+2+2+1+2+9-10	4	0.02
1+2+2+3+2+4-6	1+2+2+3+2+4-6	4	0.02
1+2+2+5plus+3+4-6	1+2+2+5plus+3+4-6	4	0.02
1+4+2+2+3+4-6	1+4+2+2+3+4-6	4	0.02
1+6+2+2+3+9-10	1+6+2+2+3+9-10	4	0.02
1+5+2+2+2+9-10	1+5+2+2+2+9-10	4	0.02
1+7+1+1+2+9-10	1+7+1+1+2+9-10	4	0.02
1+3+1+1+2+4-6	1+3+1+1+2+4-6	4	0.02
2+6+3+2+3+9-10	2+6+3+2+3+9-10	4	0.02
2+4+2+3+1+7-8	2+4+2+3+1+7-8	4	0.02
2+1+1+5plus+3+4-6	2+1+1+5plus+3+4-6	4	0.02
2+5+2+2+2+7-8	2+5+2+2+2+7-8	4	0.02
2+8+2+2+3+4-6	2+8+2+2+3+4-6	4	0.02
1+5+2+3+1+7-8	1+5+2+3+1+7-8	4	0.02
6+6+1+2+1+7-8	6+6+1+2+1+7-8	4	0.02
2+8+2+2+2+2-3	2+8+2+2+2+2-3	4	0.02
2+4+1+1+1+7-8	2+4+1+1+1+7-8	4	0.02
2+5+2+1+3+9-10	2+5+2+1+3+9-10	4	0.02
2+4+2+3+3+7-8	2+4+2+3+3+7-8	4	0.02
2+4+2+1+3+4-6	2+4+2+1+3+4-6	4	0.02
2+8+1+4+2+9-10	2+8+1+4+2+9-10	4	0.02
2+8+1+5plus+1+7-8	2+8+1+5plus+1+7-8	4	0.02
3+2+1+5plus+2+7-8	3+2+1+5plus+2+7-8	4	0.02
3+2+1+4+3+7-8	3+2+1+4+3+7-8	4	0.02
3+2+2+1+1+7-8	3+2+2+1+1+7-8	4	0.02
3+5+2+5plus+3+7-8	3+5+2+5plus+3+7-8	4	0.02
3+8+2+2+2+9-10	3+8+2+2+2+9-10	4	0.02
3+4+1+3+2+4-6	3+4+1+3+2+4-6	4	0.02
4+1+1+1+1+4-6	4+1+1+1+1+4-6	4	0.02
4+3+2+2+2+4-6	4+3+2+2+2+4-6	4	0.02
4+4+2+5plus+3+4-6	4+4+2+5plus+3+4-6	4	0.02
4+1+1+2+1+4-6	4+1+1+2+1+4-6	4	0.02
4+4+2+2+3+7-8	4+4+2+2+3+7-8	4	0.02
4+4+1+2+1+4-6	4+4+1+2+1+4-6	4	0.02
4+2+1+1+2+2-3	4+2+1+1+2+2-3	4	0.02
5+1+1+4+3+7-8	5+1+1+4+3+7-8	4	0.02
5+1+1+1+2+2-3	5+1+1+1+2+2-3	4	0.02
5+8+2+5plus+3+9-10	5+8+2+5plus+3+9-10	4	0.02
5+8+1+3+2+4-6	5+8+1+3+2+4-6	4	0.02
5+7+2+3+2+9-10	5+7+2+3+2+9-10	4	0.02
5+5+2+3+2+7-8	5+5+2+3+2+7-8	4	0.02
5+5+1+3+2+7-8	5+5+1+3+2+7-8	4	0.02
5+5+3+3+2+7-8	5+5+3+3+2+7-8	4	0.02
5+5+1+2+2+9-10	5+5+1+2+2+9-10	4	0.02
5+6+2+2+2+4-6	5+6+2+2+2+4-6	4	0.02
5+5+1+1+2+4-6	5+5+1+1+2+4-6	4	0.02
5+6+2+3+2+4-6	5+6+2+3+2+4-6	4	0.02
6+2+1+4+2+4-6	6+2+1+4+2+4-6	4	0.02
6+8+2+4+3+4-6	6+8+2+4+3+4-6	4	0.02
6+1+1+5plus+3+4-6	6+1+1+5plus+3+4-6	4	0.02
6+4+2+4+3+7-8	6+4+2+4+3+7-8	4	0.02
6+9+2+2+3+9-10	6+9+2+2+3+9-10	4	0.02
6+2+1+2+2+2-3	6+2+1+2+2+2-3	4	0.02
6+6+2+4+3+7-8	6+6+2+4+3+7-8	4	0.02
7+2+2+3+3+7-8	7+2+2+3+3+7-8	4	0.02

7+4+1+2+2+7-8	7+4+1+2+2+7-8	4	0.02
2+2+2+3+3+4-6	2+2+2+3+3+4-6	4	0.02
2+9+2+2+2+4-6	2+9+2+2+2+4-6	4	0.02
2+8+3+2+3+7-8	2+8+3+2+3+7-8	4	0.02
2+7+1+4+2+9-10	2+7+1+4+2+9-10	4	0.02
2+9+2+5plus+3+4-6	2+9+2+5plus+3+4-6	4	0.02
2+7+2+1+2+4-6	2+7+2+1+2+4-6	4	0.02
3+6+1+3+2+4-6	3+6+1+3+2+4-6	4	0.02
3+3+2+4+2+4-6	3+3+2+4+2+4-6	4	0.02
3+3+2+5plus+2+4-6	3+3+2+5plus+2+4-6	4	0.02
3+3+2+1+2+9-10	3+3+2+1+2+9-10	4	0.02
1+6+2+3+2+7-8	1+6+2+3+2+7-8	4	0.02
1+8+2+4+3+9-10	1+8+2+4+3+9-10	4	0.02
1+8+2+2+1+2-3	1+8+2+2+1+2-3	4	0.02
1+6+1+3+2+7-8	1+6+1+3+2+7-8	4	0.02
2+8+1+2+3+4-6	2+8+1+2+3+4-6	4	0.02
4+4+2+4+3+4-6	4+4+2+4+3+4-6	4	0.02
4+4+2+3+2+4-6	4+4+2+3+2+4-6	4	0.02
5+4+2+2+3+9-10	5+4+2+2+3+9-10	4	0.02
5+6+3+2+2+4-6	5+6+3+2+2+4-6	4	0.02
5+4+1+3+1+7-8	5+4+1+3+1+7-8	4	0.02
5+2+1+3+2+9-10	5+2+1+3+2+9-10	4	0.02
6+2+2+1+3+7-8	6+2+2+1+3+7-8	4	0.02
6+7+2+4+3+4-6	6+7+2+4+3+4-6	4	0.02
6+1+2+1+3+9-10	6+1+2+1+3+9-10	4	0.02
2+8+3+2+2+7-8	2+8+3+2+2+7-8	4	0.02
7+2+2+5plus+3+7-8	7+2+2+5plus+3+7-8	4	0.02
1+1+1+4+1+4-6	1+1+1+4+1+4-6	3	0.01
1+4+2+5plus+1+9-10	1+4+2+5plus+1+9-10	3	0.01
1+2+2+4+2+2-3	1+2+2+4+2+2-3	3	0.01
1+2+2+2+2+2-3	1+2+2+2+2+2-3	3	0.01
1+7+1+2+3+9-10	1+7+1+2+3+9-10	3	0.01
1+4+2+5plus+3+4-6	1+4+2+5plus+3+4-6	3	0.01
1+1+1+2+1+9-10	1+1+1+2+1+9-10	3	0.01
1+3+1+3+2+2-3	1+3+1+3+2+2-3	3	0.01
1+2+1+2+2+2-3	1+2+1+2+2+2-3	3	0.01
1+2+1+4+2+9-10	1+2+1+4+2+9-10	3	0.01
1+6+1+1+3+4-6	1+6+1+1+3+4-6	3	0.01
1+6+1+2+3+4-6	1+6+1+2+3+4-6	3	0.01
1+6+1+2+2+7-8	1+6+1+2+2+7-8	3	0.01
1+1+2+5plus+3+9-10	1+1+2+5plus+3+9-10	3	0.01
1+2+1+2+1+9-10	1+2+1+2+1+9-10	3	0.01
1+2+2+1+1+2-3	1+2+2+1+1+2-3	3	0.01
1+2+1+1+1+4-6	1+2+1+1+1+4-6	3	0.01
1+2+2+3+3+7-8	1+2+2+3+3+7-8	3	0.01
1+3+2+3+2+9-10	1+3+2+3+2+9-10	3	0.01
1+5+2+3+2+7-8	1+5+2+3+2+7-8	3	0.01
1+1+1+1+3+9-10	1+1+1+1+3+9-10	3	0.01
1+4+2+1+3+4-6	1+4+2+1+3+4-6	3	0.01
2+4+1+3+2+2-3	2+4+1+3+2+2-3	3	0.01
2+6+1+2+3+4-6	2+6+1+2+3+4-6	3	0.01
2+4+2+2+3+4-6	2+4+2+2+3+4-6	3	0.01
2+8+2+4+3+9-10	2+8+2+4+3+9-10	3	0.01
2+2+3+2+2+9-10	2+2+3+2+2+9-10	3	0.01
2+4+1+1+3+4-6	2+4+1+1+3+4-6	3	0.01
2+2+1+1+1+7-8	2+2+1+1+1+7-8	3	0.01

2+7+1+5plus+3+7-8	2+7+1+5plus+3+7-8	3	0.01
1+8+2+4+2+4-6	1+8+2+4+2+4-6	3	0.01
2+7+1+2+3+4-6	2+7+1+2+3+4-6	3	0.01
2+8+2+4+1+7-8	2+8+2+4+1+7-8	3	0.01
2+5+2+5plus+2+9-10	2+5+2+5plus+2+9-10	3	0.01
2+7+2+5plus+3+4-6	2+7+2+5plus+3+4-6	3	0.01
2+6+1+1+3+4-6	2+6+1+1+3+4-6	3	0.01
2+4+2+1+2+9-10	2+4+2+1+2+9-10	3	0.01
2+6+2+5plus+1+9-10	2+6+2+5plus+1+9-10	3	0.01
2+1+2+4+3+4-6	2+1+2+4+3+4-6	3	0.01
2+2+2+3+1+4-6	2+2+2+3+1+4-6	3	0.01
3+1+1+2+3+9-10	3+1+1+2+3+9-10	3	0.01
3+1+1+4+3+7-8	3+1+1+4+3+7-8	3	0.01
2+1+1+2+3+9-10	2+1+1+2+3+9-10	3	0.01
2+1+1+2+1+2-3	2+1+1+2+1+2-3	3	0.01
2+6+1+4+3+7-8	2+6+1+4+3+7-8	3	0.01
2+6+1+4+3+9-10	2+6+1+4+3+9-10	3	0.01
3+6+2+2+3+4-6	3+6+2+2+3+4-6	3	0.01
3+3+1+1+2+9-10	3+3+1+1+2+9-10	3	0.01
3+2+1+3+2+2-3	3+2+1+3+2+2-3	3	0.01
3+7+1+1+2+9-10	3+7+1+1+2+9-10	3	0.01
3+1+2+2+3+4-6	3+1+2+2+3+4-6	3	0.01
3+6+2+1+2+7-8	3+6+2+1+2+7-8	3	0.01
3+6+2+2+1+7-8	3+6+2+2+1+7-8	3	0.01
3+6+1+4+3+4-6	3+6+1+4+3+4-6	3	0.01
3+6+2+4+2+4-6	3+6+2+4+2+4-6	3	0.01
3+2+2+2+3+4-6	3+2+2+2+3+4-6	3	0.01
3+7+2+2+3+9-10	3+7+2+2+3+9-10	3	0.01
3+2+2+3+1+7-8	3+2+2+3+1+7-8	3	0.01
3+4+2+2+3+2-3	3+4+2+2+3+2-3	3	0.01
5+1+1+1+3+4-6	5+1+1+1+3+4-6	3	0.01
3+4+1+1+3+7-8	3+4+1+1+3+7-8	3	0.01
7+4+2+4+3+7-8	7+4+2+4+3+7-8	3	0.01
4+2+1+5plus+1+9-10	4+2+1+5plus+1+9-10	3	0.01
4+1+1+4+1+4-6	4+1+1+4+1+4-6	3	0.01
6+1+1+3+3+7-8	6+1+1+3+3+7-8	3	0.01
6+7+2+3+3+7-8	6+7+2+3+3+7-8	3	0.01
4+7+2+2+3+7-8	4+7+2+2+3+7-8	3	0.01
4+1+1+3+1+4-6	4+1+1+3+1+4-6	3	0.01
4+2+1+5plus+1+4-6	4+2+1+5plus+1+4-6	3	0.01
4+3+2+3+2+7-8	4+3+2+3+2+7-8	3	0.01
4+1+1+5plus+1+7-8	4+1+1+5plus+1+7-8	3	0.01
4+2+2+2+1+7-8	4+2+2+2+1+7-8	3	0.01
4+7+2+3+2+9-10	4+7+2+3+2+9-10	3	0.01
5+6+1+4+3+9-10	5+6+1+4+3+9-10	3	0.01
5+2+1+2+1+7-8	5+2+1+2+1+7-8	3	0.01
5+1+1+5plus+2+7-8	5+1+1+5plus+2+7-8	3	0.01
5+1+1+1+1+9-10	5+1+1+1+1+9-10	3	0.01
5+8+2+2+2+4-6	5+8+2+2+2+4-6	3	0.01
5+2+1+1+2+9-10	5+2+1+1+2+9-10	3	0.01
6+4+1+1+2+4-6	6+4+1+1+2+4-6	3	0.01
3+4+2+5plus+3+7-8	3+4+2+5plus+3+7-8	3	0.01
3+2+2+5plus+3+7-8	3+2+2+5plus+3+7-8	3	0.01
5+1+2+1+3+4-6	5+1+2+1+3+4-6	3	0.01
5+1+2+1+3+7-8	5+1+2+1+3+7-8	3	0.01
6+2+1+5plus+1+7-8	6+2+1+5plus+1+7-8	3	0.01

6+2+2+1+3+9-10	6+2+2+1+3+9-10	3	0.01
6+1+1+1+3+9-10	6+1+1+1+3+9-10	3	0.01
6+5+2+4+2+4-6	6+5+2+4+2+4-6	3	0.01
6+2+2+4+3+4-6	6+2+2+4+3+4-6	3	0.01
6+8+2+2+3+7-8	6+8+2+2+3+7-8	3	0.01
7+4+1+3+2+7-8	7+4+1+3+2+7-8	3	0.01
7+2+1+1+3+4-6	7+2+1+1+3+4-6	3	0.01
7+7+2+5plus+2+9-10	7+7+2+5plus+2+9-10	3	0.01
7+4+2+2+1+9-10	7+4+2+2+1+9-10	3	0.01
7+2+2+4+2+4-6	7+2+2+4+2+4-6	3	0.01
7+1+1+1+2+9-10	7+1+1+1+2+9-10	3	0.01
7+3+1+4+2+7-8	7+3+1+4+2+7-8	3	0.01
7+5+2+3+3+7-8	7+5+2+3+3+7-8	3	0.01
1+4+1+2+3+4-6	1+4+1+2+3+4-6	3	0.01
1+4+1+2+2+7-8	1+4+1+2+2+7-8	3	0.01
1+1+1+2+2+0-1	1+1+1+2+2+0-1	3	0.01
1+2+1+4+1+4-6	1+2+1+4+1+4-6	3	0.01
1+2+1+4+3+7-8	1+2+1+4+3+7-8	3	0.01
1+1+2+1+3+9-10	1+1+2+1+3+9-10	3	0.01
1+1+2+4+1+9-10	1+1+2+4+1+9-10	3	0.01
2+6+2+5plus+3+4-6	2+6+2+5plus+3+4-6	3	0.01
2+6+1+1+3+7-8	2+6+1+1+3+7-8	3	0.01
2+5+2+2+3+4-6	2+5+2+2+3+4-6	3	0.01
2+7+2+3+3+7-8	2+7+2+3+3+7-8	3	0.01
2+6+1+2+3+9-10	2+6+1+2+3+9-10	3	0.01
6+6+1+2+1+4-6	6+6+1+2+1+4-6	3	0.01
2+4+1+1+2+9-10	2+4+1+1+2+9-10	3	0.01
2+7+2+1+3+9-10	2+7+2+1+3+9-10	3	0.01
2+2+1+4+3+7-8	2+2+1+4+3+7-8	3	0.01
2+2+1+3+3+4-6	2+2+1+3+3+4-6	3	0.01
2+4+2+1+3+7-8	2+4+2+1+3+7-8	3	0.01
2+8+1+4+3+9-10	2+8+1+4+3+9-10	3	0.01
2+7+2+2+2+0-1	2+7+2+2+2+0-1	3	0.01
3+7+2+3+3+7-8	3+7+2+3+3+7-8	3	0.01
6+8+1+4+3+7-8	6+8+1+4+3+7-8	3	0.01
3+2+1+5plus+2+9-10	3+2+1+5plus+2+9-10	3	0.01
3+2+1+4+2+9-10	3+2+1+4+2+9-10	3	0.01
3+4+2+1+3+9-10	3+4+2+1+3+9-10	3	0.01
3+7+2+5plus+3+7-8	3+7+2+5plus+3+7-8	3	0.01
3+1+1+2+3+4-6	3+1+1+2+3+4-6	3	0.01
3+2+2+1+3+7-8	3+2+2+1+3+7-8	3	0.01
3+1+1+2+2+0-1	3+1+1+2+2+0-1	3	0.01
3+4+1+1+3+9-10	3+4+1+1+3+9-10	3	0.01
3+7+1+4+1+7-8	3+7+1+4+1+7-8	3	0.01
3+8+1+5plus+2+9-10	3+8+1+5plus+2+9-10	3	0.01
3+8+2+4+3+9-10	3+8+2+4+3+9-10	3	0.01
3+7+2+4+3+4-6	3+7+2+4+3+4-6	3	0.01
3+4+1+4+2+7-8	3+4+1+4+2+7-8	3	0.01
3+1+1+2+1+4-6	3+1+1+2+1+4-6	3	0.01
4+1+1+3+1+7-8	4+1+1+3+1+7-8	3	0.01
4+4+2+3+3+9-10	4+4+2+3+3+9-10	3	0.01
4+1+1+3+1+2-3	4+1+1+3+1+2-3	3	0.01
4+4+1+2+1+9-10	4+4+1+2+1+9-10	3	0.01
4+1+1+1+1+2-3	4+1+1+1+1+2-3	3	0.01
4+2+2+5plus+1+9-10	4+2+2+5plus+1+9-10	3	0.01
5+8+2+4+1+7-8	5+8+2+4+1+7-8	3	0.01

5+1+2+4+2+4-6	5+1+2+4+2+4-6	3	0.01
5+8+2+1+2+7-8	5+8+2+1+2+7-8	3	0.01
5+1+1+2+1+9-10	5+1+1+2+1+9-10	3	0.01
5+1+2+5plus+2+4-6	5+1+2+5plus+2+4-6	3	0.01
5+5+3+3+2+9-10	5+5+3+3+2+9-10	3	0.01
5+5+2+3+2+9-10	5+5+2+3+2+9-10	3	0.01
5+1+2+2+3+9-10	5+1+2+2+3+9-10	3	0.01
5+1+1+5plus+3+9-10	5+1+1+5plus+3+9-10	3	0.01
5+6+1+5plus+2+7-8	5+6+1+5plus+2+7-8	3	0.01
5+4+2+5plus+3+7-8	5+4+2+5plus+3+7-8	3	0.01
6+6+1+4+2+9-10	6+6+1+4+2+9-10	3	0.01
4+1+1+3+3+2-3	4+1+1+3+3+2-3	3	0.01
6+2+1+1+1+7-8	6+2+1+1+1+7-8	3	0.01
6+7+2+4+3+9-10	6+7+2+4+3+9-10	3	0.01
6+7+2+4+2+4-6	6+7+2+4+2+4-6	3	0.01
6+4+1+3+2+9-10	6+4+1+3+2+9-10	3	0.01
6+2+1+3+3+2-3	6+2+1+3+3+2-3	3	0.01
6+4+2+4+3+4-6	6+4+2+4+3+4-6	3	0.01
6+1+1+3+2+4-6	6+1+1+3+2+4-6	3	0.01
6+9+2+2+2+9-10	6+9+2+2+2+9-10	3	0.01
6+7+2+5plus+2+7-8	6+7+2+5plus+2+7-8	3	0.01
6+7+2+5plus+2+4-6	6+7+2+5plus+2+4-6	3	0.01
6+7+2+5plus+3+7-8	6+7+2+5plus+3+7-8	3	0.01
6+4+2+5plus+2+4-6	6+4+2+5plus+2+4-6	3	0.01
6+4+2+3+3+7-8	6+4+2+3+3+7-8	3	0.01
6+4+2+2+3+4-6	6+4+2+2+3+4-6	3	0.01
6+6+1+2+2+4-6	6+6+1+2+2+4-6	3	0.01
6+6+1+2+2+7-8	6+6+1+2+2+7-8	3	0.01
7+4+2+2+3+7-8	7+4+2+2+3+7-8	3	0.01
7+3+2+2+2+9-10	7+3+2+2+2+9-10	3	0.01
7+3+2+2+3+9-10	7+3+2+2+3+9-10	3	0.01
7+3+2+4+3+4-6	7+3+2+4+3+4-6	3	0.01
2+8+2+5plus+2+2-3	2+8+2+5plus+2+2-3	3	0.01
2+9+2+3+2+7-8	2+9+2+3+2+7-8	3	0.01
2+9+2+2+2+7-8	2+9+2+2+2+7-8	3	0.01
2+8+1+4+1+4-6	2+8+1+4+1+4-6	3	0.01
5+5+1+3+3+7-8	5+5+1+3+3+7-8	3	0.01
3+8+2+1+2+7-8	3+8+2+1+2+7-8	3	0.01
3+6+1+4+2+7-8	3+6+1+4+2+7-8	3	0.01
3+1+1+1+1+7-8	3+1+1+1+1+7-8	3	0.01
3+3+2+4+3+7-8	3+3+2+4+3+7-8	3	0.01
3+3+1+5plus+2+4-6	3+3+1+5plus+2+4-6	3	0.01
3+1+1+1+1+2-3	3+1+1+1+1+2-3	3	0.01
3+3+1+5plus+3+9-10	3+3+1+5plus+3+9-10	3	0.01
3+4+2+1+3+7-8	3+4+2+1+3+7-8	3	0.01
3+1+2+3+3+7-8	3+1+2+3+3+7-8	3	0.01
4+3+2+5plus+2+4-6	4+3+2+5plus+2+4-6	3	0.01
1+6+2+1+3+4-6	1+6+2+1+3+4-6	3	0.01
1+5+2+4+2+4-6	1+5+2+4+2+4-6	3	0.01
1+5+2+4+2+7-8	1+5+2+4+2+7-8	3	0.01
1+3+2+5plus+3+9-10	1+3+2+5plus+3+9-10	3	0.01
1+4+1+4+2+7-8	1+4+1+4+2+7-8	3	0.01
1+2+2+1+3+9-10	1+2+2+1+3+9-10	3	0.01
1+8+2+2+2+0-1	1+8+2+2+2+0-1	3	0.01
2+1+1+3+1+7-8	2+1+1+3+1+7-8	3	0.01
1+9+2+4+3+7-8	1+9+2+4+3+7-8	3	0.01

1+9+2+4+2+7-8	1+9+2+4+2+7-8	3	0.01
1+2+2+5plus+2+4-6	1+2+2+5plus+2+4-6	3	0.01
1+2+2+5plus+1+2-3	1+2+2+5plus+1+2-3	3	0.01
1+2+2+5plus+1+0-1	1+2+2+5plus+1+0-1	3	0.01
4+3+2+1+3+7-8	4+3+2+1+3+7-8	3	0.01
4+4+1+2+3+9-10	4+4+1+2+3+9-10	3	0.01
5+7+1+4+2+7-8	5+7+1+4+2+7-8	3	0.01
5+6+3+2+2+7-8	5+6+3+2+2+7-8	3	0.01
5+4+1+3+3+7-8	5+4+1+3+3+7-8	3	0.01
5+2+2+3+3+7-8	5+2+2+3+3+7-8	3	0.01
5+1+2+1+3+0-1	5+1+2+1+3+0-1	3	0.01
6+7+2+3+2+4-6	6+7+2+3+2+4-6	3	0.01
6+2+2+3+3+9-10	6+2+2+3+3+9-10	3	0.01
6+6+1+2+2+9-10	6+6+1+2+2+9-10	3	0.01
6+8+2+3+2+7-8	6+8+2+3+2+7-8	3	0.01
6+8+2+3+3+9-10	6+8+2+3+3+9-10	3	0.01
7+4+2+1+2+2-3	7+4+2+1+2+2-3	3	0.01
5+7+1+2+2+4-6	5+7+1+2+2+4-6	3	0.01
5+7+2+3+2+4-6	5+7+2+3+2+4-6	3	0.01
5+7+2+3+2+7-8	5+7+2+3+2+7-8	3	0.01
7+2+1+2+2+7-8	7+2+1+2+2+7-8	3	0.01
1+1+1+5plus+2+0-1	1+1+1+5plus+2+0-1	2	0.01
1+2+1+5plus+3+4-6	1+2+1+5plus+3+4-6	2	0.01
1+2+1+5plus+3+7-8	1+2+1+5plus+3+7-8	2	0.01
1+8+2+1+2+7-8	1+8+2+1+2+7-8	2	0.01
1+6+1+4+3+4-6	1+6+1+4+3+4-6	2	0.01
1+6+1+3+2+4-6	1+6+1+3+2+4-6	2	0.01
1+6+1+3+1+4-6	1+6+1+3+1+4-6	2	0.01
1+1+2+2+1+7-8	1+1+2+2+1+7-8	2	0.01
1+8+2+3+3+9-10	1+8+2+3+3+9-10	2	0.01
1+2+1+3+3+7-8	1+2+1+3+3+7-8	2	0.01
1+6+1+1+1+9-10	1+6+1+1+1+9-10	2	0.01
1+2+2+3+1+4-6	1+2+2+3+1+4-6	2	0.01
1+2+1+4+2+0-1	1+2+1+4+2+0-1	2	0.01
1+7+2+3+3+4-6	1+7+2+3+3+4-6	2	0.01
1+2+2+1+2+0-1	1+2+2+1+2+0-1	2	0.01
1+2+2+1+1+0-1	1+2+2+1+1+0-1	2	0.01
2+4+2+5plus+3+7-8	2+4+2+5plus+3+7-8	2	0.01
1+3+2+2+2+4-6	1+3+2+2+2+4-6	2	0.01
1+5+2+3+2+4-6	1+5+2+3+2+4-6	2	0.01
1+6+2+4+1+7-8	1+6+2+4+1+7-8	2	0.01
1+7+2+4+3+7-8	1+7+2+4+3+7-8	2	0.01
2+4+1+4+2+4-6	2+4+1+4+2+4-6	2	0.01
2+7+2+3+3+9-10	2+7+2+3+3+9-10	2	0.01
2+7+2+5plus+3+9-10	2+7+2+5plus+3+9-10	2	0.01
2+1+1+2+2+2-3	2+1+1+2+2+2-3	2	0.01
2+1+1+3+3+7-8	2+1+1+3+3+7-8	2	0.01
2+1+1+3+3+4-6	2+1+1+3+3+4-6	2	0.01
2+8+2+2+1+7-8	2+8+2+2+1+7-8	2	0.01
2+8+2+5plus+3+9-10	2+8+2+5plus+3+9-10	2	0.01
2+2+2+2+3+2-3	2+2+2+2+3+2-3	2	0.01
2+6+1+1+2+2-3	2+6+1+1+2+2-3	2	0.01
2+1+1+1+3+2-3	2+1+1+1+3+2-3	2	0.01
2+7+1+5plus+2+7-8	2+7+1+5plus+2+7-8	2	0.01
2+7+1+4+3+7-8	2+7+1+4+3+7-8	2	0.01
2+1+2+2+1+9-10	2+1+2+2+1+9-10	2	0.01

2+2+2+4+1+9-10	2+2+2+4+1+9-10	2	0.01
2+8+1+2+1+7-8	2+8+1+2+1+7-8	2	0.01
2+8+1+2+2+9-10	2+8+1+2+2+9-10	2	0.01
2+6+2+1+3+7-8	2+6+2+1+3+7-8	2	0.01
2+2+2+1+1+7-8	2+2+2+1+1+7-8	2	0.01
2+6+2+4+3+4-6	2+6+2+4+3+4-6	2	0.01
2+5+2+5plus+2+4-6	2+5+2+5plus+2+4-6	2	0.01
2+7+1+3+3+4-6	2+7+1+3+3+4-6	2	0.01
2+6+1+3+1+4-6	2+6+1+3+1+4-6	2	0.01
2+8+2+1+3+7-8	2+8+2+1+3+7-8	2	0.01
2+7+2+3+1+2-3	2+7+2+3+1+2-3	2	0.01
2+9+1+3+2+7-8	2+9+1+3+2+7-8	2	0.01
2+9+1+4+2+7-8	2+9+1+4+2+7-8	2	0.01
2+8+2+2+1+4-6	2+8+2+2+1+4-6	2	0.01
1+8+1+1+3+7-8	1+8+1+1+3+7-8	2	0.01
2+2+2+5plus+1+7-8	2+2+2+5plus+1+7-8	2	0.01
2+8+1+1+3+4-6	2+8+1+1+3+4-6	2	0.01
2+1+1+3+3+9-10	2+1+1+3+3+9-10	2	0.01
2+6+2+4+1+7-8	2+6+2+4+1+7-8	2	0.01
2+1+2+3+2+2-3	2+1+2+3+2+2-3	2	0.01
2+2+2+2+1+7-8	2+2+2+2+1+7-8	2	0.01
2+2+1+3+1+4-6	2+2+1+3+1+4-6	2	0.01
2+7+1+2+3+9-10	2+7+1+2+3+9-10	2	0.01
2+7+1+2+2+4-6	2+7+1+2+2+4-6	2	0.01
3+3+1+3+2+7-8	3+3+1+3+2+7-8	2	0.01
3+3+1+3+2+4-6	3+3+1+3+2+4-6	2	0.01
3+3+1+1+1+4-6	3+3+1+1+1+4-6	2	0.01
3+3+1+1+1+9-10	3+3+1+1+1+9-10	2	0.01
3+3+1+1+3+7-8	3+3+1+1+3+7-8	2	0.01
2+6+1+3+3+7-8	2+6+1+3+3+7-8	2	0.01
2+7+1+1+2+2-3	2+7+1+1+2+2-3	2	0.01
6+6+1+1+2+4-6	6+6+1+1+2+4-6	2	0.01
3+2+1+2+1+9-10	3+2+1+2+1+9-10	2	0.01
3+2+3+2+2+9-10	3+2+3+2+2+9-10	2	0.01
3+2+1+2+3+4-6	3+2+1+2+3+4-6	2	0.01
3+4+2+5plus+3+9-10	3+4+2+5plus+3+9-10	2	0.01
3+6+2+2+2+4-6	3+6+2+2+2+4-6	2	0.01
3+7+1+3+2+4-6	3+7+1+3+2+4-6	2	0.01
3+6+1+1+1+7-8	3+6+1+1+1+7-8	2	0.01
3+6+1+1+3+4-6	3+6+1+1+3+4-6	2	0.01
3+2+2+2+1+9-10	3+2+2+2+1+9-10	2	0.01
3+4+2+4+3+9-10	3+4+2+4+3+9-10	2	0.01
3+7+2+1+2+2-3	3+7+2+1+2+2-3	2	0.01
3+7+1+2+1+9-10	3+7+1+2+1+9-10	2	0.01
3+2+2+4+1+9-10	3+2+2+4+1+9-10	2	0.01
3+1+1+3+1+7-8	3+1+1+3+1+7-8	2	0.01
3+2+1+3+3+4-6	3+2+1+3+3+4-6	2	0.01
3+2+1+2+3+7-8	3+2+1+2+3+7-8	2	0.01
3+2+1+1+1+7-8	3+2+1+1+1+7-8	2	0.01
3+4+1+1+1+7-8	3+4+1+1+1+7-8	2	0.01
4+2+2+1+3+7-8	4+2+2+1+3+7-8	2	0.01
4+4+2+4+1+9-10	4+4+2+4+1+9-10	2	0.01
4+2+1+5plus+1+7-8	4+2+1+5plus+1+7-8	2	0.01
4+2+1+3+3+9-10	4+2+1+3+3+9-10	2	0.01
4+1+2+5plus+2+0-1	4+1+2+5plus+2+0-1	2	0.01
4+1+2+5plus+2+2-3	4+1+2+5plus+2+2-3	2	0.01

4+2+1+5plus+3+9-10	4+2+1+5plus+3+9-10	2	0.01
4+4+1+4+2+9-10	4+4+1+4+2+9-10	2	0.01
4+4+1+3+2+7-8	4+4+1+3+2+7-8	2	0.01
3+1+2+3+3+9-10	3+1+2+3+3+9-10	2	0.01
4+6+2+3+2+4-6	4+6+2+3+2+4-6	2	0.01
4+1+1+2+1+7-8	4+1+1+2+1+7-8	2	0.01
4+7+2+4+3+7-8	4+7+2+4+3+7-8	2	0.01
4+3+1+2+1+9-10	4+3+1+2+1+9-10	2	0.01
4+2+2+5plus+3+4-6	4+2+2+5plus+3+4-6	2	0.01
6+1+1+5plus+2+4-6	6+1+1+5plus+2+4-6	2	0.01
6+1+1+5plus+2+2-3	6+1+1+5plus+2+2-3	2	0.01
4+1+2+1+3+7-8	4+1+2+1+3+7-8	2	0.01
4+2+1+5plus+3+4-6	4+2+1+5plus+3+4-6	2	0.01
4+2+1+5plus+3+7-8	4+2+1+5plus+3+7-8	2	0.01
4+4+2+5plus+3+9-10	4+4+2+5plus+3+9-10	2	0.01
4+3+2+3+2+4-6	4+3+2+3+2+4-6	2	0.01
4+2+2+4+1+7-8	4+2+2+4+1+7-8	2	0.01
4+2+1+1+1+4-6	4+2+1+1+1+4-6	2	0.01
4+1+2+2+1+7-8	4+1+2+2+1+7-8	2	0.01
4+3+1+1+3+4-6	4+3+1+1+3+4-6	2	0.01
4+2+1+1+2+0-1	4+2+1+1+2+0-1	2	0.01
4+4+2+1+1+9-10	4+4+2+1+1+9-10	2	0.01
4+2+1+2+1+9-10	4+2+1+2+1+9-10	2	0.01
4+7+2+5plus+3+4-6	4+7+2+5plus+3+4-6	2	0.01
4+7+2+5plus+2+4-6	4+7+2+5plus+2+4-6	2	0.01
4+7+2+2+2+9-10	4+7+2+2+2+9-10	2	0.01
4+7+2+2+3+9-10	4+7+2+2+3+9-10	2	0.01
4+7+2+2+3+4-6	4+7+2+2+3+4-6	2	0.01
4+1+2+3+3+4-6	4+1+2+3+3+4-6	2	0.01
5+2+2+2+1+7-8	5+2+2+2+1+7-8	2	0.01
5+6+1+3+1+9-10	5+6+1+3+1+9-10	2	0.01
5+6+1+4+2+7-8	5+6+1+4+2+7-8	2	0.01
5+2+2+5plus+3+4-6	5+2+2+5plus+3+4-6	2	0.01
5+8+1+3+3+4-6	5+8+1+3+3+4-6	2	0.01
5+5+2+1+2+7-8	5+5+2+1+2+7-8	2	0.01
5+5+2+1+3+4-6	5+5+2+1+3+4-6	2	0.01
5+4+2+3+3+4-6	5+4+2+3+3+4-6	2	0.01
5+8+2+4+2+0-1	5+8+2+4+2+0-1	2	0.01
2+3+1+4+2+9-10	2+3+1+4+2+9-10	2	0.01
2+3+1+3+2+4-6	2+3+1+3+2+4-6	2	0.01
5+1+1+5plus+3+4-6	5+1+1+5plus+3+4-6	2	0.01
5+1+1+1+1+4-6	5+1+1+1+1+4-6	2	0.01
5+5+1+5plus+2+9-10	5+5+1+5plus+2+9-10	2	0.01
5+5+1+3+2+9-10	5+5+1+3+2+9-10	2	0.01
5+4+2+3+3+7-8	5+4+2+3+3+7-8	2	0.01
5+1+1+4+2+9-10	5+1+1+4+2+9-10	2	0.01
5+4+1+4+2+9-10	5+4+1+4+2+9-10	2	0.01
5+2+2+4+1+7-8	5+2+2+4+1+7-8	2	0.01
6+1+1+1+1+2-3	6+1+1+1+1+2-3	2	0.01
6+6+2+1+3+7-8	6+6+2+1+3+7-8	2	0.01
6+6+2+1+3+9-10	6+6+2+1+3+9-10	2	0.01
6+1+2+3+2+4-6	6+1+2+3+2+4-6	2	0.01
6+2+1+5plus+2+9-10	6+2+1+5plus+2+9-10	2	0.01
6+1+1+1+1+9-10	6+1+1+1+1+9-10	2	0.01
6+5+2+4+2+9-10	6+5+2+4+2+9-10	2	0.01
6+8+1+2+2+4-6	6+8+1+2+2+4-6	2	0.01

6+4+1+2+2+4-6	6+4+1+2+2+4-6	2	0.01
6+3+2+1+2+7-8	6+3+2+1+2+7-8	2	0.01
6+6+1+4+2+7-8	6+6+1+4+2+7-8	2	0.01
6+6+2+3+2+9-10	6+6+2+3+2+9-10	2	0.01
7+1+2+4+3+7-8	7+1+2+4+3+7-8	2	0.01
7+4+2+5plus+2+9-10	7+4+2+5plus+2+9-10	2	0.01
7+4+2+5plus+2+4-6	7+4+2+5plus+2+4-6	2	0.01
7+1+2+2+3+9-10	7+1+2+2+3+9-10	2	0.01
7+2+2+2+3+7-8	7+2+2+2+3+7-8	2	0.01
7+2+1+1+3+7-8	7+2+1+1+3+7-8	2	0.01
7+4+2+1+2+9-10	7+4+2+1+2+9-10	2	0.01
7+4+2+3+2+0-1	7+4+2+3+2+0-1	2	0.01
7+1+2+1+2+0-1	7+1+2+1+2+0-1	2	0.01
7+7+2+5plus+2+7-8	7+7+2+5plus+2+7-8	2	0.01
7+1+1+1+3+9-10	7+1+1+1+3+9-10	2	0.01
7+3+1+4+3+7-8	7+3+1+4+3+7-8	2	0.01
7+2+1+4+3+4-6	7+2+1+4+3+4-6	2	0.01
7+5+2+4+3+7-8	7+5+2+4+3+7-8	2	0.01
7+5+2+3+3+9-10	7+5+2+3+3+9-10	2	0.01
7+5+2+3+2+4-6	7+5+2+3+2+4-6	2	0.01
7+5+2+3+2+7-8	7+5+2+3+2+7-8	2	0.01
7+5+2+3+3+4-6	7+5+2+3+3+4-6	2	0.01
7+2+2+1+2+4-6	7+2+2+1+2+4-6	2	0.01
7+2+2+1+3+7-8	7+2+2+1+3+7-8	2	0.01
1+1+1+4+1+2-3	1+1+1+4+1+2-3	2	0.01
1+1+1+3+2+2-3	1+1+1+3+2+2-3	2	0.01
1+1+1+4+1+7-8	1+1+1+4+1+7-8	2	0.01
1+1+1+1+3+2-3	1+1+1+1+3+2-3	2	0.01
1+6+1+2+3+7-8	1+6+1+2+3+7-8	2	0.01
1+3+2+1+3+7-8	1+3+2+1+3+7-8	2	0.01
1+1+2+3+3+4-6	1+1+2+3+3+4-6	2	0.01
1+2+1+4+3+9-10	1+2+1+4+3+9-10	2	0.01
1+5+2+2+3+9-10	1+5+2+2+3+9-10	2	0.01
1+4+1+1+3+7-8	1+4+1+1+3+7-8	2	0.01
1+1+1+3+3+2-3	1+1+1+3+3+2-3	2	0.01
1+2+2+1+1+4-6	1+2+2+1+1+4-6	2	0.01
2+4+1+2+1+7-8	2+4+1+2+1+7-8	2	0.01
2+7+1+3+3+9-10	2+7+1+3+3+9-10	2	0.01
2+6+1+1+3+9-10	2+6+1+1+3+9-10	2	0.01
2+4+2+3+3+4-6	2+4+2+3+3+4-6	2	0.01
2+1+1+5plus+3+9-10	2+1+1+5plus+3+9-10	2	0.01
2+6+1+1+1+7-8	2+6+1+1+1+7-8	2	0.01
2+5+2+2+3+7-8	2+5+2+2+3+7-8	2	0.01
2+7+1+1+1+4-6	2+7+1+1+1+4-6	2	0.01
2+4+2+2+1+7-8	2+4+2+2+1+7-8	2	0.01
2+3+2+2+2+7-8	2+3+2+2+2+7-8	2	0.01
2+1+1+1+1+9-10	2+1+1+1+1+9-10	2	0.01
1+5+2+4+2+9-10	1+5+2+4+2+9-10	2	0.01
2+6+2+5plus+3+9-10	2+6+2+5plus+3+9-10	2	0.01
2+4+1+1+1+4-6	2+4+1+1+1+4-6	2	0.01
2+7+2+1+3+7-8	2+7+2+1+3+7-8	2	0.01
2+2+1+3+1+9-10	2+2+1+3+1+9-10	2	0.01
2+2+2+4+3+4-6	2+2+2+4+3+4-6	2	0.01
3+1+1+2+1+7-8	3+1+1+2+1+7-8	2	0.01
2+8+1+5plus+2+7-8	2+8+1+5plus+2+7-8	2	0.01
2+8+2+5plus+3+4-6	2+8+2+5plus+3+4-6	2	0.01

2+2+2+5plus+2+4-6	2+2+2+5plus+2+4-6	2	0.01
2+2+1+2+1+4-6	2+2+1+2+1+4-6	2	0.01
3+4+2+2+1+9-10	3+4+2+2+1+9-10	2	0.01
3+6+1+3+2+9-10	3+6+1+3+2+9-10	2	0.01
3+2+1+4+1+7-8	3+2+1+4+1+7-8	2	0.01
3+2+1+4+3+4-6	3+2+1+4+3+4-6	2	0.01
3+4+2+1+3+4-6	3+4+2+1+3+4-6	2	0.01
3+1+1+2+2+2-3	3+1+1+2+2+2-3	2	0.01
3+7+1+4+2+7-8	3+7+1+4+2+7-8	2	0.01
3+5+2+4+3+7-8	3+5+2+4+3+7-8	2	0.01
3+5+2+5plus+2+9-10	3+5+2+5plus+2+9-10	2	0.01
3+7+1+4+1+4-6	3+7+1+4+1+4-6	2	0.01
3+2+2+4+1+4-6	3+2+2+4+1+4-6	2	0.01
3+8+1+5plus+2+7-8	3+8+1+5plus+2+7-8	2	0.01
3+8+2+2+3+7-8	3+8+2+2+3+7-8	2	0.01
4+3+2+4+3+7-8	4+3+2+4+3+7-8	2	0.01
4+2+2+1+3+9-10	4+2+2+1+3+9-10	2	0.01
4+4+2+5plus+2+2-3	4+4+2+5plus+2+2-3	2	0.01
4+3+1+2+3+4-6	4+3+1+2+3+4-6	2	0.01
4+3+1+2+3+9-10	4+3+1+2+3+9-10	2	0.01
4+2+1+2+1+4-6	4+2+1+2+1+4-6	2	0.01
4+1+2+4+3+2-3	4+1+2+4+3+2-3	2	0.01
4+1+1+2+3+2-3	4+1+1+2+3+2-3	2	0.01
4+4+1+3+2+4-6	4+4+1+3+2+4-6	2	0.01
4+2+1+3+3+4-6	4+2+1+3+3+4-6	2	0.01
4+1+2+3+2+9-10	4+1+2+3+2+9-10	2	0.01
4+4+1+2+1+2-3	4+4+1+2+1+2-3	2	0.01
4+4+2+5plus+1+9-10	4+4+2+5plus+1+9-10	2	0.01
4+3+1+3+3+4-6	4+3+1+3+3+4-6	2	0.01
5+2+1+3+1+7-8	5+2+1+3+1+7-8	2	0.01
5+6+1+2+1+9-10	5+6+1+2+1+9-10	2	0.01
5+4+1+2+3+9-10	5+4+1+2+3+9-10	2	0.01
5+8+2+3+3+4-6	5+8+2+3+3+4-6	2	0.01
5+8+1+1+2+9-10	5+8+1+1+2+9-10	2	0.01
5+8+1+1+2+4-6	5+8+1+1+2+4-6	2	0.01
5+1+1+2+3+4-6	5+1+1+2+3+4-6	2	0.01
5+8+2+5plus+3+7-8	5+8+2+5plus+3+7-8	2	0.01
5+7+2+3+3+7-8	5+7+2+3+3+7-8	2	0.01
5+2+3+2+2+7-8	5+2+3+2+2+7-8	2	0.01
5+6+1+2+1+7-8	5+6+1+2+1+7-8	2	0.01
5+9+2+4+3+7-8	5+9+2+4+3+7-8	2	0.01
5+9+2+5plus+2+4-6	5+9+2+5plus+2+4-6	2	0.01
5+9+2+5plus+2+7-8	5+9+2+5plus+2+7-8	2	0.01
5+6+1+5plus+3+7-8	5+6+1+5plus+3+7-8	2	0.01
5+6+2+4+1+7-8	5+6+2+4+1+7-8	2	0.01
5+5+1+1+2+7-8	5+5+1+1+2+7-8	2	0.01
5+2+1+4+3+9-10	5+2+1+4+3+9-10	2	0.01
6+6+1+1+3+9-10	6+6+1+1+3+9-10	2	0.01
5+2+1+4+2+7-8	5+2+1+4+2+7-8	2	0.01
5+8+2+4+1+4-6	5+8+2+4+1+4-6	2	0.01
5+8+2+5plus+3+4-6	5+8+2+5plus+3+4-6	2	0.01
6+3+1+2+2+7-8	6+3+1+2+2+7-8	2	0.01
5+3+1+2+2+9-10	5+3+1+2+2+9-10	2	0.01
5+4+2+4+2+4-6	5+4+2+4+2+4-6	2	0.01
6+4+1+3+2+7-8	6+4+1+3+2+7-8	2	0.01
6+8+2+5plus+3+9-10	6+8+2+5plus+3+9-10	2	0.01

6+1+1+5plus+3+7-8	6+1+1+5plus+3+7-8	2	0.01
6+2+2+2+1+7-8	6+2+2+2+1+7-8	2	0.01
6+2+3+2+2+7-8	6+2+3+2+2+7-8	2	0.01
6+2+1+3+3+0-1	6+2+1+3+3+0-1	2	0.01
6+2+1+3+2+0-1	6+2+1+3+2+0-1	2	0.01
6+6+2+4+3+9-10	6+6+2+4+3+9-10	2	0.01
6+2+2+2+3+7-8	6+2+2+2+3+7-8	2	0.01
6+2+2+3+2+4-6	6+2+2+3+2+4-6	2	0.01
6+9+2+5plus+2+7-8	6+9+2+5plus+2+7-8	2	0.01
6+9+2+3+2+7-8	6+9+2+3+2+7-8	2	0.01
6+9+2+3+2+9-10	6+9+2+3+2+9-10	2	0.01
6+9+2+2+2+7-8	6+9+2+2+2+7-8	2	0.01
6+1+2+2+2+4-6	6+1+2+2+2+4-6	2	0.01
6+2+1+2+2+0-1	6+2+1+2+2+0-1	2	0.01
6+4+1+5plus+2+7-8	6+4+1+5plus+2+7-8	2	0.01
5+4+1+5plus+2+7-8	5+4+1+5plus+2+7-8	2	0.01
3+1+1+2+3+7-8	3+1+1+2+3+7-8	2	0.01
6+6+2+4+3+4-6	6+6+2+4+3+4-6	2	0.01
7+2+1+3+2+9-10	7+2+1+3+2+9-10	2	0.01
7+4+1+1+2+4-6	7+4+1+1+2+4-6	2	0.01
7+3+2+3+2+7-8	7+3+2+3+2+7-8	2	0.01
7+2+1+2+2+2-3	7+2+1+2+2+2-3	2	0.01
1+8+2+2+2+7-8	1+8+2+2+2+7-8	2	0.01
2+9+1+2+2+7-8	2+9+1+2+2+7-8	2	0.01
2+9+3+2+2+9-10	2+9+3+2+2+9-10	2	0.01
2+9+2+2+3+7-8	2+9+2+2+3+7-8	2	0.01
2+7+1+4+2+7-8	2+7+1+4+2+7-8	2	0.01
2+9+2+5plus+3+7-8	2+9+2+5plus+3+7-8	2	0.01
2+2+1+4+3+9-10	2+2+1+4+3+9-10	2	0.01
2+8+2+1+2+2-3	2+8+2+1+2+2-3	2	0.01
2+9+2+3+2+4-6	2+9+2+3+2+4-6	2	0.01
2+9+2+3+3+4-6	2+9+2+3+3+4-6	2	0.01
2+2+1+3+1+7-8	2+2+1+3+1+7-8	2	0.01
3+8+2+5plus+2+4-6	3+8+2+5plus+2+4-6	2	0.01
3+8+2+3+2+4-6	3+8+2+3+2+4-6	2	0.01
3+8+2+3+2+7-8	3+8+2+3+2+7-8	2	0.01
3+8+2+1+3+7-8	3+8+2+1+3+7-8	2	0.01
3+7+2+2+3+4-6	3+7+2+2+3+4-6	2	0.01
3+4+2+5plus+2+4-6	3+4+2+5plus+2+4-6	2	0.01
3+2+2+3+3+4-6	3+2+2+3+3+4-6	2	0.01
3+3+1+5plus+3+4-6	3+3+1+5plus+3+4-6	2	0.01
3+3+1+5plus+3+7-8	3+3+1+5plus+3+7-8	2	0.01
3+1+1+3+1+9-10	3+1+1+3+1+9-10	2	0.01
3+3+1+4+3+9-10	3+3+1+4+3+9-10	2	0.01
3+2+2+1+3+4-6	3+2+2+1+3+4-6	2	0.01
3+3+1+4+2+4-6	3+3+1+4+2+4-6	2	0.01
3+3+1+4+2+7-8	3+3+1+4+2+7-8	2	0.01
4+3+1+3+2+9-10	4+3+1+3+2+9-10	2	0.01
4+4+2+4+1+7-8	4+4+2+4+1+7-8	2	0.01
1+1+1+3+1+7-8	1+1+1+3+1+7-8	2	0.01
1+4+2+4+1+9-10	1+4+2+4+1+9-10	2	0.01
1+8+1+3+2+4-6	1+8+1+3+2+4-6	2	0.01
1+8+1+3+3+4-6	1+8+1+3+3+4-6	2	0.01
1+8+1+2+2+4-6	1+8+1+2+2+4-6	2	0.01
1+7+2+3+2+9-10	1+7+2+3+2+9-10	2	0.01
1+4+1+3+2+4-6	1+4+1+3+2+4-6	2	0.01

1+3+1+5plus+2+4-6	1+3+1+5plus+2+4-6	2	0.01
1+3+2+5plus+3+4-6	1+3+2+5plus+3+4-6	2	0.01
1+8+2+3+2+9-10	1+8+2+3+2+9-10	2	0.01
1+4+2+2+1+9-10	1+4+2+2+1+9-10	2	0.01
1+5+1+4+2+4-6	1+5+1+4+2+4-6	2	0.01
1+5+1+3+2+7-8	1+5+1+3+2+7-8	2	0.01
1+8+2+2+3+9-10	1+8+2+2+3+9-10	2	0.01
1+7+2+3+1+0-1	1+7+2+3+1+0-1	2	0.01
2+2+1+1+3+2-3	2+2+1+1+3+2-3	2	0.01
1+6+1+3+3+7-8	1+6+1+3+3+7-8	2	0.01
2+3+2+2+3+7-8	2+3+2+2+3+7-8	2	0.01
2+6+2+3+2+2-3	2+6+2+3+2+2-3	2	0.01
2+8+3+1+2+4-6	2+8+3+1+2+4-6	2	0.01
2+7+1+5plus+2+2-3	2+7+1+5plus+2+2-3	2	0.01
2+7+2+3+1+4-6	2+7+2+3+1+4-6	2	0.01
2+7+2+3+1+0-1	2+7+2+3+1+0-1	2	0.01
1+9+2+3+3+9-10	1+9+2+3+3+9-10	2	0.01
2+3+2+2+2+9-10	2+3+2+2+2+9-10	2	0.01
4+4+1+1+1+7-8	4+4+1+1+1+7-8	2	0.01
4+7+2+5plus+3+7-8	4+7+2+5plus+3+7-8	2	0.01
4+3+2+3+2+9-10	4+3+2+3+2+9-10	2	0.01
5+7+2+5plus+3+9-10	5+7+2+5plus+3+9-10	2	0.01
5+7+2+2+2+4-6	5+7+2+2+2+4-6	2	0.01
5+7+2+1+2+9-10	5+7+2+1+2+9-10	2	0.01
5+5+2+4+2+7-8	5+5+2+4+2+7-8	2	0.01
5+5+2+4+2+9-10	5+5+2+4+2+9-10	2	0.01
5+1+1+4+3+4-6	5+1+1+4+3+4-6	2	0.01
5+4+1+3+1+9-10	5+4+1+3+1+9-10	2	0.01
5+1+1+1+2+0-1	5+1+1+1+2+0-1	2	0.01
5+2+2+1+2+7-8	5+2+2+1+2+7-8	2	0.01
3+2+1+1+2+2-3	3+2+1+1+2+2-3	2	0.01
3+2+1+3+1+2-3	3+2+1+3+1+2-3	2	0.01
3+2+1+1+1+2-3	3+2+1+1+1+2-3	2	0.01
5+6+2+3+1+7-8	5+6+2+3+1+7-8	2	0.01
5+2+1+1+3+7-8	5+2+1+1+3+7-8	2	0.01
6+8+3+1+2+4-6	6+8+3+1+2+4-6	2	0.01
6+6+1+2+3+7-8	6+6+1+2+3+7-8	2	0.01
6+8+2+2+1+9-10	6+8+2+2+1+9-10	2	0.01
6+6+2+5plus+2+4-6	6+6+2+5plus+2+4-6	2	0.01
6+6+2+5plus+3+7-8	6+6+2+5plus+3+7-8	2	0.01
6+3+1+2+3+7-8	6+3+1+2+3+7-8	2	0.01
6+2+2+5plus+3+4-6	6+2+2+5plus+3+4-6	2	0.01
7+4+2+3+2+2-3	7+4+2+3+2+2-3	2	0.01
2+8+3+2+3+9-10	2+8+3+2+3+9-10	2	0.01
4+1+1+5plus+3+7-8	4+1+1+5plus+3+7-8	2	0.01
5+7+1+2+2+9-10	5+7+1+2+2+9-10	2	0.01
7+2+1+2+2+9-10	7+2+1+2+2+9-10	2	0.01
7+2+2+2+2+4-6	7+2+2+2+2+4-6	2	0.01
7+2+1+2+1+4-6	7+2+1+2+1+4-6	2	0.01
7+2+1+2+3+4-6	7+2+1+2+3+4-6	2	0.01
7+4+1+2+2+4-6	7+4+1+2+2+4-6	2	0.01
7+4+1+4+3+4-6	7+4+1+4+3+4-6	2	0.01
7+2+1+3+2+4-6	7+2+1+3+2+4-6	2	0.01
7+2+1+1+3+2-3	7+2+1+1+3+2-3	2	0.01
7+2+1+1+2+2-3	7+2+1+1+2+2-3	2	0.01
1+1+1+5plus+1+4-6	1+1+1+5plus+1+4-6	1	0.00

1+1+1+5plus+3+4-6	1+1+1+5plus+3+4-6	1	0.00
1+1+1+5plus+3+2-3	1+1+1+5plus+3+2-3	1	0.00
1+3+2+4+3+4-6	1+3+2+4+3+4-6	1	0.00
1+2+1+5plus+3+2-3	1+2+1+5plus+3+2-3	1	0.00
1+2+3+1+2+9-10	1+2+3+1+2+9-10	1	0.00
1+7+1+4+3+4-6	1+7+1+4+3+4-6	1	0.00
1+7+2+4+2+2-3	1+7+2+4+2+2-3	1	0.00
1+7+2+4+1+4-6	1+7+2+4+1+4-6	1	0.00
1+6+1+4+2+4-6	1+6+1+4+2+4-6	1	0.00
1+6+1+4+1+4-6	1+6+1+4+1+4-6	1	0.00
1+6+1+4+1+9-10	1+6+1+4+1+9-10	1	0.00
1+7+1+4+3+7-8	1+7+1+4+3+7-8	1	0.00
1+1+1+4+3+2-3	1+1+1+4+3+2-3	1	0.00
1+6+2+4+2+9-10	1+6+2+4+2+9-10	1	0.00
1+2+2+4+1+7-8	1+2+2+4+1+7-8	1	0.00
1+2+2+4+3+2-3	1+2+2+4+3+2-3	1	0.00
1+2+2+2+3+2-3	1+2+2+2+3+2-3	1	0.00
1+7+1+2+1+7-8	1+7+1+2+1+7-8	1	0.00
1+9+2+3+2+9-10	1+9+2+3+2+9-10	1	0.00
1+8+2+3+3+7-8	1+8+2+3+3+7-8	1	0.00
1+4+2+5plus+3+2-3	1+4+2+5plus+3+2-3	1	0.00
1+3+1+2+2+9-10	1+3+1+2+2+9-10	1	0.00
1+3+1+3+1+9-10	1+3+1+3+1+9-10	1	0.00
1+7+1+3+1+7-8	1+7+1+3+1+7-8	1	0.00
1+7+1+3+2+4-6	1+7+1+3+2+4-6	1	0.00
1+1+1+3+1+9-10	1+1+1+3+1+9-10	1	0.00
1+2+1+3+2+0-1	1+2+1+3+2+0-1	1	0.00
1+7+2+4+2+9-10	1+7+2+4+2+9-10	1	0.00
1+5+1+2+2+9-10	1+5+1+2+2+9-10	1	0.00
1+5+1+3+2+9-10	1+5+1+3+2+9-10	1	0.00
1+5+1+4+2+9-10	1+5+1+4+2+9-10	1	0.00
1+1+2+5plus+3+4-6	1+1+2+5plus+3+4-6	1	0.00
1+1+2+5plus+2+7-8	1+1+2+5plus+2+7-8	1	0.00
1+2+2+3+3+4-6	1+2+2+3+3+4-6	1	0.00
1+4+1+5plus+3+7-8	1+4+1+5plus+3+7-8	1	0.00
1+4+1+5plus+1+4-6	1+4+1+5plus+1+4-6	1	0.00
1+4+1+5plus+2+7-8	1+4+1+5plus+2+7-8	1	0.00
2+4+2+4+2+4-6	2+4+2+4+2+4-6	1	0.00
1+8+1+2+3+9-10	1+8+1+2+3+9-10	1	0.00
1+2+1+2+1+4-6	1+2+1+2+1+4-6	1	0.00
1+7+2+1+3+9-10	1+7+2+1+3+9-10	1	0.00
1+6+2+1+3+9-10	1+6+2+1+3+9-10	1	0.00
2+4+1+3+3+4-6	2+4+1+3+3+4-6	1	0.00
2+4+1+2+3+4-6	2+4+1+2+3+4-6	1	0.00
2+7+1+3+2+2-3	2+7+1+3+2+2-3	1	0.00
2+7+1+5plus+2+9-10	2+7+1+5plus+2+9-10	1	0.00
2+1+1+4+3+4-6	2+1+1+4+3+4-6	1	0.00
2+1+1+4+1+7-8	2+1+1+4+1+7-8	1	0.00
2+7+2+4+1+9-10	2+7+2+4+1+9-10	1	0.00
2+1+2+1+2+2-3	2+1+2+1+2+2-3	1	0.00
2+1+1+4+3+7-8	2+1+1+4+3+7-8	1	0.00
2+7+1+5plus+3+9-10	2+7+1+5plus+3+9-10	1	0.00
2+7+1+5plus+2+4-6	2+7+1+5plus+2+4-6	1	0.00
2+6+1+4+1+9-10	2+6+1+4+1+9-10	1	0.00
2+2+2+2+1+9-10	2+2+2+2+1+9-10	1	0.00
2+2+1+2+3+9-10	2+2+1+2+3+9-10	1	0.00

2+5+2+4+3+7-8	2+5+2+4+3+7-8	1	0.00
2+5+2+3+3+7-8	2+5+2+3+3+7-8	1	0.00
2+5+3+3+2+7-8	2+5+3+3+2+7-8	1	0.00
2+7+2+3+3+2-3	2+7+2+3+3+2-3	1	0.00
2+2+1+2+1+9-10	2+2+1+2+1+9-10	1	0.00
2+7+1+1+3+2-3	2+7+1+1+3+2-3	1	0.00
2+6+1+1+3+2-3	2+6+1+1+3+2-3	1	0.00
2+6+2+2+1+7-8	2+6+2+2+1+7-8	1	0.00
2+1+2+5plus+3+4-6	2+1+2+5plus+3+4-6	1	0.00
2+9+1+3+3+7-8	2+9+1+3+3+7-8	1	0.00
2+9+1+4+3+4-6	2+9+1+4+3+4-6	1	0.00
1+8+1+1+3+4-6	1+8+1+1+3+4-6	1	0.00
2+2+2+1+1+9-10	2+2+2+1+1+9-10	1	0.00
2+1+1+4+1+4-6	2+1+1+4+1+4-6	1	0.00
2+1+2+4+1+4-6	2+1+2+4+1+4-6	1	0.00
2+1+2+3+1+7-8	2+1+2+3+1+7-8	1	0.00
2+1+1+5plus+1+9-10	2+1+1+5plus+1+9-10	1	0.00
2+2+2+2+1+2-3	2+2+2+2+1+2-3	1	0.00
2+2+2+2+2+2-3	2+2+2+2+2+2-3	1	0.00
2+8+1+1+1+4-6	2+8+1+1+1+4-6	1	0.00
2+4+1+2+3+9-10	2+4+1+2+3+9-10	1	0.00
2+6+1+3+1+9-10	2+6+1+3+1+9-10	1	0.00
3+3+1+2+3+4-6	3+3+1+2+3+4-6	1	0.00
3+3+1+2+3+7-8	3+3+1+2+3+7-8	1	0.00
3+5+1+1+2+7-8	3+5+1+1+2+7-8	1	0.00
3+7+1+1+3+4-6	3+7+1+1+3+4-6	1	0.00
2+7+1+1+3+7-8	2+7+1+1+3+7-8	1	0.00
2+7+1+1+2+0-1	2+7+1+1+2+0-1	1	0.00
3+2+3+2+2+7-8	3+2+3+2+2+7-8	1	0.00
3+2+1+4+2+2-3	3+2+1+4+2+2-3	1	0.00
3+7+1+1+2+4-6	3+7+1+1+2+4-6	1	0.00
3+1+1+3+2+2-3	3+1+1+3+2+2-3	1	0.00
3+1+2+5plus+3+4-6	3+1+2+5plus+3+4-6	1	0.00
3+1+2+5plus+2+4-6	3+1+2+5plus+2+4-6	1	0.00
3+1+2+2+1+4-6	3+1+2+2+1+4-6	1	0.00
3+7+2+1+3+9-10	3+7+2+1+3+9-10	1	0.00
3+6+2+1+2+9-10	3+6+2+1+2+9-10	1	0.00
3+6+2+1+3+7-8	3+6+2+1+3+7-8	1	0.00
3+6+2+3+3+4-6	3+6+2+3+3+4-6	1	0.00
3+6+2+1+3+4-6	3+6+2+1+3+4-6	1	0.00
3+7+1+3+2+7-8	3+7+1+3+2+7-8	1	0.00
3+6+1+4+3+2-3	3+6+1+4+3+2-3	1	0.00
3+6+2+4+2+2-3	3+6+2+4+2+2-3	1	0.00
3+1+2+4+2+2-3	3+1+2+4+2+2-3	1	0.00
3+1+2+2+2+2-3	3+1+2+2+2+2-3	1	0.00
3+1+2+2+1+9-10	3+1+2+2+1+9-10	1	0.00
3+1+2+1+2+2-3	3+1+2+1+2+2-3	1	0.00
3+1+1+1+1+9-10	3+1+1+1+1+9-10	1	0.00
3+8+1+2+1+7-8	3+8+1+2+1+7-8	1	0.00
3+8+1+2+3+7-8	3+8+1+2+3+7-8	1	0.00
3+2+2+2+2+2-3	3+2+2+2+2+2-3	1	0.00
3+2+2+2+2+0-1	3+2+2+2+2+0-1	1	0.00
3+3+1+1+1+7-8	3+3+1+1+1+7-8	1	0.00
3+7+2+3+3+9-10	3+7+2+3+3+9-10	1	0.00
3+1+1+3+3+4-6	3+1+1+3+3+4-6	1	0.00
3+1+1+3+1+4-6	3+1+1+3+1+4-6	1	0.00

4+2+2+1+3+2-3	4+2+2+1+3+2-3	1	0.00
4+1+2+4+1+9-10	4+1+2+4+1+9-10	1	0.00
4+2+1+5plus+2+2-3	4+2+1+5plus+2+2-3	1	0.00
4+1+2+4+1+7-8	4+1+2+4+1+7-8	1	0.00
4+2+1+3+1+9-10	4+2+1+3+1+9-10	1	0.00
4+2+2+4+3+4-6	4+2+2+4+3+4-6	1	0.00
4+1+2+5plus+3+7-8	4+1+2+5plus+3+7-8	1	0.00
4+2+1+1+1+9-10	4+2+1+1+1+9-10	1	0.00
4+4+1+3+2+9-10	4+4+1+3+2+9-10	1	0.00
4+6+2+4+1+7-8	4+6+2+4+1+7-8	1	0.00
4+6+2+3+2+9-10	4+6+2+3+2+9-10	1	0.00
4+7+2+1+2+7-8	4+7+2+1+2+7-8	1	0.00
4+1+1+1+1+9-10	4+1+1+1+1+9-10	1	0.00
4+7+2+5plus+1+9-10	4+7+2+5plus+1+9-10	1	0.00
4+1+2+1+1+7-8	4+1+2+1+1+7-8	1	0.00
4+1+2+1+2+4-6	4+1+2+1+2+4-6	1	0.00
4+1+2+5plus+1+9-10	4+1+2+5plus+1+9-10	1	0.00
4+4+2+5plus+3+7-8	4+4+2+5plus+3+7-8	1	0.00
4+1+1+5plus+3+4-6	4+1+1+5plus+3+4-6	1	0.00
4+2+1+4+3+9-10	4+2+1+4+3+9-10	1	0.00
4+1+2+2+2+2-3	4+1+2+2+2+2-3	1	0.00
4+3+1+1+1+9-10	4+3+1+1+1+9-10	1	0.00
4+3+1+1+3+9-10	4+3+1+1+3+9-10	1	0.00
4+3+1+1+2+2-3	4+3+1+1+2+2-3	1	0.00
4+7+2+4+2+4-6	4+7+2+4+2+4-6	1	0.00
4+7+2+4+2+9-10	4+7+2+4+2+9-10	1	0.00
4+1+2+3+3+7-8	4+1+2+3+3+7-8	1	0.00
5+5+2+2+3+9-10	5+5+2+2+3+9-10	1	0.00
5+6+1+3+3+7-8	5+6+1+3+3+7-8	1	0.00
5+2+1+1+3+4-6	5+2+1+1+3+4-6	1	0.00
5+4+1+2+2+4-6	5+4+1+2+2+4-6	1	0.00
5+1+1+5plus+3+7-8	5+1+1+5plus+3+7-8	1	0.00
5+2+1+4+1+7-8	5+2+1+4+1+7-8	1	0.00
5+2+1+5plus+2+9-10	5+2+1+5plus+2+9-10	1	0.00
5+2+2+5plus+3+9-10	5+2+2+5plus+3+9-10	1	0.00
5+8+1+3+1+2-3	5+8+1+3+1+2-3	1	0.00
5+5+2+1+1+4-6	5+5+2+1+1+4-6	1	0.00
5+5+2+1+2+2-3	5+5+2+1+2+2-3	1	0.00
5+5+1+3+2+0-1	5+5+1+3+2+0-1	1	0.00
5+6+1+1+3+4-6	5+6+1+1+3+4-6	1	0.00
5+8+1+3+1+9-10	5+8+1+3+1+9-10	1	0.00
5+8+2+4+1+9-10	5+8+2+4+1+9-10	1	0.00
2+3+1+2+2+9-10	2+3+1+2+2+9-10	1	0.00
5+1+1+5plus+2+0-1	5+1+1+5plus+2+0-1	1	0.00
5+1+1+5plus+2+2-3	5+1+1+5plus+2+2-3	1	0.00
5+1+1+4+2+2-3	5+1+1+4+2+2-3	1	0.00
5+5+1+5plus+2+4-6	5+5+1+5plus+2+4-6	1	0.00
5+5+2+3+3+9-10	5+5+2+3+3+9-10	1	0.00
5+8+2+1+2+9-10	5+8+2+1+2+9-10	1	0.00
5+4+1+4+3+7-8	5+4+1+4+3+7-8	1	0.00
5+4+1+4+3+9-10	5+4+1+4+3+9-10	1	0.00
5+1+1+4+1+9-10	5+1+1+4+1+9-10	1	0.00
5+1+1+3+3+7-8	5+1+1+3+3+7-8	1	0.00
6+1+1+2+1+4-6	6+1+1+2+1+4-6	1	0.00
6+1+1+2+3+4-6	6+1+1+2+3+4-6	1	0.00
6+1+1+1+2+0-1	6+1+1+1+2+0-1	1	0.00

6+2+1+1+3+9-10	6+2+1+1+3+9-10	1	0.00
6+6+2+1+1+7-8	6+6+2+1+1+7-8	1	0.00
6+6+2+1+2+4-6	6+6+2+1+2+4-6	1	0.00
6+1+1+5plus+1+7-8	6+1+1+5plus+1+7-8	1	0.00
6+1+2+5plus+3+9-10	6+1+2+5plus+3+9-10	1	0.00
6+1+2+4+3+7-8	6+1+2+4+3+7-8	1	0.00
6+1+2+4+3+9-10	6+1+2+4+3+9-10	1	0.00
6+1+2+3+3+9-10	6+1+2+3+3+9-10	1	0.00
6+1+2+1+1+4-6	6+1+2+1+1+4-6	1	0.00
6+1+2+1+3+2-3	6+1+2+1+3+2-3	1	0.00
6+6+2+2+2+4-6	6+6+2+2+2+4-6	1	0.00
6+1+1+4+3+2-3	6+1+1+4+3+2-3	1	0.00
6+2+2+2+3+9-10	6+2+2+2+3+9-10	1	0.00
6+2+1+3+1+4-6	6+2+1+3+1+4-6	1	0.00
6+5+2+2+2+4-6	6+5+2+2+2+4-6	1	0.00
6+5+2+2+2+7-8	6+5+2+2+2+7-8	1	0.00
6+8+2+5plus+2+2-3	6+8+2+5plus+2+2-3	1	0.00
6+8+2+5plus+3+2-3	6+8+2+5plus+3+2-3	1	0.00
6+8+1+2+2+0-1	6+8+1+2+2+0-1	1	0.00
6+8+1+1+2+2-3	6+8+1+1+2+2-3	1	0.00
6+3+2+1+1+7-8	6+3+2+1+1+7-8	1	0.00
5+3+2+1+3+4-6	5+3+2+1+3+4-6	1	0.00
5+3+2+1+3+7-8	5+3+2+1+3+7-8	1	0.00
6+6+1+3+2+9-10	6+6+1+3+2+9-10	1	0.00
6+2+2+4+1+4-6	6+2+2+4+1+4-6	1	0.00
6+7+1+1+2+7-8	6+7+1+1+2+7-8	1	0.00
6+7+2+1+3+7-8	6+7+2+1+3+7-8	1	0.00
7+4+2+5plus+3+9-10	7+4+2+5plus+3+9-10	1	0.00
7+4+2+5plus+1+9-10	7+4+2+5plus+1+9-10	1	0.00
7+4+2+5plus+3+4-6	7+4+2+5plus+3+4-6	1	0.00
7+1+2+5plus+2+4-6	7+1+2+5plus+2+4-6	1	0.00
7+8+2+2+2+9-10	7+8+2+2+2+9-10	1	0.00
7+8+2+2+3+7-8	7+8+2+2+3+7-8	1	0.00
7+1+2+2+1+7-8	7+1+2+2+1+7-8	1	0.00
7+4+2+3+1+7-8	7+4+2+3+1+7-8	1	0.00
7+2+2+3+3+9-10	7+2+2+3+3+9-10	1	0.00
7+3+1+3+2+7-8	7+3+1+3+2+7-8	1	0.00
7+3+1+3+2+4-6	7+3+1+3+2+4-6	1	0.00
7+3+1+5plus+2+9-10	7+3+1+5plus+2+9-10	1	0.00
7+2+1+1+1+9-10	7+2+1+1+1+9-10	1	0.00
7+4+1+3+3+9-10	7+4+1+3+3+9-10	1	0.00
7+4+1+2+1+7-8	7+4+1+2+1+7-8	1	0.00
7+1+2+2+2+4-6	7+1+2+2+2+4-6	1	0.00
7+1+2+2+3+7-8	7+1+2+2+3+7-8	1	0.00
7+1+2+3+3+4-6	7+1+2+3+3+4-6	1	0.00
7+1+2+3+2+4-6	7+1+2+3+2+4-6	1	0.00
7+1+2+3+2+9-10	7+1+2+3+2+9-10	1	0.00
7+4+2+1+3+0-1	7+4+2+1+3+0-1	1	0.00
7+1+2+1+2+2-3	7+1+2+1+2+2-3	1	0.00
7+7+2+4+2+9-10	7+7+2+4+2+9-10	1	0.00
7+2+2+4+1+7-8	7+2+2+4+1+7-8	1	0.00
7+1+2+1+2+9-10	7+1+2+1+2+9-10	1	0.00
7+4+2+2+2+4-6	7+4+2+2+2+4-6	1	0.00
7+1+2+1+1+9-10	7+1+2+1+1+9-10	1	0.00
7+3+1+3+1+9-10	7+3+1+3+1+9-10	1	0.00
7+3+1+3+3+9-10	7+3+1+3+3+9-10	1	0.00

7+3+1+4+3+4-6	7+3+1+4+3+4-6	1	0.00
7+2+1+4+2+0-1	7+2+1+4+2+0-1	1	0.00
7+2+1+4+2+2-3	7+2+1+4+2+2-3	1	0.00
7+6+2+2+3+7-8	7+6+2+2+3+7-8	1	0.00
7+5+1+3+1+9-10	7+5+1+3+1+9-10	1	0.00
7+5+1+3+3+7-8	7+5+1+3+3+7-8	1	0.00
7+5+1+4+3+9-10	7+5+1+4+3+9-10	1	0.00
7+5+2+4+2+7-8	7+5+2+4+2+7-8	1	0.00
7+5+2+4+3+9-10	7+5+2+4+3+9-10	1	0.00
7+5+2+3+2+9-10	7+5+2+3+2+9-10	1	0.00
7+5+2+3+3+2-3	7+5+2+3+3+2-3	1	0.00
7+2+2+1+1+9-10	7+2+2+1+1+9-10	1	0.00
1+2+1+5plus+1+9-10	1+2+1+5plus+1+9-10	1	0.00
1+2+2+5plus+1+9-10	1+2+2+5plus+1+9-10	1	0.00
1+2+2+4+1+9-10	1+2+2+4+1+9-10	1	0.00
1+3+2+1+3+9-10	1+3+2+1+3+9-10	1	0.00
1+1+1+2+1+0-1	1+1+1+2+1+0-1	1	0.00
1+1+2+4+1+7-8	1+1+2+4+1+7-8	1	0.00
1+1+2+1+1+4-6	1+1+2+1+1+4-6	1	0.00
1+6+2+2+1+9-10	1+6+2+2+1+9-10	1	0.00
1+6+2+2+3+7-8	1+6+2+2+3+7-8	1	0.00
1+2+1+3+3+9-10	1+2+1+3+3+9-10	1	0.00
1+2+2+2+1+7-8	1+2+2+2+1+7-8	1	0.00
1+1+2+3+3+0-1	1+1+2+3+3+0-1	1	0.00
1+1+2+4+2+2-3	1+1+2+4+2+2-3	1	0.00
1+3+1+1+2+2-3	1+3+1+1+2+2-3	1	0.00
1+2+1+3+1+7-8	1+2+1+3+1+7-8	1	0.00
1+1+2+2+1+0-1	1+1+2+2+1+0-1	1	0.00
1+5+2+2+1+9-10	1+5+2+2+1+9-10	1	0.00
1+5+2+2+3+7-8	1+5+2+2+3+7-8	1	0.00
1+4+2+4+1+4-6	1+4+2+4+1+4-6	1	0.00
1+4+2+1+2+4-6	1+4+2+1+2+4-6	1	0.00
1+4+1+1+1+4-6	1+4+1+1+1+4-6	1	0.00
1+4+1+1+3+9-10	1+4+1+1+3+9-10	1	0.00
1+7+1+1+3+7-8	1+7+1+1+3+7-8	1	0.00
1+3+2+3+1+4-6	1+3+2+3+1+4-6	1	0.00
1+3+2+3+3+7-8	1+3+2+3+3+7-8	1	0.00
1+7+2+5plus+2+2-3	1+7+2+5plus+2+2-3	1	0.00
1+6+1+2+3+9-10	1+6+1+2+3+9-10	1	0.00
1+6+1+4+3+7-8	1+6+1+4+3+7-8	1	0.00
1+1+1+3+2+0-1	1+1+1+3+2+0-1	1	0.00
1+1+1+3+3+0-1	1+1+1+3+3+0-1	1	0.00
1+1+2+1+1+7-8	1+1+2+1+1+7-8	1	0.00
1+1+2+1+3+2-3	1+1+2+1+3+2-3	1	0.00
1+1+2+1+2+2-3	1+1+2+1+2+2-3	1	0.00
1+3+1+3+3+9-10	1+3+1+3+3+9-10	1	0.00
1+4+2+3+2+2-3	1+4+2+3+2+2-3	1	0.00
1+7+2+4+1+7-8	1+7+2+4+1+7-8	1	0.00
1+3+1+3+1+2-3	1+3+1+3+1+2-3	1	0.00
1+3+1+2+1+4-6	1+3+1+2+1+4-6	1	0.00
2+7+2+3+1+9-10	2+7+2+3+1+9-10	1	0.00
2+7+2+2+3+2-3	2+7+2+2+3+2-3	1	0.00
1+4+1+2+2+9-10	1+4+1+2+2+9-10	1	0.00
1+4+1+2+3+9-10	1+4+1+2+3+9-10	1	0.00
2+1+1+1+2+2-3	2+1+1+1+2+2-3	1	0.00
2+1+1+5plus+2+9-10	2+1+1+5plus+2+9-10	1	0.00

2+6+1+1+1+4-6	2+6+1+1+1+4-6	1	0.00
2+6+1+3+2+2-3	2+6+1+3+2+2-3	1	0.00
2+8+3+2+2+4-6	2+8+3+2+2+4-6	1	0.00
2+1+1+5plus+1+4-6	2+1+1+5plus+1+4-6	1	0.00
2+1+2+1+1+9-10	2+1+2+1+1+9-10	1	0.00
1+2+1+2+1+7-8	1+2+1+2+1+7-8	1	0.00
1+5+2+3+2+9-10	1+5+2+3+2+9-10	1	0.00
1+5+2+1+2+7-8	1+5+2+1+2+7-8	1	0.00
2+8+2+2+1+9-10	2+8+2+2+1+9-10	1	0.00
2+6+1+2+3+2-3	2+6+1+2+3+2-3	1	0.00
2+8+1+3+2+9-10	2+8+1+3+2+9-10	1	0.00
2+5+1+1+2+9-10	2+5+1+1+2+9-10	1	0.00
2+7+1+4+1+7-8	2+7+1+4+1+7-8	1	0.00
2+7+1+5plus+1+7-8	2+7+1+5plus+1+7-8	1	0.00
2+6+2+5plus+1+7-8	2+6+2+5plus+1+7-8	1	0.00
2+4+1+3+1+4-6	2+4+1+3+1+4-6	1	0.00
2+8+1+4+1+9-10	2+8+1+4+1+9-10	1	0.00
2+2+1+4+1+7-8	2+2+1+4+1+7-8	1	0.00
2+8+1+5plus+3+4-6	2+8+1+5plus+3+4-6	1	0.00
2+8+1+5plus+3+2-3	2+8+1+5plus+3+2-3	1	0.00
2+2+2+2+3+0-1	2+2+2+2+3+0-1	1	0.00
2+8+2+5plus+1+4-6	2+8+2+5plus+1+4-6	1	0.00
2+8+2+3+1+4-6	2+8+2+3+1+4-6	1	0.00
2+4+2+4+1+4-6	2+4+2+4+1+4-6	1	0.00
3+6+1+2+1+7-8	3+6+1+2+1+7-8	1	0.00
6+8+1+4+3+9-10	6+8+1+4+3+9-10	1	0.00
3+6+1+2+3+7-8	3+6+1+2+3+7-8	1	0.00
3+2+2+3+1+4-6	3+2+2+3+1+4-6	1	0.00
3+3+2+1+2+4-6	3+3+2+1+2+4-6	1	0.00
3+2+1+4+3+9-10	3+2+1+4+3+9-10	1	0.00
3+4+2+4+1+4-6	3+4+2+4+1+4-6	1	0.00
3+7+2+5plus+1+9-10	3+7+2+5plus+1+9-10	1	0.00
3+7+2+5plus+3+4-6	3+7+2+5plus+3+4-6	1	0.00
3+1+2+4+1+9-10	3+1+2+4+1+9-10	1	0.00
3+1+2+4+1+7-8	3+1+2+4+1+7-8	1	0.00
3+7+1+2+1+7-8	3+7+1+2+1+7-8	1	0.00
3+7+1+3+2+9-10	3+7+1+3+2+9-10	1	0.00
3+5+2+5plus+3+9-10	3+5+2+5plus+3+9-10	1	0.00
3+2+2+3+1+2-3	3+2+2+3+1+2-3	1	0.00
3+2+2+1+1+4-6	3+2+2+1+1+4-6	1	0.00
3+7+1+2+3+7-8	3+7+1+2+3+7-8	1	0.00
3+7+1+2+2+2-3	3+7+1+2+2+2-3	1	0.00
3+7+1+2+2+4-6	3+7+1+2+2+4-6	1	0.00
3+8+1+5plus+3+7-8	3+8+1+5plus+3+7-8	1	0.00
3+8+2+5plus+3+9-10	3+8+2+5plus+3+9-10	1	0.00
3+8+2+4+3+7-8	3+8+2+4+3+7-8	1	0.00
3+8+2+3+3+9-10	3+8+2+3+3+9-10	1	0.00
3+8+2+3+3+4-6	3+8+2+3+3+4-6	1	0.00
3+7+2+2+1+7-8	3+7+2+2+1+7-8	1	0.00
3+7+2+3+2+0-1	3+7+2+3+2+0-1	1	0.00
3+4+1+3+2+2-3	3+4+1+3+2+2-3	1	0.00
3+4+1+4+3+4-6	3+4+1+4+3+4-6	1	0.00
3+4+1+4+3+2-3	3+4+1+4+3+2-3	1	0.00
3+7+1+2+3+9-10	3+7+1+2+3+9-10	1	0.00
3+7+2+5plus+3+2-3	3+7+2+5plus+3+2-3	1	0.00
3+7+2+5plus+2+2-3	3+7+2+5plus+2+2-3	1	0.00

4+4+1+1+1+9-10	4+4+1+1+1+9-10	1	0.00
4+1+1+5plus+1+9-10	4+1+1+5plus+1+9-10	1	0.00
4+2+2+4+2+2-3	4+2+2+4+2+2-3	1	0.00
4+1+2+2+3+2-3	4+1+2+2+3+2-3	1	0.00
4+2+2+2+2+0-1	4+2+2+2+2+0-1	1	0.00
4+2+2+2+3+2-3	4+2+2+2+3+2-3	1	0.00
4+2+1+3+1+7-8	4+2+1+3+1+7-8	1	0.00
4+4+2+3+3+4-6	4+4+2+3+3+4-6	1	0.00
4+4+1+5plus+2+4-6	4+4+1+5plus+2+4-6	1	0.00
2+1+3+1+2+7-8	2+1+3+1+2+7-8	1	0.00
4+4+1+1+3+9-10	4+4+1+1+3+9-10	1	0.00
4+4+1+4+1+4-6	4+4+1+4+1+4-6	1	0.00
4+1+2+4+3+4-6	4+1+2+4+3+4-6	1	0.00
4+1+2+3+2+2-3	4+1+2+3+2+2-3	1	0.00
4+1+1+3+2+0-1	4+1+1+3+2+0-1	1	0.00
4+4+1+1+3+4-6	4+4+1+1+3+4-6	1	0.00
4+4+2+2+2+2-3	4+4+2+2+2+2-3	1	0.00
4+7+2+2+2+2-3	4+7+2+2+2+2-3	1	0.00
4+7+1+2+1+7-8	4+7+1+2+1+7-8	1	0.00
4+7+1+2+3+9-10	4+7+1+2+3+9-10	1	0.00
4+7+1+2+3+7-8	4+7+1+2+3+7-8	1	0.00
4+7+1+2+3+4-6	4+7+1+2+3+4-6	1	0.00
4+4+1+3+3+2-3	4+4+1+3+3+2-3	1	0.00
4+4+1+3+1+4-6	4+4+1+3+1+4-6	1	0.00
4+4+1+3+1+2-3	4+4+1+3+1+2-3	1	0.00
4+4+1+3+1+0-1	4+4+1+3+1+0-1	1	0.00
4+4+1+2+2+2-3	4+4+1+2+2+2-3	1	0.00
4+3+1+3+3+2-3	4+3+1+3+3+2-3	1	0.00
4+3+1+4+1+4-6	4+3+1+4+1+4-6	1	0.00
4+3+1+4+2+2-3	4+3+1+4+2+2-3	1	0.00
5+2+1+3+2+4-6	5+2+1+3+2+4-6	1	0.00
5+2+1+1+1+4-6	5+2+1+1+1+4-6	1	0.00
5+1+3+1+2+4-6	5+1+3+1+2+4-6	1	0.00
5+2+2+2+1+9-10	5+2+2+2+1+9-10	1	0.00
5+6+1+2+3+4-6	5+6+1+2+3+4-6	1	0.00
5+1+2+4+3+7-8	5+1+2+4+3+7-8	1	0.00
5+8+1+4+1+4-6	5+8+1+4+1+4-6	1	0.00
5+8+2+4+3+4-6	5+8+2+4+3+4-6	1	0.00
5+4+1+2+3+7-8	5+4+1+2+3+7-8	1	0.00
5+4+1+1+2+7-8	5+4+1+1+2+7-8	1	0.00
5+1+1+3+1+4-6	5+1+1+3+1+4-6	1	0.00
5+1+1+3+1+2-3	5+1+1+3+1+2-3	1	0.00
5+1+1+2+1+4-6	5+1+1+2+1+4-6	1	0.00
5+1+1+2+2+2-3	5+1+1+2+2+2-3	1	0.00
5+6+2+1+2+7-8	5+6+2+1+2+7-8	1	0.00
5+8+1+5plus+2+9-10	5+8+1+5plus+2+9-10	1	0.00
5+7+2+2+3+9-10	5+7+2+2+3+9-10	1	0.00
5+5+3+2+2+9-10	5+5+3+2+2+9-10	1	0.00
5+5+3+1+2+9-10	5+5+3+1+2+9-10	1	0.00
5+2+3+2+2+9-10	5+2+3+2+2+9-10	1	0.00
5+7+2+4+2+7-8	5+7+2+4+2+7-8	1	0.00
5+7+2+2+3+7-8	5+7+2+2+3+7-8	1	0.00
5+9+2+3+1+7-8	5+9+2+3+1+7-8	1	0.00
5+9+2+3+3+7-8	5+9+2+3+3+7-8	1	0.00
5+1+2+3+3+7-8	5+1+2+3+3+7-8	1	0.00
5+1+2+3+3+9-10	5+1+2+3+3+9-10	1	0.00

5+4+2+5plus+1+7-8	5+4+2+5plus+1+7-8	1	0.00
5+5+2+2+1+9-10	5+5+2+2+1+9-10	1	0.00
5+5+1+1+3+9-10	5+5+1+1+3+9-10	1	0.00
5+8+2+3+3+7-8	5+8+2+3+3+7-8	1	0.00
5+8+2+3+3+9-10	5+8+2+3+3+9-10	1	0.00
5+2+1+4+1+9-10	5+2+1+4+1+9-10	1	0.00
6+6+1+4+3+7-8	6+6+1+4+3+7-8	1	0.00
6+6+1+1+3+2-3	6+6+1+1+3+2-3	1	0.00
5+6+2+5plus+1+9-10	5+6+2+5plus+1+9-10	1	0.00
5+6+2+5plus+3+9-10	5+6+2+5plus+3+9-10	1	0.00
5+4+2+4+3+4-6	5+4+2+4+3+4-6	1	0.00
5+1+2+1+1+9-10	5+1+2+1+1+9-10	1	0.00
5+1+2+1+3+9-10	5+1+2+1+3+9-10	1	0.00
5+3+1+2+2+4-6	5+3+1+2+2+4-6	1	0.00
5+3+1+2+2+7-8	5+3+1+2+2+7-8	1	0.00
5+8+2+4+1+0-1	5+8+2+4+1+0-1	1	0.00
5+8+2+4+2+2-3	5+8+2+4+2+2-3	1	0.00
4+1+1+2+3+0-1	4+1+1+2+3+0-1	1	0.00
5+8+2+3+2+4-6	5+8+2+3+2+4-6	1	0.00
6+6+2+5plus+1+9-10	6+6+2+5plus+1+9-10	1	0.00
6+3+1+4+1+7-8	6+3+1+4+1+7-8	1	0.00
6+3+2+4+3+7-8	6+3+2+4+3+7-8	1	0.00
6+3+1+4+3+9-10	6+3+1+4+3+9-10	1	0.00
6+3+1+4+3+7-8	6+3+1+4+3+7-8	1	0.00
6+2+1+1+1+4-6	6+2+1+1+1+4-6	1	0.00
6+3+1+2+2+9-10	6+3+1+2+2+9-10	1	0.00
6+3+2+2+2+7-8	6+3+2+2+2+7-8	1	0.00
6+7+2+4+1+7-8	6+7+2+4+1+7-8	1	0.00
6+2+1+3+1+9-10	6+2+1+3+1+9-10	1	0.00
6+7+2+2+3+9-10	6+7+2+2+3+9-10	1	0.00
6+8+2+3+1+9-10	6+8+2+3+1+9-10	1	0.00
6+1+2+2+1+7-8	6+1+2+2+1+7-8	1	0.00
6+1+1+2+2+2-3	6+1+1+2+2+2-3	1	0.00
6+1+1+5plus+3+9-10	6+1+1+5plus+3+9-10	1	0.00
6+2+2+2+2+4-6	6+2+2+2+2+4-6	1	0.00
6+2+1+4+1+4-6	6+2+1+4+1+4-6	1	0.00
6+2+1+3+3+9-10	6+2+1+3+3+9-10	1	0.00
6+2+1+3+3+4-6	6+2+1+3+3+4-6	1	0.00
6+9+2+5plus+3+7-8	6+9+2+5plus+3+7-8	1	0.00
6+9+2+3+2+4-6	6+9+2+3+2+4-6	1	0.00
6+9+2+4+2+9-10	6+9+2+4+2+9-10	1	0.00
6+9+2+3+3+7-8	6+9+2+3+3+7-8	1	0.00
6+1+1+2+1+7-8	6+1+1+2+1+7-8	1	0.00
6+2+1+4+3+9-10	6+2+1+4+3+9-10	1	0.00
6+2+1+3+2+2-3	6+2+1+3+2+2-3	1	0.00
6+2+1+5plus+3+7-8	6+2+1+5plus+3+7-8	1	0.00
6+2+2+5plus+3+9-10	6+2+2+5plus+3+9-10	1	0.00
6+4+1+5plus+2+9-10	6+4+1+5plus+2+9-10	1	0.00
6+4+2+5plus+3+4-6	6+4+2+5plus+3+4-6	1	0.00
6+4+2+3+3+9-10	6+4+2+3+3+9-10	1	0.00
6+4+2+3+2+4-6	6+4+2+3+2+4-6	1	0.00
6+1+1+4+1+9-10	6+1+1+4+1+9-10	1	0.00
6+1+1+4+3+9-10	6+1+1+4+3+9-10	1	0.00
5+1+1+3+2+2-3	5+1+1+3+2+2-3	1	0.00
6+1+1+3+2+0-1	6+1+1+3+2+0-1	1	0.00
6+1+1+3+2+2-3	6+1+1+3+2+2-3	1	0.00

6+6+2+4+2+2-3	6+6+2+4+2+2-3	1	0.00
6+1+2+3+3+7-8	6+1+2+3+3+7-8	1	0.00
7+2+2+1+2+2-3	7+2+2+1+2+2-3	1	0.00
7+2+2+3+3+4-6	7+2+2+3+3+4-6	1	0.00
7+2+1+4+2+9-10	7+2+1+4+2+9-10	1	0.00
7+1+1+3+2+9-10	7+1+1+3+2+9-10	1	0.00
7+1+1+2+2+7-8	7+1+1+2+2+7-8	1	0.00
7+1+2+3+3+9-10	7+1+2+3+3+9-10	1	0.00
7+4+1+1+1+7-8	7+4+1+1+1+7-8	1	0.00
7+4+1+1+3+7-8	7+4+1+1+3+7-8	1	0.00
7+4+1+2+3+7-8	7+4+1+2+3+7-8	1	0.00
7+4+1+1+2+7-8	7+4+1+1+2+7-8	1	0.00
7+1+2+1+3+7-8	7+1+2+1+3+7-8	1	0.00
7+1+1+2+1+7-8	7+1+1+2+1+7-8	1	0.00
7+3+2+3+1+0-1	7+3+2+3+1+0-1	1	0.00
7+3+2+4+3+2-3	7+3+2+4+3+2-3	1	0.00
7+3+2+4+2+7-8	7+3+2+4+2+7-8	1	0.00
7+3+2+3+3+2-3	7+3+2+3+3+2-3	1	0.00
7+3+2+3+2+0-1	7+3+2+3+2+0-1	1	0.00
7+3+2+3+2+2-3	7+3+2+3+2+2-3	1	0.00
7+2+1+2+3+7-8	7+2+1+2+3+7-8	1	0.00
7+2+2+2+3+4-6	7+2+2+2+3+4-6	1	0.00
1+1+1+4+1+9-10	1+1+1+4+1+9-10	1	0.00
1+8+3+1+2+9-10	1+8+3+1+2+9-10	1	0.00
2+1+2+4+1+9-10	2+1+2+4+1+9-10	1	0.00
2+8+2+4+1+9-10	2+8+2+4+1+9-10	1	0.00
2+9+2+1+2+2-3	2+9+2+1+2+2-3	1	0.00
2+4+1+5plus+1+4-6	2+4+1+5plus+1+4-6	1	0.00
2+4+2+5plus+3+4-6	2+4+2+5plus+3+4-6	1	0.00
2+9+3+2+3+7-8	2+9+3+2+3+7-8	1	0.00
2+9+3+2+2+7-8	2+9+3+2+2+7-8	1	0.00
2+9+3+2+2+4-6	2+9+3+2+2+4-6	1	0.00
2+9+3+1+2+9-10	2+9+3+1+2+9-10	1	0.00
2+2+1+2+2+2-3	2+2+1+2+2+2-3	1	0.00
2+2+2+3+1+7-8	2+2+2+3+1+7-8	1	0.00
2+9+2+4+3+0-1	2+9+2+4+3+0-1	1	0.00
2+9+2+4+2+2-3	2+9+2+4+2+2-3	1	0.00
2+9+2+2+2+9-10	2+9+2+2+2+9-10	1	0.00
2+8+2+2+2+0-1	2+8+2+2+2+0-1	1	0.00
2+7+1+4+1+4-6	2+7+1+4+1+4-6	1	0.00
2+7+2+4+3+2-3	2+7+2+4+3+2-3	1	0.00
2+7+2+4+2+2-3	2+7+2+4+2+2-3	1	0.00
2+9+2+4+1+9-10	2+9+2+4+1+9-10	1	0.00
2+9+2+4+2+7-8	2+9+2+4+2+7-8	1	0.00
2+9+2+5plus+1+7-8	2+9+2+5plus+1+7-8	1	0.00
2+9+2+5plus+2+9-10	2+9+2+5plus+2+9-10	1	0.00
2+8+2+5plus+3+2-3	2+8+2+5plus+3+2-3	1	0.00
2+8+1+4+3+7-8	2+8+1+4+3+7-8	1	0.00
2+8+1+3+1+7-8	2+8+1+3+1+7-8	1	0.00
2+7+2+2+1+9-10	2+7+2+2+1+9-10	1	0.00
2+7+3+1+2+4-6	2+7+3+1+2+4-6	1	0.00
2+7+3+1+2+7-8	2+7+3+1+2+7-8	1	0.00
2+6+1+3+3+9-10	2+6+1+3+3+9-10	1	0.00
2+2+1+5plus+2+9-10	2+2+1+5plus+2+9-10	1	0.00
2+2+2+5plus+3+7-8	2+2+2+5plus+3+7-8	1	0.00
2+2+2+4+2+2-3	2+2+2+4+2+2-3	1	0.00

2+7+2+1+2+2-3	2+7+2+1+2+2-3	1	0.00
2+9+2+1+2+9-10	2+9+2+1+2+9-10	1	0.00
2+9+2+1+3+9-10	2+9+2+1+3+9-10	1	0.00
2+8+2+1+3+4-6	2+8+2+1+3+4-6	1	0.00
2+9+2+1+3+4-6	2+9+2+1+3+4-6	1	0.00
2+9+2+1+3+7-8	2+9+2+1+3+7-8	1	0.00
2+8+2+3+1+9-10	2+8+2+3+1+9-10	1	0.00
2+6+2+3+1+7-8	2+6+2+3+1+7-8	1	0.00
2+2+1+4+1+4-6	2+2+1+4+1+4-6	1	0.00
2+7+1+4+3+4-6	2+7+1+4+3+4-6	1	0.00
3+2+2+5plus+2+4-6	3+2+2+5plus+2+4-6	1	0.00
3+4+2+4+3+4-6	3+4+2+4+3+4-6	1	0.00
3+8+2+4+2+4-6	3+8+2+4+2+4-6	1	0.00
3+8+2+5plus+3+7-8	3+8+2+5plus+3+7-8	1	0.00
3+8+2+4+3+4-6	3+8+2+4+3+4-6	1	0.00
3+2+2+2+1+7-8	3+2+2+2+1+7-8	1	0.00
3+6+1+3+2+7-8	3+6+1+3+2+7-8	1	0.00
3+2+2+1+1+9-10	3+2+2+1+1+9-10	1	0.00
3+2+2+1+3+9-10	3+2+2+1+3+9-10	1	0.00
3+2+2+1+2+2-3	3+2+2+1+2+2-3	1	0.00
3+7+1+5plus+1+9-10	3+7+1+5plus+1+9-10	1	0.00
3+2+1+1+1+4-6	3+2+1+1+1+4-6	1	0.00
3+4+2+5plus+1+9-10	3+4+2+5plus+1+9-10	1	0.00
3+4+2+5plus+1+4-6	3+4+2+5plus+1+4-6	1	0.00
3+4+2+3+3+0-1	3+4+2+3+3+0-1	1	0.00
3+3+2+3+3+4-6	3+3+2+3+3+4-6	1	0.00
3+3+2+4+3+9-10	3+3+2+4+3+9-10	1	0.00
3+3+1+5plus+1+4-6	3+3+1+5plus+1+4-6	1	0.00
3+3+1+5plus+2+2-3	3+3+1+5plus+2+2-3	1	0.00
2+4+1+3+1+7-8	2+4+1+3+1+7-8	1	0.00
3+4+1+3+1+4-6	3+4+1+3+1+4-6	1	0.00
3+2+1+3+3+2-3	3+2+1+3+3+2-3	1	0.00
3+3+2+1+3+7-8	3+3+2+1+3+7-8	1	0.00
3+3+2+1+3+9-10	3+3+2+1+3+9-10	1	0.00
3+2+2+1+3+2-3	3+2+2+1+3+2-3	1	0.00
3+4+2+1+1+9-10	3+4+2+1+1+9-10	1	0.00
3+4+2+2+1+7-8	3+4+2+2+1+7-8	1	0.00
3+2+1+2+1+7-8	3+2+1+2+1+7-8	1	0.00
3+6+2+2+1+4-6	3+6+2+2+1+4-6	1	0.00
3+3+1+4+2+0-1	3+3+1+4+2+0-1	1	0.00
3+3+1+4+3+4-6	3+3+1+4+3+4-6	1	0.00
3+3+1+4+3+7-8	3+3+1+4+3+7-8	1	0.00
3+8+1+1+3+7-8	3+8+1+1+3+7-8	1	0.00
3+6+2+5plus+2+7-8	3+6+2+5plus+2+7-8	1	0.00
3+6+2+3+2+7-8	3+6+2+3+2+7-8	1	0.00
3+3+2+3+3+9-10	3+3+2+3+3+9-10	1	0.00
3+3+2+2+1+9-10	3+3+2+2+1+9-10	1	0.00
3+3+2+2+2+4-6	3+3+2+2+2+4-6	1	0.00
3+4+1+2+2+4-6	3+4+1+2+2+4-6	1	0.00
3+4+1+2+2+9-10	3+4+1+2+2+9-10	1	0.00
3+4+1+3+3+7-8	3+4+1+3+3+7-8	1	0.00
4+3+1+5plus+1+7-8	4+3+1+5plus+1+7-8	1	0.00
4+2+2+1+1+7-8	4+2+2+1+1+7-8	1	0.00
4+1+1+4+2+2-3	4+1+1+4+2+2-3	1	0.00
4+2+2+5plus+1+7-8	4+2+2+5plus+1+7-8	1	0.00
4+2+1+4+1+9-10	4+2+1+4+1+9-10	1	0.00

4+2+1+4+1+4-6	4+2+1+4+1+4-6	1	0.00
4+3+2+5plus+2+9-10	4+3+2+5plus+2+9-10	1	0.00
4+3+2+5plus+3+9-10	4+3+2+5plus+3+9-10	1	0.00
4+2+1+3+3+7-8	4+2+1+3+3+7-8	1	0.00
1+6+2+2+1+4-6	1+6+2+2+1+4-6	1	0.00
1+6+3+1+2+7-8	1+6+3+1+2+7-8	1	0.00
1+1+1+5plus+2+9-10	1+1+1+5plus+2+9-10	1	0.00
1+5+2+3+3+4-6	1+5+2+3+3+4-6	1	0.00
1+2+1+5plus+2+9-10	1+2+1+5plus+2+9-10	1	0.00
1+2+2+1+3+4-6	1+2+2+1+3+4-6	1	0.00
1+7+1+3+1+9-10	1+7+1+3+1+9-10	1	0.00
1+4+1+1+2+2-3	1+4+1+1+2+2-3	1	0.00
1+4+1+1+2+0-1	1+4+1+1+2+0-1	1	0.00
1+4+1+1+3+4-6	1+4+1+1+3+4-6	1	0.00
1+4+1+2+2+0-1	1+4+1+2+2+0-1	1	0.00
1+4+1+2+1+4-6	1+4+1+2+1+4-6	1	0.00
1+4+1+2+2+2-3	1+4+1+2+2+2-3	1	0.00
1+8+1+2+1+4-6	1+8+1+2+1+4-6	1	0.00
1+8+1+2+3+4-6	1+8+1+2+3+4-6	1	0.00
1+8+1+3+3+2-3	1+8+1+3+3+2-3	1	0.00
1+8+1+3+2+2-3	1+8+1+3+2+2-3	1	0.00
1+8+1+1+2+2-3	1+8+1+1+2+2-3	1	0.00
1+8+1+1+2+0-1	1+8+1+1+2+0-1	1	0.00
1+8+2+1+2+0-1	1+8+2+1+2+0-1	1	0.00
1+4+2+3+1+7-8	1+4+2+3+1+7-8	1	0.00
1+6+2+3+2+4-6	1+6+2+3+2+4-6	1	0.00
1+6+2+3+3+4-6	1+6+2+3+3+4-6	1	0.00
1+2+1+4+2+2-3	1+2+1+4+2+2-3	1	0.00
1+2+1+4+3+4-6	1+2+1+4+3+4-6	1	0.00
1+2+1+1+3+0-1	1+2+1+1+3+0-1	1	0.00
1+2+1+1+1+9-10	1+2+1+1+1+9-10	1	0.00
1+7+2+3+1+9-10	1+7+2+3+1+9-10	1	0.00
1+7+2+3+3+7-8	1+7+2+3+3+7-8	1	0.00
1+3+2+1+3+4-6	1+3+2+1+3+4-6	1	0.00
1+2+1+4+3+0-1	1+2+1+4+3+0-1	1	0.00
1+4+1+4+3+4-6	1+4+1+4+3+4-6	1	0.00
1+2+1+1+1+2-3	1+2+1+1+1+2-3	1	0.00
1+4+2+5plus+1+4-6	1+4+2+5plus+1+4-6	1	0.00
1+2+2+3+1+7-8	1+2+2+3+1+7-8	1	0.00
1+2+2+5plus+1+7-8	1+2+2+5plus+1+7-8	1	0.00
1+2+2+5plus+2+9-10	1+2+2+5plus+2+9-10	1	0.00
1+3+1+4+1+4-6	1+3+1+4+1+4-6	1	0.00
1+3+1+4+3+9-10	1+3+1+4+3+9-10	1	0.00
1+3+1+1+3+4-6	1+3+1+1+3+4-6	1	0.00
1+3+1+3+3+4-6	1+3+1+3+3+4-6	1	0.00
1+1+1+3+1+4-6	1+1+1+3+1+4-6	1	0.00
1+8+2+4+1+7-8	1+8+2+4+1+7-8	1	0.00
1+8+1+4+3+9-10	1+8+1+4+3+9-10	1	0.00
1+8+2+2+2+4-6	1+8+2+2+2+4-6	1	0.00
1+1+1+1+3+0-1	1+1+1+1+3+0-1	1	0.00
1+5+1+4+3+4-6	1+5+1+4+3+4-6	1	0.00
1+4+1+4+1+4-6	1+4+1+4+1+4-6	1	0.00
1+4+1+4+2+9-10	1+4+1+4+2+9-10	1	0.00
1+4+1+4+1+7-8	1+4+1+4+1+7-8	1	0.00
1+4+2+4+1+7-8	1+4+2+4+1+7-8	1	0.00
1+2+2+1+1+9-10	1+2+2+1+1+9-10	1	0.00

1+2+3+1+3+9-10	1+2+3+1+3+9-10	1	0.00
1+7+2+3+3+2-3	1+7+2+3+3+2-3	1	0.00
1+7+2+3+2+2-3	1+7+2+3+2+2-3	1	0.00
1+7+2+3+1+2-3	1+7+2+3+1+2-3	1	0.00
1+8+2+2+2+2-3	1+8+2+2+2+2-3	1	0.00
1+8+2+2+1+4-6	1+8+2+2+1+4-6	1	0.00
2+7+2+5plus+2+0-1	2+7+2+5plus+2+0-1	1	0.00
2+6+1+5plus+1+9-10	2+6+1+5plus+1+9-10	1	0.00
2+1+1+5plus+1+7-8	2+1+1+5plus+1+7-8	1	0.00
2+7+2+4+1+7-8	2+7+2+4+1+7-8	1	0.00
2+1+2+3+3+2-3	2+1+2+3+3+2-3	1	0.00
2+6+2+2+1+9-10	2+6+2+2+1+9-10	1	0.00
2+6+1+3+1+7-8	2+6+1+3+1+7-8	1	0.00
1+6+1+4+2+9-10	1+6+1+4+2+9-10	1	0.00
1+6+1+4+2+7-8	1+6+1+4+2+7-8	1	0.00
2+1+2+5plus+1+7-8	2+1+2+5plus+1+7-8	1	0.00
2+3+2+2+3+9-10	2+3+2+2+3+9-10	1	0.00
2+1+2+2+2+2-3	2+1+2+2+2+2-3	1	0.00
2+7+2+5plus+1+4-6	2+7+2+5plus+1+4-6	1	0.00
2+7+1+1+3+0-1	2+7+1+1+3+0-1	1	0.00
2+6+2+1+3+2-3	2+6+2+1+3+2-3	1	0.00
2+6+2+1+2+2-3	2+6+2+1+2+2-3	1	0.00
2+6+2+2+2+2-3	2+6+2+2+2+2-3	1	0.00
2+8+1+3+3+4-6	2+8+1+3+3+4-6	1	0.00
2+8+1+3+2+4-6	2+8+1+3+2+4-6	1	0.00
2+8+1+3+2+0-1	2+8+1+3+2+0-1	1	0.00
2+2+2+3+1+2-3	2+2+2+3+1+2-3	1	0.00
2+4+1+2+2+2-3	2+4+1+2+2+2-3	1	0.00
2+2+2+5plus+1+4-6	2+2+2+5plus+1+4-6	1	0.00
2+2+2+5plus+3+9-10	2+2+2+5plus+3+9-10	1	0.00
2+2+2+4+1+7-8	2+2+2+4+1+7-8	1	0.00
2+7+1+2+1+7-8	2+7+1+2+1+7-8	1	0.00
2+1+1+1+3+0-1	2+1+1+1+3+0-1	1	0.00
2+7+1+3+1+4-6	2+7+1+3+1+4-6	1	0.00
2+7+1+5plus+1+0-1	2+7+1+5plus+1+0-1	1	0.00
2+7+1+5plus+1+4-6	2+7+1+5plus+1+4-6	1	0.00
2+7+1+5plus+1+2-3	2+7+1+5plus+1+2-3	1	0.00
2+6+2+3+3+4-6	2+6+2+3+3+4-6	1	0.00
2+4+1+4+1+7-8	2+4+1+4+1+7-8	1	0.00
1+2+1+3+2+2-3	1+2+1+3+2+2-3	1	0.00
1+9+1+2+2+9-10	1+9+1+2+2+9-10	1	0.00
1+9+1+2+3+9-10	1+9+1+2+3+9-10	1	0.00
1+9+2+4+3+9-10	1+9+2+4+3+9-10	1	0.00
1+9+2+4+2+9-10	1+9+2+4+2+9-10	1	0.00
1+2+1+3+1+4-6	1+2+1+3+1+4-6	1	0.00
1+6+2+2+3+4-6	1+6+2+2+3+4-6	1	0.00
1+2+2+5plus+2+2-3	1+2+2+5plus+2+2-3	1	0.00
1+2+2+4+1+4-6	1+2+2+4+1+4-6	1	0.00
1+2+2+3+2+2-3	1+2+2+3+2+2-3	1	0.00
1+7+1+1+1+9-10	1+7+1+1+1+9-10	1	0.00
4+3+1+3+1+7-8	4+3+1+3+1+7-8	1	0.00
4+6+2+4+1+9-10	4+6+2+4+1+9-10	1	0.00
4+3+2+3+1+7-8	4+3+2+3+1+7-8	1	0.00
4+3+2+3+3+7-8	4+3+2+3+3+7-8	1	0.00
4+2+2+5plus+2+0-1	4+2+2+5plus+2+0-1	1	0.00
4+2+2+3+2+0-1	4+2+2+3+2+0-1	1	0.00

4+3+2+1+1+9-10	4+3+2+1+1+9-10	1	0.00
4+3+2+1+3+9-10	4+3+2+1+3+9-10	1	0.00
4+1+1+3+3+9-10	4+1+1+3+3+9-10	1	0.00
4+6+2+2+1+7-8	4+6+2+2+1+7-8	1	0.00
4+1+1+2+2+0-1	4+1+1+2+2+0-1	1	0.00
4+2+2+1+3+4-6	4+2+2+1+3+4-6	1	0.00
4+7+2+5plus+3+9-10	4+7+2+5plus+3+9-10	1	0.00
4+3+1+1+1+7-8	4+3+1+1+1+7-8	1	0.00
4+2+1+2+2+0-1	4+2+1+2+2+0-1	1	0.00
4+2+1+2+2+2-3	4+2+1+2+2+2-3	1	0.00
4+2+1+3+3+2-3	4+2+1+3+3+2-3	1	0.00
4+1+3+1+3+9-10	4+1+3+1+3+9-10	1	0.00
4+1+3+1+2+9-10	4+1+3+1+2+9-10	1	0.00
4+1+3+1+2+7-8	4+1+3+1+2+7-8	1	0.00
4+4+2+2+1+4-6	4+4+2+2+1+4-6	1	0.00
5+7+1+4+2+9-10	5+7+1+4+2+9-10	1	0.00
5+2+1+4+1+4-6	5+2+1+4+1+4-6	1	0.00
5+7+2+5plus+1+9-10	5+7+2+5plus+1+9-10	1	0.00
5+7+2+5plus+3+7-8	5+7+2+5plus+3+7-8	1	0.00
5+7+2+5plus+2+4-6	5+7+2+5plus+2+4-6	1	0.00
5+2+2+5plus+1+7-8	5+2+2+5plus+1+7-8	1	0.00
5+7+2+1+2+2-3	5+7+2+1+2+2-3	1	0.00
5+6+1+2+3+9-10	5+6+1+2+3+9-10	1	0.00
5+5+2+5plus+2+9-10	5+5+2+5plus+2+9-10	1	0.00
5+5+1+4+2+4-6	5+5+1+4+2+4-6	1	0.00
5+2+2+4+1+9-10	5+2+2+4+1+9-10	1	0.00
5+6+3+2+2+9-10	5+6+3+2+2+9-10	1	0.00
5+2+1+1+2+2-3	5+2+1+1+2+2-3	1	0.00
5+2+1+3+2+0-1	5+2+1+3+2+0-1	1	0.00
5+2+1+3+2+2-3	5+2+1+3+2+2-3	1	0.00
5+2+2+3+3+9-10	5+2+2+3+3+9-10	1	0.00
5+2+2+3+3+4-6	5+2+2+3+3+4-6	1	0.00
5+2+2+1+3+9-10	5+2+2+1+3+9-10	1	0.00
5+6+1+1+2+7-8	5+6+1+1+2+7-8	1	0.00
5+6+1+1+2+4-6	5+6+1+1+2+4-6	1	0.00
5+6+1+3+2+4-6	5+6+1+3+2+4-6	1	0.00
5+1+1+2+2+0-1	5+1+1+2+2+0-1	1	0.00
3+2+1+2+1+0-1	3+2+1+2+1+0-1	1	0.00
5+6+2+5plus+2+4-6	5+6+2+5plus+2+4-6	1	0.00
5+6+1+3+1+7-8	5+6+1+3+1+7-8	1	0.00
5+1+1+2+3+2-3	5+1+1+2+3+2-3	1	0.00
5+6+2+5plus+1+7-8	5+6+2+5plus+1+7-8	1	0.00
5+4+2+2+3+7-8	5+4+2+2+3+7-8	1	0.00
6+2+3+2+2+9-10	6+2+3+2+2+9-10	1	0.00
5+2+2+1+3+7-8	5+2+2+1+3+7-8	1	0.00
6+4+2+4+1+9-10	6+4+2+4+1+9-10	1	0.00
6+7+2+2+3+7-8	6+7+2+2+3+7-8	1	0.00
5+8+3+1+2+4-6	5+8+3+1+2+4-6	1	0.00
5+8+3+1+3+7-8	5+8+3+1+3+7-8	1	0.00
6+8+2+1+2+7-8	6+8+2+1+2+7-8	1	0.00
6+8+3+1+3+9-10	6+8+3+1+3+9-10	1	0.00
5+2+1+5plus+1+7-8	5+2+1+5plus+1+7-8	1	0.00
5+2+1+5plus+2+4-6	5+2+1+5plus+2+4-6	1	0.00
6+2+1+5plus+1+4-6	6+2+1+5plus+1+4-6	1	0.00
6+2+1+5plus+3+4-6	6+2+1+5plus+3+4-6	1	0.00
6+5+2+3+3+7-8	6+5+2+3+3+7-8	1	0.00

6+6+1+2+3+9-10	6+6+1+2+3+9-10	1	0.00
6+1+1+2+1+9-10	6+1+1+2+1+9-10	1	0.00
6+4+2+4+1+4-6	6+4+2+4+1+4-6	1	0.00
6+4+2+3+2+9-10	6+4+2+3+2+9-10	1	0.00
6+4+2+2+2+9-10	6+4+2+2+2+9-10	1	0.00
6+7+2+5plus+1+7-8	6+7+2+5plus+1+7-8	1	0.00
6+7+2+5plus+3+9-10	6+7+2+5plus+3+9-10	1	0.00
6+2+2+3+1+9-10	6+2+2+3+1+9-10	1	0.00
6+8+2+4+3+7-8	6+8+2+4+3+7-8	1	0.00
6+6+2+5plus+2+2-3	6+6+2+5plus+2+2-3	1	0.00
6+6+2+2+1+9-10	6+6+2+2+1+9-10	1	0.00
6+3+1+2+3+4-6	6+3+1+2+3+4-6	1	0.00
6+8+2+2+2+4-6	6+8+2+2+2+4-6	1	0.00
6+2+2+3+3+7-8	6+2+2+3+3+7-8	1	0.00
6+2+2+5plus+2+4-6	6+2+2+5plus+2+4-6	1	0.00
6+2+2+4+1+9-10	6+2+2+4+1+9-10	1	0.00
6+8+2+5plus+1+9-10	6+8+2+5plus+1+9-10	1	0.00
6+8+2+5plus+1+7-8	6+8+2+5plus+1+7-8	1	0.00
6+1+2+1+3+7-8	6+1+2+1+3+7-8	1	0.00
2+2+2+1+3+9-10	2+2+2+1+3+9-10	1	0.00
2+8+3+2+1+9-10	2+8+3+2+1+9-10	1	0.00
2+8+3+1+2+7-8	2+8+3+1+2+7-8	1	0.00
3+2+1+4+1+9-10	3+2+1+4+1+9-10	1	0.00
3+4+2+1+1+4-6	3+4+2+1+1+4-6	1	0.00
3+4+2+1+1+7-8	3+4+2+1+1+7-8	1	0.00
4+3+2+2+3+4-6	4+3+2+2+3+4-6	1	0.00
5+7+1+3+2+7-8	5+7+1+3+2+7-8	1	0.00
6+1+2+4+2+9-10	6+1+2+4+2+9-10	1	0.00
6+1+1+4+2+2-3	6+1+1+4+2+2-3	1	0.00
7+2+1+2+1+7-8	7+2+1+2+1+7-8	1	0.00
7+2+1+2+1+9-10	7+2+1+2+1+9-10	1	0.00
7+4+1+2+3+9-10	7+4+1+2+3+9-10	1	0.00
7+4+1+4+3+9-10	7+4+1+4+3+9-10	1	0.00
7+4+1+5plus+3+9-10	7+4+1+5plus+3+9-10	1	0.00
7+2+2+5plus+3+9-10	7+2+2+5plus+3+9-10	1	0.00
7+2+2+5plus+2+4-6	7+2+2+5plus+2+4-6	1	0.00
7+6+2+2+1+7-8	7+6+2+2+1+7-8	1	0.00
7+6+2+2+3+4-6	7+6+2+2+3+4-6	1	0.00
7+6+2+1+3+7-8	7+6+2+1+3+7-8	1	0.00
7+6+2+1+3+4-6	7+6+2+1+3+4-6	1	0.00
7+2+1+1+1+7-8	7+2+1+1+1+7-8	1	0.00
7+1+1+1+3+7-8	7+1+1+1+3+7-8	1	0.00
7+2+1+4+1+4-6	7+2+1+4+1+4-6	1	0.00
7+2+2+4+3+7-8	7+2+2+4+3+7-8	1	0.00

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['4+2+2+2+2+7-8', '4+2+2+2+2+9-10', '4+1+1+2+2+7-8', '4+1+1+1+2+7-8', '4
+2+1+1+2+7-8', '2+2+2+2+2+9-10', '1+1+1+1+2+7-8', '2+1+1+2+2+7-8', '3+2+2+2+2+7-8',
'2+4+2+2+2+7-8', '4+2+1+2+2+7-8', '2+2+1+1+2+7-8', '2+2+2+2+2+7-8', '1+1+1+2+2+7-8',
'2+1+1+1+2+7-8', '2+1+2+2+2+7-8', '1+2+1+1+2+7-8', '2+2+2+4+2+7-8', '1+2+1+2+2+7-8',
'6+1+2+2+2+9-10', '4+1+1+2+2+9-10', '1+1+2+2+2+7-8', '5+2+2+2+2+7-8', '6+2+1+2+2
+7-8', '2+2+1+2+2+7-8', '3+1+2+2+2+9-10', '3+4+2+2+2+9-10', '3+2+2+2+2+9-10', '2+1+2
+2+2+9-10', '3+2+2+4+2+7-8', '1+4+2+2+2+9-10', '1+1+1+2+2+4-6', '2+7+2+2+2+7-8', '4
```

```
+2+2+2+3+9-10', '4+2+2+4+2+7-8', '4+2+2+2+2+4-6', '5+6+2+2+2+9-10', '2+2+1+1+2+9-1
0', '3+2+1+1+2+7-8', '1+2+2+2+2+7-8', '3+1+1+1+2+7-8', '3+2+1+2+2+7-8', '5+2+2+4+2
+7-8', '3+4+2+2+2+7-8', '2+8+2+2+2+7-8', '2+1+2+1+2+7-8', '3+1+2+2+2+7-8', '3+7+2+2
+2+7-8', '3+7+2+2+2+9-10', '4+2+1+2+2+9-10', '4+2+1+1+2+9-10', '4+1+1+1+2+9-10', '4
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'1+1+1+1+2+4-6', '4+2+2+4+2+9-10', '4+2+2+1+2+7-8', '4+1+1+4+2+7-8', '4+2+1+2+2
+4-6', '6+2+1+2+2+4-6', '1+1+1+2+2+9-10', '2+1+1+1+2+4-6', '3+4+2+3+2+7-8', '5+1+1+1
+2+7-8', '1+2+1+2+2+9-10', '1+4+2+2+2+7-8', '1+2+1+1+2+4-6', '2+1+2+5plus+2+7-8', '3
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+7-8', '6+1+1+2+2+7-8', '3+7+2+4+2+7-8', '4+4+2+4+2+7-8', '5+6+2+4+2+7-8', '2+8+2+2
+2+9-10', '2+6+2+3+2+7-8', '4+2+1+3+2+9-10', '5+1+1+1+2+9-10', '2+2+2+3+2+7-8', '2+7
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'2+1+1+2+2+9-10', '2+6+2+2+2+9-10', '3+2+2+3+2+7-8', '3+4+2+2+3+9-10', '5+8+2+2+2
+9-10', '5+2+2+2+2+9-10', '2+1+2+1+2+9-10', '5+1+1+2+2+7-8', '1+1+1+4+2+4-6', '2+7+2
+4+2+7-8', '2+7+2+2+2+4-6', '6+1+1+1+2+9-10', '4+2+2+3+2+7-8', '4+2+1+1+3+7-8', '6+8
+2+5plus+2+7-8', '5+6+2+2+2+7-8', '1+4+2+5plus+2+7-8', '1+2+2+2+2+9-10', '1+1+1+3+2
+7-8', '2+8+2+4+2+7-8', '3+4+2+1+2+7-8', '4+1+1+3+2+7-8', '4+2+1+4+2+7-8', '4+3+2+2
+2+7-8', '6+2+1+1+2+7-8', '2+6+1+4+2+7-8', '4+1+1+1+2+4-6', '4+1+1+2+2+4-6', '2+4+2
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+2+5plus+2+7-8', '1+4+2+4+2+7-8', '2+4+2+2+2+9-10', '2+8+2+5plus+2+9-10', '2+2+1+2+2
+4-6', '3+4+1+1+2+7-8', '4+2+2+5plus+2+7-8', '4+1+2+1+2+7-8', '5+1+1+2+2+9-10', '1+1
+1+1+2+9-10', '2+2+2+1+2+7-8', '2+2+2+5plus+2+7-8', '3+1+1+1+2+9-10', '3+3+2+2+2+9-1
0', '6+1+1+2+2+9-10', '5+1+1+1+2+4-6', '7+4+2+2+2+9-10', '1+2+2+1+2+7-8', '2+2+2+4+2
+9-10', '3+2+1+2+2+9-10', '4+1+2+2+2+9-10', '5+2+1+2+2+7-8', '6+1+1+1+2+7-8', '1+2+1
+3+2+7-8', '2+1+1+4+2+7-8', '2+6+2+4+2+7-8', '2+1+2+2+2+4-6', '4+1+2+4+2+7-8', '4+4
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[>] Labels: ['4+2+2+2+2+7-8', '4+2+2+2+2+9-10', '4+1+1+2+2+7-8', '4+1+1+1+2+7-8', '4
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```

```
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+4-6', '5+1+2+1+1+9-10', '5+1+2+1+3+9-10', '5+3+1+2+2+4-6', '5+3+1+2+2+7-8', '5+8+2
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'6+3+1+4+1+7-8', '6+3+2+4+3+7-8', '6+3+1+4+3+9-10', '6+3+1+4+3+7-8', '6+2+1+1+1
+4-6', '6+3+1+2+2+9-10', '6+3+2+2+2+7-8', '6+7+2+4+1+7-8', '6+2+1+3+1+9-10', '6+7+2
+2+3+9-10', '6+8+2+3+1+9-10', '6+1+2+2+1+7-8', '6+1+1+2+2+2-3', '6+1+1+5plus+3+9-1
0', '6+2+2+2+2+4-6', '6+2+1+4+1+4-6', '6+2+1+3+3+9-10', '6+2+1+3+3+4-6', '6+9+2+5plu
s+3+7-8', '6+9+2+3+2+4-6', '6+9+2+4+2+9-10', '6+9+2+3+3+7-8', '6+1+1+2+1+7-8', '6+2
+1+4+3+9-10', '6+2+1+3+2+2-3', '6+2+1+5plus+3+7-8', '6+2+2+5plus+3+9-10', '6+4+1+5pl
us+2+9-10', '6+4+2+5plus+3+4-6', '6+4+2+3+3+9-10', '6+4+2+3+2+4-6', '6+1+1+4+1+9-1
0', '6+1+1+4+3+9-10', '5+1+1+3+2+2-3', '6+1+1+3+2+0-1', '6+1+1+3+2+2-3', '6+6+2+4+2
+2-3', '6+1+2+3+3+7-8', '7+2+2+1+2+2-3', '7+2+2+3+3+4-6', '7+2+1+4+2+9-10', '7+1+1+3
+2+9-10', '7+1+1+2+2+7-8', '7+1+2+3+3+9-10', '7+4+1+1+1+7-8', '7+4+1+1+3+7-8', '7+4
+1+2+3+7-8', '7+4+1+1+2+7-8', '7+1+2+1+3+7-8', '7+1+1+2+1+7-8', '7+3+2+3+1+0-1', '7
+3+2+4+3+2-3', '7+3+2+4+2+7-8', '7+3+2+3+3+2-3', '7+3+2+3+2+0-1', '7+3+2+3+2+2-3',
'7+2+1+2+3+7-8', '7+2+2+2+3+4-6', '1+1+1+4+1+9-10', '1+8+3+1+2+9-10', '2+1+2+4+1+9-1
0', '2+8+2+4+1+9-10', '2+9+2+1+2+2-3', '2+4+1+5plus+1+4-6', '2+4+2+5plus+3+4-6', '2
+9+3+2+3+7-8', '2+9+3+2+2+7-8', '2+9+3+2+2+4-6', '2+9+3+1+2+9-10', '2+2+1+2+2+2-3',
'2+2+2+3+1+7-8', '2+9+2+4+3+0-1', '2+9+2+4+2+2-3', '2+9+2+2+2+9-10', '2+8+2+2+2
+0-1', '2+7+1+4+1+4-6', '2+7+2+4+3+2-3', '2+7+2+4+2+2-3', '2+9+2+4+1+9-10', '2+9+2+4
+2+7-8', '2+9+2+5plus+1+7-8', '2+9+2+5plus+2+9-10', '2+8+2+5plus+3+2-3', '2+8+1+4+3
+7-8', '2+8+1+3+1+7-8', '2+7+2+2+1+9-10', '2+7+3+1+2+4-6', '2+7+3+1+2+7-8', '2+6+1+3
+3+9-10', '2+2+1+5plus+2+9-10', '2+2+2+5plus+3+7-8', '2+2+2+4+2+2-3', '2+7+2+1+2
+2-3', '2+9+2+1+2+9-10', '2+9+2+1+3+9-10', '2+8+2+1+3+4-6', '2+9+2+1+3+4-6', '2+9+2
+1+3+7-8', '2+8+2+3+1+9-10', '2+6+2+3+1+7-8', '2+2+1+4+1+4-6', '2+7+1+4+3+4-6', '3+2
+2+5plus+2+4-6', '3+4+2+4+3+4-6', '3+8+2+4+2+4-6', '3+8+2+5plus+3+7-8', '3+8+2+4+3
+4-6', '3+2+2+2+1+7-8', '3+6+1+3+2+7-8', '3+2+2+1+1+9-10', '3+2+2+1+3+9-10', '3+2+2
+1+2+2-3', '3+7+1+5plus+1+9-10', '3+2+1+1+1+4-6', '3+4+2+5plus+1+9-10', '3+4+2+5plus
+1+4-6', '3+4+2+3+3+0-1', '3+3+2+3+3+4-6', '3+3+2+4+3+9-10', '3+3+1+5plus+1+4-6', '3
+3+1+5plus+2+2-3', '2+4+1+3+1+7-8', '3+4+1+3+1+4-6', '3+2+1+3+3+2-3', '3+3+2+1+3
+7-8', '3+3+2+1+3+9-10', '3+2+2+1+3+2-3', '3+4+2+1+1+9-10', '3+4+2+2+1+7-8', '3+2+1
+2+1+7-8', '3+6+2+2+1+4-6', '3+3+1+4+2+0-1', '3+3+1+4+3+4-6', '3+3+1+4+3+7-8', '3+8
+1+1+3+7-8', '3+6+2+5plus+2+7-8', '3+6+2+3+2+7-8', '3+3+2+3+3+9-10', '3+3+2+2+1+9-1
0', '3+3+2+2+2+4-6', '3+4+1+2+2+4-6', '3+4+1+2+2+9-10', '3+4+1+3+3+7-8', '4+3+1+5plu
s+1+7-8', '4+2+2+1+1+7-8', '4+1+1+4+2+2-3', '4+2+2+5plus+1+7-8', '4+2+1+4+1+9-10',
'4+2+1+4+1+4-6', '4+3+2+5plus+2+9-10', '4+3+2+5plus+3+9-10', '4+2+1+3+3+7-8', '1+6+2
+2+1+4-6', '1+6+3+1+2+7-8', '1+1+1+5plus+2+9-10', '1+5+2+3+3+4-6', '1+2+1+5plus+2
+9-10', '1+2+2+1+3+4-6', '1+7+1+3+1+9-10', '1+4+1+1+2+2-3', '1+4+1+1+2+0-1', '1+4+1
+1+3+4-6', '1+4+1+2+2+0-1', '1+4+1+2+1+4-6', '1+4+1+2+2+2-3', '1+8+1+2+1+4-6', '1+8
+1+2+3+4-6', '1+8+1+3+3+2-3', '1+8+1+3+2+2-3', '1+8+1+1+2+2-3', '1+8+1+1+2+0-1', '1
+8+2+1+2+0-1', '1+4+2+3+1+7-8', '1+6+2+3+2+4-6', '1+6+2+3+3+4-6', '1+2+1+4+2+2-3',
'1+2+1+4+3+4-6', '1+2+1+1+3+0-1', '1+2+1+1+1+9-10', '1+7+2+3+1+9-10', '1+7+2+3+3
+7-8', '1+3+2+1+3+4-6', '1+2+1+4+3+0-1', '1+4+1+4+3+4-6', '1+2+1+1+1+2-3', '1+4+2+5p
lus+1+4-6', '1+2+2+3+1+7-8', '1+2+2+5plus+1+7-8', '1+2+2+5plus+2+9-10', '1+3+1+4+1
+4-6', '1+3+1+4+3+9-10', '1+3+1+1+3+4-6', '1+3+1+3+3+4-6', '1+1+1+3+1+4-6', '1+8+2+4
+1+7-8', '1+8+1+4+3+9-10', '1+8+2+2+2+4-6', '1+1+1+1+3+0-1', '1+5+1+4+3+4-6', '1+4+1
+4+1+4-6', '1+4+1+4+2+9-10', '1+4+1+4+1+7-8', '1+4+2+4+1+7-8', '1+2+2+1+1+9-10', '1
+2+3+1+3+9-10', '1+7+2+3+3+2-3', '1+7+2+3+2+2-3', '1+7+2+3+1+2-3', '1+8+2+2+2+2-3',
'1+8+2+2+1+4-6', '2+7+2+5plus+2+0-1', '2+6+1+5plus+1+9-10', '2+1+1+5plus+1+7-8', '2
+7+2+4+1+7-8', '2+1+2+3+3+2-3', '2+6+2+2+1+9-10', '2+6+1+3+1+7-8', '1+6+1+4+2+9-10',
```

```
'1+6+1+4+2+7-8', '2+1+2+5plus+1+7-8', '2+3+2+2+3+9-10', '2+1+2+2+2+2-3', '2+7+2+5plu
s+1+4-6', '2+7+1+1+3+0-1', '2+6+2+1+3+2-3', '2+6+2+1+2+2-3', '2+6+2+2+2+2-3', '2+8+1
+3+3+4-6', '2+8+1+3+2+4-6', '2+8+1+3+2+0-1', '2+2+2+3+1+2-3', '2+4+1+2+2+2-3', '2+2
+2+5plus+1+4-6', '2+2+2+5plus+3+9-10', '2+2+2+4+1+7-8', '2+7+1+2+1+7-8', '2+1+1+1+3
+0-1', '2+7+1+3+1+4-6', '2+7+1+5plus+1+0-1', '2+7+1+5plus+1+4-6', '2+7+1+5plus+1
+2-3', '2+6+2+3+3+4-6', '2+4+1+4+1+7-8', '1+2+1+3+2+2-3', '1+9+1+2+2+9-10', '1+9+1+2
+3+9-10', '1+9+2+4+3+9-10', '1+9+2+4+2+9-10', '1+2+1+3+1+4-6', '1+6+2+2+3+4-6', '1+2
+2+5plus+2+2-3', '1+2+2+4+1+4-6', '1+2+2+3+2+2-3', '1+7+1+1+1+9-10', '4+3+1+3+1
+7-8', '4+6+2+4+1+9-10', '4+3+2+3+1+7-8', '4+3+2+3+3+7-8', '4+2+2+5plus+2+0-1', '4+2
+2+3+2+0-1', '4+3+2+1+1+9-10', '4+3+2+1+3+9-10', '4+1+1+3+3+9-10', '4+6+2+2+1+7-8',
'4+1+1+2+2+0-1', '4+2+2+1+3+4-6', '4+7+2+5plus+3+9-10', '4+3+1+1+1+7-8', '4+2+1+2+2
+0-1', '4+2+1+2+2+2-3', '4+2+1+3+3+2-3', '4+1+3+1+3+9-10', '4+1+3+1+2+9-10', '4+1+3
+1+2+7-8', '4+4+2+2+1+4-6', '5+7+1+4+2+9-10', '5+2+1+4+1+4-6', '5+7+2+5plus+1+9-10',
'5+7+2+5plus+3+7-8', '5+7+2+5plus+2+4-6', '5+2+2+5plus+1+7-8', '5+7+2+1+2+2-3', '5+6
+1+2+3+9-10', '5+5+2+5plus+2+9-10', '5+5+1+4+2+4-6', '5+2+2+4+1+9-10', '5+6+3+2+2
+9-10', '5+2+1+1+2+2-3', '5+2+1+3+2+0-1', '5+2+1+3+2+2-3', '5+2+2+3+3+9-10', '5+2+2
+3+3+4-6', '5+2+2+1+3+9-10', '5+6+1+1+2+7-8', '5+6+1+1+2+4-6', '5+6+1+3+2+4-6', '5+1
+1+2+2+0-1', '3+2+1+2+1+0-1', '5+6+2+5plus+2+4-6', '5+6+1+3+1+7-8', '5+1+1+2+3+2-3',
'5+6+2+5plus+1+7-8', '5+4+2+2+3+7-8', '6+2+3+2+2+9-10', '5+2+2+1+3+7-8', '6+4+2+4+1
+9-10', '6+7+2+2+3+7-8', '5+8+3+1+2+4-6', '5+8+3+1+3+7-8', '6+8+2+1+2+7-8', '6+8+3+1
+3+9-10', '5+2+1+5plus+1+7-8', '5+2+1+5plus+2+4-6', '6+2+1+5plus+1+4-6', '6+2+1+5plu
s+3+4-6', '6+5+2+3+3+7-8', '6+6+1+2+3+9-10', '6+1+1+2+1+9-10', '6+4+2+4+1+4-6', '6+4
+2+3+2+9-10', '6+4+2+2+2+9-10', '6+7+2+5plus+1+7-8', '6+7+2+5plus+3+9-10', '6+2+2+3
+1+9-10', '6+8+2+4+3+7-8', '6+6+2+5plus+2+2-3', '6+6+2+2+1+9-10', '6+3+1+2+3+4-6',
'6+8+2+2+2+4-6', '6+2+2+3+3+7-8', '6+2+2+5plus+2+4-6', '6+2+2+4+1+9-10', '6+8+2+5plu
s+1+9-10', '6+8+2+5plus+1+7-8', '6+1+2+1+3+7-8', '2+2+2+1+3+9-10', '2+8+3+2+1+9-10',
'2+8+3+1+2+7-8', '3+2+1+4+1+9-10', '3+4+2+1+1+4-6', '3+4+2+1+1+7-8', '4+3+2+2+3
+4-6', '5+7+1+3+2+7-8', '6+1+2+4+2+9-10', '6+1+1+4+2+2-3', '7+2+1+2+1+7-8', '7+2+1+2
+1+9-10', '7+4+1+2+3+9-10', '7+4+1+4+3+9-10', '7+4+1+5plus+3+9-10', '7+2+2+5plus+3
+9-10', '7+2+2+5plus+2+4-6', '7+6+2+2+1+7-8', '7+6+2+2+3+4-6', '7+6+2+1+3+7-8', '7+6
+2+1+3+4-6', '7+2+1+1+1+7-8', '7+1+1+1+3+7-8', '7+2+1+4+1+4-6', '7+2+2+4+3+7-8']
[>] Weights: Not provided
[>] Processing 1032 sequences with 2562 unique states.
  - Creating 2563x2563 substitution-cost matrix using 2 as constant value
[>] generated an indel of type number

[>] Identified 1032 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
Temps d'exécution : 3.04 s
```

Cost Additive Trick

In [58]:
```
extract_shp_sts_region = extract_shp_sts_region.astype(str)
extract_shp_sts_community = extract_shp_sts_community.astype(str)
extract_shp_sts_type_logement = extract_shp_sts_type_logement.astype(str)
extract_shp_sts_nbpers = extract_shp_sts_nbpers.astype(str)
extract_shp_sts_condlieuvie = extract_shp_sts_condlieuvie.astype(str)
extract_shp_sts_situatuionfinance = extract_shp_sts_situatuionfinance.astype(str)
```

In [55]:
```
start_time = time.time()
# Region (Conversion des états en STR)
shp_region_seq = sequenzo.SequenceData(
    data=extract_shp_sts_region,
```

```
    time=["region_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7]], # Force str
    id_col="idhous"
)
# Community
shp_community_seq = sequenzo.SequenceData(
    data=extract_shp_sts_community,
    time=["com2__"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7,8,9]], # Force str
    id_col="idhous"
)
# Type logement
shp_type_logement_seq = sequenzo.SequenceData(
    data=extract_shp_sts_type_logement,
    time=["hh29_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
# Condition de lieu de vie
shp_condlieuvie_seq = sequenzo.SequenceData(
    data=extract_shp_sts_condlieuvie,
    time=["hh14_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
# Nb personnes dans le foyer
shp_nbpers_seq = sequenzo.SequenceData(
    data=extract_shp_sts_nbpers,
    time=["nbpers_"+str(year) for year in range(1999,2019)],
    states=["1","2","3","4","5plus"],
    id_col="idhous"
)
# Situation financière
shp_situation_finance_seq = sequenzo.SequenceData(
    data=extract_shp_sts_situatuionfinance,
    time=["hi30_"+str(year) for year in range(1999,2019)],
    states=["0-1","2-3","4-6","7-8","9-10"],
    id_col="idhous"
)
distance_matrix = sequenzo.compute_cat_distance_matrix(
    [shp_region_seq, shp_community_seq, shp_type_logement_seq, shp_nbpers_seq, shp_
    method="OM",
    sm=["CONSTANT","CONSTANT","CONSTANT","CONSTANT","CONSTANT","CONSTANT"],
    indel=[1,1,1,1,1,1],
    what="diss"
)
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels = cluster_result.get_cluster_labels(num_clusters=4)
time_taken = time.time() - start_time
print(str(time_taken)+" s")
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5plus']
[>] Labels: ['1', '2', '3', '4', '5plus']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Labels: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Weights: Not provided
[>] 6 domains with 1032 sequences.
[>] Building MD sequences of combined states.
  - OK.
[>] Computing substitution cost matrix for domain 0.
[>] Computing substitution cost matrix for domain 1.
[>] Computing substitution cost matrix for domain 2.
[>] Computing substitution cost matrix for domain 3.
[>] Computing substitution cost matrix for domain 4.
[>] Computing substitution cost matrix for domain 5.
```

```
[>] Computing MD substitution and indel costs with additive trick.
  - OK.
[>] Computing MD distances using additive trick.
  - OK.
382.71762561798096 s
```

Distance Additive Trick

In [56]:
```
start_time = time.time()
# Region (Conversion des états en STR)
shp_region_seq = sequenzo.SequenceData(
    data=extract_shp_sts_region,
    time=["region_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7]], # Force str
    id_col="idhous"
)
# Community
shp_community_seq = sequenzo.SequenceData(
    data=extract_shp_sts_community,
    time=["com2__"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7,8,9]], # Force str
    id_col="idhous"
)
# Type logement
shp_type_logement_seq = sequenzo.SequenceData(
    data=extract_shp_sts_type_logement,
    time=["hh29_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
# Condition de lieu de vie
shp_condlieuvie_seq = sequenzo.SequenceData(
    data=extract_shp_sts_condlieuvie,
    time=["hh14_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
# Nb personnes dans le foyer
shp_nbpers_seq = sequenzo.SequenceData(
    data=extract_shp_sts_nbpers,
    time=["nbpers_"+str(year) for year in range(1999,2019)],
    states=["1","2","3","4","5plus"],
    id_col="idhous"
)
# Situation financière
shp_situation_finance_seq = sequenzo.SequenceData(
    data=extract_shp_sts_situatuionfinance,
    time=["hi30_"+str(year) for year in range(1999,2019)],
    states=["0-1","2-3","4-6","7-8","9-10"],
    id_col="idhous"
)
distance_matrix = sequenzo.compute_dat_distance_matrix(
    sequence_objects=[shp_region_seq, shp_community_seq, shp_type_logement_seq, shp
    method_params=[{"method": "OM", "sm": "CONSTANT", "indel": 1},{"method": "OM",
                   {"method": "OM", "sm": "CONSTANT", "indel": 1},{"method": "OM",
```

```
                    )
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels = cluster_result.get_cluster_labels(num_clusters=4)
time_taken = time.time() - start_time
print(str(time_taken)+" s")
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5plus']
[>] Labels: ['1', '2', '3', '4', '5plus']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Labels: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Weights: Not provided
[>] Processing 1032 sequences with 7 unique states.
  - Creating 8x8 substitution-cost matrix using 2 as constant value
[>] Identified 64 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 9 unique states.
  - Creating 10x10 substitution-cost matrix using 2 as constant value
[>] Identified 245 unique sequences.
```

```
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 3 unique states.
  - Creating 4x4 substitution-cost matrix using 2 as constant value
[>] Identified 145 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 5 unique states.
  - Creating 6x6 substitution-cost matrix using 2 as constant value
[>] Identified 529 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 3 unique states.
  - Creating 4x4 substitution-cost matrix using 2 as constant value
[>] Identified 746 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 5 unique states.
  - Creating 6x6 substitution-cost matrix using 2 as constant value
[>] Identified 987 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
1.028916597366333 s
```

Combining Trajectories

In [60]:

```
start_time = time.time()
# Region (Conversion des états en STR)
shp_region_seq = sequenzo.SequenceData(
    data=extract_shp_sts_region,
    time=["region_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7]], # Force str
    id_col="idhous"
)
# Community
shp_community_seq = sequenzo.SequenceData(
    data=extract_shp_sts_community,
    time=["com2__"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7,8,9]], # Force str
    id_col="idhous"
)
# Type logement
shp_type_logement_seq = sequenzo.SequenceData(
    data=extract_shp_sts_type_logement,
    time=["hh29_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
# Condition de lieu de vie
shp_condlieuvie_seq = sequenzo.SequenceData(
```

```
    data=extract_shp_sts_condlieuvie,
    time=["hh14_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
# Nb personnes dans le foyer
shp_nbpers_seq = sequenzo.SequenceData(
    data=extract_shp_sts_nbpers,
    time=["nbpers_"+str(year) for year in range(1999,2019)],
    states=["1","2","3","4","5plus"],
    id_col="idhous"
)
# Situation financière
shp_situation_finance_seq = sequenzo.SequenceData(
    data=extract_shp_sts_situatuionfinance,
    time=["hi30_"+str(year) for year in range(1999,2019)],
    states=["0-1","2-3","4-6","7-8","9-10"],
    id_col="idhous"
)
diss_matrices, membership_df = sequenzo.get_interactive_combined_typology(
    domains=[shp_region_seq, shp_community_seq, shp_type_logement_seq, shp_nbpers_s
    method_params=[{"method": "OM", "sm": "CONSTANT", "indel": 1},{"method": "OM",
                   {"method": "OM", "sm": "CONSTANT", "indel": 1},{"method": "OM",
)
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels = cluster_result.get_cluster_labels(num_clusters=4)
time_taken = time.time() - start_time
print(str(time_taken)+" s")
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5plus']
[>] Labels: ['1', '2', '3', '4', '5plus']
[>] Weights: Not provided

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Labels: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Weights: Not provided
[>] Processing 1032 sequences with 7 unique states.
  - Creating 8x8 substitution-cost matrix using 2 as constant value
[>] Identified 64 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 9 unique states.
  - Creating 10x10 substitution-cost matrix using 2 as constant value
[>] Identified 245 unique sequences.
```

```
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 3 unique states.
  - Creating 4x4 substitution-cost matrix using 2 as constant value
[>] Identified 145 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 5 unique states.
  - Creating 6x6 substitution-cost matrix using 2 as constant value
[>] Identified 529 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 3 unique states.
  - Creating 4x4 substitution-cost matrix using 2 as constant value
[>] Identified 746 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
[>] Processing 1032 sequences with 5 unique states.
  - Creating 6x6 substitution-cost matrix using 2 as constant value
[>] Identified 987 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.

[>] Processing domain: Domain_1
[!] Warning: Could not create or save plot: 'ClusterQuality' object has no attribute
'plot_combined_scores'
[>] Continuing without visualization...

[>] Processing domain: Domain_2
[!] Warning: Could not create or save plot: 'ClusterQuality' object has no attribute
'plot_combined_scores'
[>] Continuing without visualization...

[>] Processing domain: Domain_3
[!] Warning: Could not create or save plot: 'ClusterQuality' object has no attribute
'plot_combined_scores'
[>] Continuing without visualization...

[>] Processing domain: Domain_4
[!] Warning: Could not create or save plot: 'ClusterQuality' object has no attribute
'plot_combined_scores'
[>] Continuing without visualization...

[>] Processing domain: Domain_5
[!] Warning: Could not create or save plot: 'ClusterQuality' object has no attribute
'plot_combined_scores'
[>] Continuing without visualization...

[>] Processing domain: Domain_6
[!] Warning: Could not create or save plot: 'ClusterQuality' object has no attribute
'plot_combined_scores'
```

```
[>] Continuing without visualization...

[>] Combined Typology Membership Table Preview:
    id  Domain_1_Cluster  Domain_2_Cluster  Domain_3_Cluster  \
0  271                 1                 1                 1
1  341                 1                 1                 2
2  391                 1                 2                 3
3  741                 1                 3                 1
4  991                 1                 2                 3

   Domain_4_Cluster  Domain_5_Cluster  Domain_6_Cluster        CombT
0                 1                 1                 1  1+1+1+1+1+1
1                 2                 2                 2  1+1+2+2+2+2
2                 3                 3                 2  1+2+3+3+3+2
3                 1                 3                 1  1+3+1+1+3+1
4                 4                 4                 2  1+2+3+4+4+2

[>] combt_membership_table.csv has been saved.

[>] CombT Frequency Table:
           CombT  Frequency  Proportion (%)
0    3+2+3+2+4+2          7            0.68
1    4+3+3+2+4+2          7            0.68
2    2+2+3+2+3+2          6            0.58
3    3+3+3+2+3+2          6            0.58
4    4+1+1+2+4+2          6            0.58
..           ...        ...             ...
694  4+4+3+3+4+2          1            0.10
695  4+1+4+2+3+4          1            0.10
696  3+1+1+2+4+3          1            0.10
697  3+3+2+2+3+1          1            0.10
698  3+3+2+3+2+1          1            0.10

[699 rows x 3 columns]

[>] freq_table.csv has been saved.

[>] Frequency of Combined Typologies.png has been saved.
13.17542028427124 s
```

In [57]:

```
start_time = time.time()
# Region (Conversion des états en STR)
shp_region_seq = sequenzo.SequenceData(
    data=extract_shp_sts_region,
    time=["region_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7]], # Force str
    id_col="idhous"
)
distance_matrix = sequenzo.get_distance_matrix(shp_region_seq, method="OM", sm="CON
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels_region = cluster_result.get_cluster_labels(num_clusters=4)
# Community
shp_community_seq = sequenzo.SequenceData(
    data=extract_shp_sts_community,
    time=["com2__"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3,4,5,6,7,8,9]], # Force str
```

```
    id_col="idhous"
)
distance_matrix = sequenzo.get_distance_matrix(shp_community_seq, method="OM", sm="
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels_community = cluster_result.get_cluster_labels(num_clusters=4)
# Type logement
shp_type_logement_seq = sequenzo.SequenceData(
    data=extract_shp_sts_type_logement,
    time=["hh29_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
distance_matrix = sequenzo.get_distance_matrix(shp_type_logement_seq, method="OM",
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels_type_logement = cluster_result.get_cluster_labels(num_clusters=4)
# Condition de lieu de vie
shp_condlieuvie_seq = sequenzo.SequenceData(
    data=extract_shp_sts_condlieuvie,
    time=["hh14_"+str(year) for year in range(1999,2019)],
    states=[str(s) for s in [1,2,3]], # Force str
    id_col="idhous"
)
distance_matrix = sequenzo.get_distance_matrix(shp_condlieuvie_seq, method="OM", sm
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels_condlieuvie = cluster_result.get_cluster_labels(num_clusters=4)
# Nb personnes dans le foyer
shp_nbpers_seq = sequenzo.SequenceData(
    data=extract_shp_sts_nbpers,
    time=["nbpers_"+str(year) for year in range(1999,2019)],
    states=["1","2","3","4","5plus"],
    id_col="idhous"
)
distance_matrix = sequenzo.get_distance_matrix(shp_nbpers_seq, method="OM", sm="CON
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels_nbpers = cluster_result.get_cluster_labels(num_clusters=4)
# Situation financière
shp_situation_finance_seq = sequenzo.SequenceData(
    data=extract_shp_sts_situatuionfinance,
    time=["hi30_"+str(year) for year in range(1999,2019)],
    states=["0-1","2-3","4-6","7-8","9-10"],
    id_col="idhous"
)
distance_matrix = sequenzo.get_distance_matrix(shp_situation_finance_seq, method="O
cluster_result = sequenzo.clustering.hierarchical_clustering.Cluster(matrix=distanc
labels_finance = cluster_result.get_cluster_labels(num_clusters=4)
# Combinaison des labels individuels pour chaque domaine
combined_labels = labels_region.astype(str) + "_" + labels_community.astype(str) +
time_taken = time.time() - start_time
print(str(time_taken)+" s")
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7']
[>] Weights: Not provided
[>] Processing 1032 sequences with 7 unique states.
  - Creating 8x8 substitution-cost matrix using 2 as constant value
[>] Identified 64 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Labels: ['1', '2', '3', '4', '5', '6', '7', '8', '9']
[>] Weights: Not provided
[>] Processing 1032 sequences with 9 unique states.
  - Creating 10x10 substitution-cost matrix using 2 as constant value
[>] Identified 245 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided
[>] Processing 1032 sequences with 3 unique states.
  - Creating 4x4 substitution-cost matrix using 2 as constant value
[>] Identified 145 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3']
[>] Labels: ['1', '2', '3']
[>] Weights: Not provided
[>] Processing 1032 sequences with 3 unique states.
  - Creating 4x4 substitution-cost matrix using 2 as constant value
[>] Identified 746 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
```

```
[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['1', '2', '3', '4', '5plus']
[>] Labels: ['1', '2', '3', '4', '5plus']
[>] Weights: Not provided
[>] Processing 1032 sequences with 5 unique states.
  - Creating 6x6 substitution-cost matrix using 2 as constant value
[>] Identified 529 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.

[>] SequenceData initialized successfully! Here's a summary:
[>] Number of sequences: 1032
[>] Number of time points: 20
[>] Min/Max sequence length: 20 / 20
[>] States: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Labels: ['0-1', '2-3', '4-6', '7-8', '9-10']
[>] Weights: Not provided
[>] Processing 1032 sequences with 5 unique states.
  - Creating 6x6 substitution-cost matrix using 2 as constant value
[>] Identified 987 unique sequences.
[>] Starting Optimal Matching(OM)...
[>] Computing all pairwise distances...
[>] Computed Successfully.
1.4333527088165283 s
```